Fermionic Gaussian states and Majorana covariance matrices

A system of n fermionic modes is described by $2n$ Majorana operators $c_1, \ldots, c_{2n}$ [2]: Hermitian, and satisfying $c_j c_k + c_k c_j = 2\delta_{jk}$. A state ρ assigns to each pair a second moment $\mathrm{Tr}(\rho\, c_j c_k)$; the anticommutation relation fixes the symmetric part of these numbers, and the remaining antisymmetric part,

$$\Gamma_{jk} = \tfrac{\mathrm{i}}{2}\,\mathrm{Tr}(\rho\,[c_j, c_k]),$$

is the *Majorana covariance* of ρ, a real antisymmetric $2n \times 2n$ matrix.

A state is *Gaussian* when its moments are given from Γ by Wick's rule (Definition 9.1) [1, Sec. 2]. The class contains the ground and thermal states of quadratic Hamiltonians and is closed under unitaries generated by quadratic Hamiltonians, discarding modes, and measuring quadratic observables [3]. These notes prove which matrices are covariances of states, how Γ changes under the unitaries and under discarding modes, and how the covariance of a chain is assembled from on-site covariances joined by maximally entangled pairs.

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About these notes

These notes are the mathematical account of the Lean library FermionicLean, which depends on Mathlib alone. We assume finite-dimensional linear algebra and complex inner products. The reading guide after the notation relates our conventions to those of the literature, the roadmap shows which sections use which, and an index of terms precedes the references.

The notes have two parts. Part I treats covariance matrices alone. Section 1 gives the two conditions that characterize them, their equivalent forms, and the operations that preserve them. Section 2 defines the contraction of a covariance against a pair covariance, in particular a bond, proves that it preserves admissibility, and fixes the bond covariance by a teleportation identity. Section 3 builds the covariance of a periodic chain from on-site covariances joined by bonds and evaluates a three-site example. Section 4 passes to momentum space for a chain whose on-site covariances all agree, where the chain covariance splits into one $2P \times 2P$ block per momentum. Every statement of Part I is a statement about matrices; the states whose covariances they are appear in Part II.

Part II constructs the states. Section 5 builds the Majorana operators as explicit matrices on the 2^n-dimensional Fock space, proves the anticommutation relations, and shows that the covariance of every density operator satisfies the conditions of Section 1. It computes the covariances of the occupation-number states, whose vacuum fixes the sign convention, and of the two-mode state whose covariance is the bond.

Section 6 places explicit states in the chain construction: occupation states as chains without virtual modes and the Kitaev chain; it also treats hopping gates and proves the congruence rule $\Gamma(U\rho U^*) = O^{\mathsf{T}}\Gamma(\rho)O$ (Theorem 6.9) used from Section 7 on. Section 7 proves that every admissible matrix is the covariance of some state and identifies the ground states of quadratic Hamiltonians. Section 8 treats the thermal states $e^{-\tau H_h}/\operatorname{Tr} e^{-\tau H_h}$ and the unitaries $e^{-\mathrm{i}tH_h}$ of those Hamiltonians.

Section 9 defines the Gaussian class by Wick's rule, shows that the states constructed so far are Gaussian, and proves that a Gaussian state is determined by its covariance; the results before it hold for arbitrary density operators. Section 10 performs the contraction of Section 2 on Fock space and identifies the periodic chain of Section 3 with a Gaussian state, the Gaussian fermionic matrix product state (Gaussian fMPS) of [4]; Section 11 proves that this state is pure for pure on-site covariances and that the Gaussian fMPS of the Kitaev chain is the ground state of the Kitaev Hamiltonian H_{-K_N}, with gap 1. Section 12 bounds the rank of the correlations across a cut of the chain by the number of virtual modes on the cut bonds.

Every numbered theorem, lemma, proposition and corollary carries the mark Lean ✓: it has a checked proof in FermionicLean whose correspondence with the statement here has been independently reviewed, and the declaration index names the declarations; the rest is checked on paper, or numerically by the scripts that a section's closing Checks paragraph names. A sentence tagged formalization records a choice of the formalization, or the route of its proof, that a proof on paper would not need; the mathematics does not depend on it.

Notation

Each symbol has one meaning throughout; the section in parentheses is where it is defined. Exceptions, each fixed where it occurs: S (a set of labels in Section 9, a set of sites in Section 12); A, B, C (the blocks of a matrix on $R \sqcup E$, and the mode sets A_i, B_i, C_i of a chain); p (the probabilities p_j, the vectors p_k of the normal form, the pair (p, q) of a plane rotation, the sign p_j in the proof of Theorem 5.4); and M, X, c, e, η. The sans-serif S and the calligraphic $\mathcal{K}$ of Remark 9.9 are distinct from S and K_N.

x_j, y_j	the two Majorana labels of mode j, in block order $(x_1, \dots, x_n, y_1, \dots, y_n)$ (1)
Ω_n, $\mathsf{D}(\varepsilon)$	the standard covariance and the canonical form (1, 7)
$\Gamma(\rho)$, $\mathsf{G}(\rho) = 1 - \mathrm{i}\Gamma(\rho)$	the covariance and the two-point matrix of a state (5, 9)
admissible, pure	$\Gamma^{\mathsf{T}} = -\Gamma$ with $\Gamma^{\mathsf{T}}\Gamma \preceq 1$, resp. $\Gamma^{\mathsf{T}}\Gamma = 1$ (1)
R, E; A, B, C, D	retained and eliminated labels; the blocks of a matrix on $R \sqcup E$ (2)
$\Gamma \star \Gamma_p$, β, J	the contraction, the bond covariance (in Section 10 also the covariance of the state σ on the eliminated modes), its inner block (2)
Q, L, L'	the physical modes, the left and the right virtual modes of the chain (3)
I_{site}, I_i	the Majorana labels of one site, of site i (3)
Γ_{in}, β_N, Γ_N	input matrix, bonds and covariance of the periodic chain (3)
S	the cyclic shift on $\mathbb{Z}/3$, in the proof of Proposition 3.6 (3)
χ_k, ω_k, Φ_N	characters, phases, the Fourier matrix (4)
$\Gamma^{(k)}$, $\Sigma(k)$	the twisted on-site covariance, the momentum-space covariance (4)
$\mathcal{F}_n$, $\lvert s\rangle$, $s^{(j)}$	Fock space, configurations, the flip of bit j (5)
X_j, Z_j, $Z_{<j}$; c_a	Pauli matrices, the string $Z_1 \cdots Z_{j-1}$, the Majorana operators (5)
$\varsigma(b)$, $\nu(b)$, Λ_s	the sign $(-1)^b$, the occupation sign $2b - 1$, $\mathrm{diag}(\nu(s_j))$ (5)
ρ_s, ρ_{vac}	occupation states and the vacuum (5)
K_N, Γ_K	the Kitaev chain covariance and its on-site covariance (6)
H_h, $\mathcal{E}_h(\rho)$, $\mathcal{E}_{\mathrm{gs}}$	the quadratic Hamiltonian of h, its energy, the ground energy (7, 11)
$d_\varepsilon(s)$	the diagonal entry of $H_{\mathsf{D}(\varepsilon)}$ at the configuration s (7)
$\mathsf{R}_{ab}(p, q)$, $\mathcal{R}$	a plane rotation (products of them: R; $\mathsf{R}(\theta)$ in 8), the reflection of y_1 (7)
ρ_λ, ρ_{gs}	the product mixture, the ground state (7)
U implements O	$U c_a U^* = \sum_b O_{ab} c_b$; then $\Gamma(U^* \rho U) = O\Gamma(\rho)O^{\mathsf{T}}$ (7)
$\mathcal{Z}(h)$, ρ_h, $\rho_{\tau h}$	partition function, Gibbs state, thermal state at inverse temperature τ (8)
W_{G}, c_S, Π_n	the Wick expansion, the monomial of a label set S, the parity operator (9)
$\rho[\Gamma]$	the Gaussian state with covariance Γ (9)
Tr_E, $\rho \star \sigma$, $\mathcal{N}(\rho, \sigma)$	partial trace, contraction of states, its normalization (10)
ρ_N	the Gaussian fermionic matrix product state (fMPS) (10)
Γ_S, C_S, $b(S)$	block covariance and cross block of a site set S, its number of cut bonds (12)

Covariances and conventions: a reading guide

Texts on fermionic Gaussian states, for example [1, 4, 6], describe a state of n modes by a matrix of second moments of size $2n \times 2n$, and they differ in which matrix they take, in its sign, in the order of its rows and columns, and in the definition of the Gaussian class. These notes make one choice throughout: the real antisymmetric Majorana covariance Γ in block order, with the sign for which the vacuum is $-\Omega_n$, and the Gaussian class defined by Wick's rule. The paragraphs below translate each choice into the others and name the statement where the translation is proved, or say that it is a computation on paper.

The matrix. The second moments of a density operator ρ form the two-point matrix $\mathsf{G}(\rho)_{ab} = \mathrm{Tr}(\rho\, c_a c_b) = \delta_{ab} - \mathrm{i}\,\Gamma(\rho)_{ab}$ (Section 9), which part of the literature calls the correlation matrix, while Γ itself is often called the covariance matrix. Its symmetric part is fixed by the anticommutation relations, so Γ carries the same information in real form. The matrix of the annihilation operators $a_j = \frac{1}{2}(c_{x_j} + \mathrm{i}c_{y_j})$ follows from G by expanding the products, a computation on paper:

$$\mathrm{Tr}(\rho\, a_j^* a_k) = \tfrac{1}{2}\delta_{jk} + \tfrac{1}{4}(\Gamma_{x_j y_k} + \Gamma_{x_k y_j}) - \tfrac{\mathrm{i}}{4}(\Gamma_{x_j x_k} + \Gamma_{y_j y_k}),$$

whose diagonal $j = k$ is the occupation $\frac{1}{2}(1 + \Gamma_{x_j y_j})$ of mode j (Section 5).

The sign. The vacuum has $\Gamma = -\Omega_n$ (Corollary 5.9), and $\Gamma_{x_j y_j}$ is -1 for an empty and $+1$ for an occupied mode. The sign comes from the factor $+\frac{\mathrm{i}}{2}$ in Definition 5.5 together with the choice $c_{y_j} = \mathrm{i}Z_{<j}X_jZ_j$ of Definition 5.3, under which a_j annihilates the vacuum. Replacing the factor by $-\frac{\mathrm{i}}{2}$ replaces Γ by $-\Gamma$, so that the vacuum has $+\Omega_n$, the convention of part of the literature, and the ground covariance $\mathrm{i}\,\mathrm{sgn}(\mathrm{i}h)$ of H_h (after Theorem 7.20) becomes $-\mathrm{i}\,\mathrm{sgn}(\mathrm{i}h)$. Replacing c_{y_j} by $-c_{y_j}$ conjugates Γ by the reflection $\mathrm{diag}(1_n, -1_n)$ of the y-labels; then a_j annihilates the state with every mode occupied, whose covariance is again $-\Omega_n$, and with H_h written in the new operators the ground covariance remains $\mathrm{i}\,\mathrm{sgn}(\mathrm{i}h)$, by a computation on paper. The sign of the hopping in the Kitaev chain is a gauge (Remark 11.11).

Order of the labels. In the interleaved numbering of [1] and [4], $c_{2j-1} = x_j$ and $c_{2j} = y_j$. A matrix in block order becomes the matrix in interleaved order by permuting rows and columns by $x_j \mapsto 2j-1$, $y_j \mapsto 2j$, which leaves admissibility and purity unchanged (Lemma 3.3). In interleaved order Ω_n becomes $\bigoplus_j \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and the canonical form $\mathsf{D}(\varepsilon)$ of Definition 7.1 becomes $\bigoplus_j \varepsilon_j \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.

Gaussian unitaries. A unitary U implements a real matrix O when $U c_a U^* = \sum_b O_{ab} c_b$; then O is orthogonal and $\Gamma(U\rho U^*) = O^{\mathsf{T}}\Gamma(\rho)O$ (Theorem 6.9). A text that writes the Heisenberg form $U^* c_a U = \sum_b O'_{ab} c_b$ has $O' = O^{\mathsf{T}}$, and the same rule reads $\Gamma(U\rho U^*) = O'\,\Gamma(\rho)O'^{\mathsf{T}}$. The evolution $e^{-\mathrm{i}tH_h}$ implements e^{-th} and moves covariances by $\Gamma \mapsto e^{th}\Gamma e^{-th}$ (Theorem 8.13).

Contraction. The contraction $\Gamma \star \Gamma_p = A + B\,\Gamma_p(1 - D\Gamma_p)^{-1}C$ of Definition 2.1 is written for any pair covariance Γ_p. On Fock space, for a Gaussian state ρ with covariance Γ, a Gaussian state σ on the eliminated modes with covariance Γ_p, and $1 - D\Gamma_p$ invertible, it is the covariance of the normalized state ρ traced against σ (Corollary 10.11 and Theorem 10.14). For antisymmetric Γ and pure Γ_p, as for the bonds of a chain, it equals $A + B(D + \Gamma_p)^{-1}B^{\mathsf{T}}$ (Theorem 2.6), the form of [4, Sec. II]. The sign of that formula depends on the orientation of the pair, which Definition 2.7 fixes and Theorem 2.9 checks; reversing the orientation replaces Γ_p by $-\Gamma_p$. On Fock space the bond is the state $(|00\rangle + \mathrm{i}|11\rangle)/\sqrt{2}$ (Theorem 5.10); it differs from $|00\rangle + |11\rangle$ by a phase on one mode, and that phase makes the bond pair x with x and y with y.

The Gaussian class. A density operator is Gaussian here when its moments are given by Wick's rule from its own two-point matrix (Definition 9.1); Wick's rule is a condition beyond the second moments (Remark 9.3), and its recursion is the first-row expansion of the Pfaffian form of [1] (after Definition 9.1). Gaussian states are often defined instead as normalized exponentials of quadratic Hamiltonians together with their limits. The notes connect the two descriptions: the Gibbs and thermal states are Gaussian (Corollary 9.11), every Gaussian state is a rotated product mixture $U^*\rho_\lambda U$, a pure one is $U^*|s\rangle\langle s|U$, and one with strictly admissible covariance is a Gibbs state (Corollary 9.16). A Gaussian state is determined by its covariance (Theorem 9.15).

Momentum space. Section 4 uses the characters $\chi_k(i) = e^{-2\pi\mathrm{i}\,ki/N}$ and expands the chain covariance in $e^{+2\pi\mathrm{i}\,k(i-j)/N}\,\Sigma(k)$; in the opposite convention the block at momentum k is $\Sigma(-k) = \overline{\Sigma(k)}$ (Theorem 4.10).

Roadmap

The sections are ordered by logical dependence. Part I is a theory of real antisymmetric matrices: no statement or proof in it uses Part II, which it cites only in unnumbered text: to read its matrices as covariances of states, and for the purity of the chain covariance (Corollary 11.5). Part II builds the states on Fock space and uses Part I throughout; its last three sections return to the chain of Section 3 as a state. Figure 1 shows which sections use which.

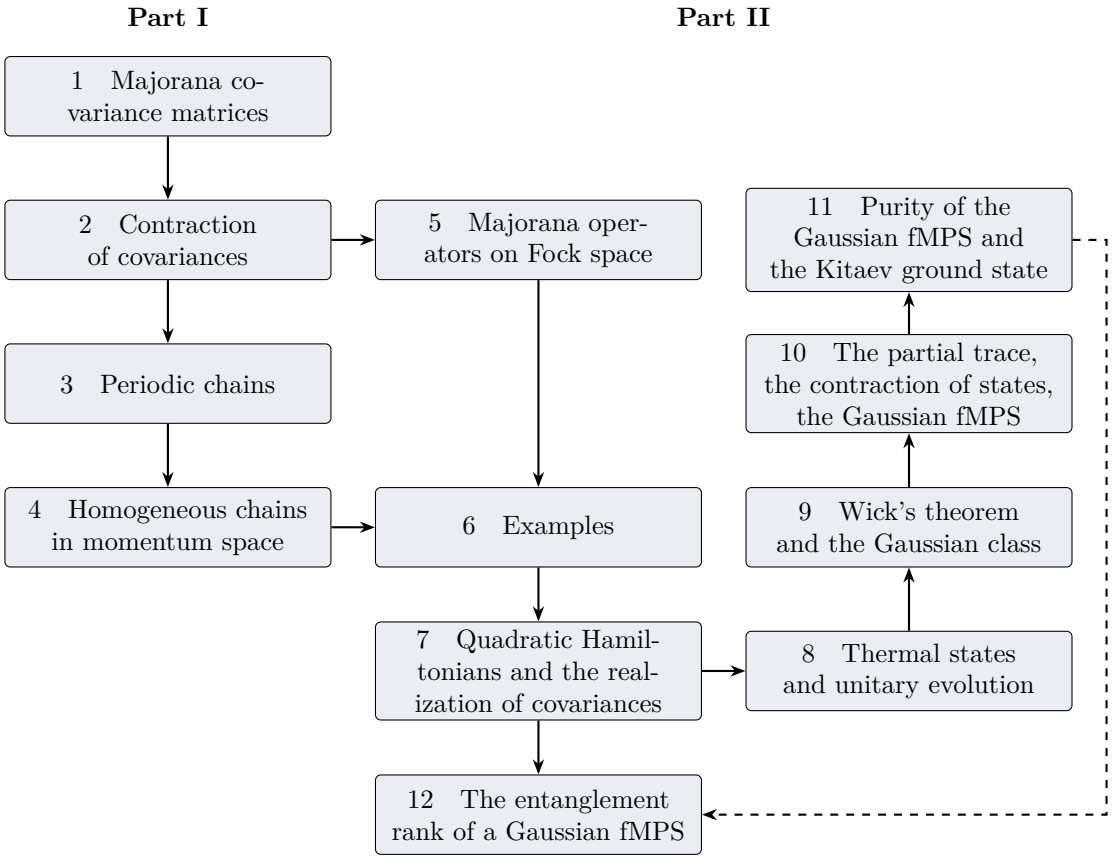


Figure 1: Dependence of the sections. An arrow means that the later section uses the earlier one, and arrows implied by chains of solid arrows are omitted; a chain records a possible dependence through intermediate sections, not a use of every section on it. The dashed arrow is a use in an example only: Corollary 12.5 names the pure on-site covariances of Corollary 11.5 as one case of its hypothesis.

Several uses are narrower than the chains suggest. Sections 7 to 9 use none of Sections 2 to 4, and Section 7 uses Section 6 only for Theorems 6.8 and 6.9, which rest on Section 5 alone; Section 5 uses Section 2 only for Definition 2.7. Section 10 uses Section 3 for the definitions of the chain and Lemma 3.3. Theorem 12.4 uses only Sections 2 and 3; Section 12 uses Section 7 only for the normal form (Theorem 7.13) and Section 6 only in remarks and asides.

Part I
Covariance matrices

1 Majorana covariance matrices

For $n \geq 0$, use coordinates $(x_1, \ldots, x_n, y_1, \ldots, y_n)$ on $\mathbb{R}^{2n}$; for each mode j we label its two Majorana operators x_j and y_j. All matrices are real unless stated otherwise, and $\|v\|^2 = v^{\mathsf{T}} v$ is the squared Euclidean norm. For a real symmetric matrix A, the notation $A \succeq 0$ means $v^{\mathsf{T}} A v \geq 0$ for every real column vector v. For a complex matrix H, $H \succeq 0$ means that $H = H^*$ and $z^* H z \geq 0$ for every complex column vector z, where H^* is the conjugate transpose. We regard real matrices as complex matrices when they enter the complex inequality. The case $n = 0$ is included throughout: the empty matrix satisfies every condition below.

The two conditions of Definition 1.1 are the ones a state imposes. If $c_1, \ldots, c_{2n}$ are Hermitian operators with $c_j c_k + c_k c_j = 2\delta_{jk}$ and ρ is a density operator, the numbers $\Gamma_{jk} = \frac{\mathrm{i}}{2} \operatorname{Tr}(\rho[c_j, c_k])$ are real and antisymmetric, and $1 + \mathrm{i}\Gamma \succeq 0$ is exactly the statement $\operatorname{Tr}(\rho A^* A) \geq 0$ for the operators $A = \sum_j \overline{z_j}\, c_j$. Section 5 proves both facts (Theorems 5.6 and 5.7); this section derives only what follows from the matrix conditions.

Definition 1.1 (Admissible covariance)**.** A real $2n \times 2n$ matrix Γ is called *admissible* (a physical covariance) if

$$\Gamma^{\mathsf{T}} = -\Gamma, \qquad 1 - \Gamma^{\mathsf{T}} \Gamma \succeq 0. \tag{1.1}$$

The matrix $1 - \Gamma^{\mathsf{T}}\Gamma$ is symmetric for every Γ, so the second condition is the real positive semidefiniteness of a symmetric matrix.

A second condition singles out the covariances of pure states: for a Gaussian state, $\Gamma^2 = -1$ holds exactly when the state is pure [1, Sec. 2] (Theorem 11.2 in Section 11). Here the condition is taken as the definition.

Definition 1.2 (Pure covariance)**.** A real $2n \times 2n$ matrix Γ is called *pure* if $\Gamma^{\mathsf{T}} = -\Gamma$ and $\Gamma^2 = -1$.

A pure covariance is orthogonal, since $\Gamma^{\mathsf{T}}\Gamma = -\Gamma^2 = 1$, and it is a complex structure on $\mathbb{R}^{2n}$: since $(\Gamma - \mathrm{i})(\Gamma + \mathrm{i}) = \Gamma^2 + 1 = 0$, Γ is diagonalizable over $\mathbb{C}$ with eigenvalues $\pm\mathrm{i}$ (a matrix annihilated by a polynomial with distinct roots is diagonalizable), and complex conjugation maps the i-eigenspace onto the $-$i-eigenspace because Γ is real, so each has dimension n.

For any real matrix A and real vector v, $v^{\mathsf{T}} A v = v^{\mathsf{T}} A^{\mathsf{T}} v$; hence $v^{\mathsf{T}}\Gamma v = 0$ when Γ is antisymmetric, and $a^{\mathsf{T}}\Gamma b = -b^{\mathsf{T}}\Gamma a$ for all real a, b. For any real matrix O and real symmetric $A \succeq 0$, the congruence OAO^{T} is symmetric and $v^{\mathsf{T}} OAO^{\mathsf{T}} v = (O^{\mathsf{T}} v)^{\mathsf{T}} A (O^{\mathsf{T}} v) \geq 0$, so $OAO^{\mathsf{T}} \succeq 0$.

1.1 Equivalent forms of admissibility

Theorem 1.3 (Norm form)**.** Lean ✓ *Let $\Gamma^{\mathsf{T}} = -\Gamma$. Then Γ is admissible if and only if $\|\Gamma v\| \leq \|v\|$ for every $v \in \mathbb{R}^{2n}$.*

Proof. For real v, $v^{\mathsf{T}}(1 - \Gamma^{\mathsf{T}}\Gamma)v = \|v\|^2 - \|\Gamma v\|^2$. □

The real form is the one the operations of Section 1.2 preserve (Theorems 1.7–1.10); the Hermitian form is the one a state delivers, since $z^*(1 + \mathrm{i}\Gamma)z = \operatorname{Tr}(\rho A^* A)$ for $A = \sum_j \overline{z_j} c_j$ (Theorem 5.7).

Theorem 1.4 (Hermitian form)**.** Lean ✓ *Let $\Gamma^{\mathsf{T}} = -\Gamma$. Then Γ is admissible if and only if $1 + \mathrm{i}\Gamma \succeq 0$ as a complex matrix.*

Proof. The matrix $1 + \mathrm{i}\Gamma$ is Hermitian, since $(\mathrm{i}\Gamma)^* = -\mathrm{i}\Gamma^{\mathsf{T}} = \mathrm{i}\Gamma$. Write $z = a + \mathrm{i}b$ with a, b real. Then $z^* = a^{\mathsf{T}} - \mathrm{i}b^{\mathsf{T}}$ and

$$z^*(1 + \mathrm{i}\Gamma)z = (a^{\mathsf{T}} - \mathrm{i}b^{\mathsf{T}})(a + \mathrm{i}b) + \mathrm{i}(a^{\mathsf{T}} - \mathrm{i}b^{\mathsf{T}})\Gamma(a + \mathrm{i}b) = \|a\|^2 + \|b\|^2 + \mathrm{i}(\mathrm{i}\, a^{\mathsf{T}}\Gamma b - \mathrm{i}\, b^{\mathsf{T}}\Gamma a),$$

using $a^{\mathsf{T}}\Gamma a = b^{\mathsf{T}}\Gamma b = 0$. With $a^{\mathsf{T}}\Gamma b = -b^{\mathsf{T}}\Gamma a$ this gives

$$z^*(1 + \mathrm{i}\Gamma)z = \|a\|^2 + \|b\|^2 + 2\, b^{\mathsf{T}}\Gamma a. \tag{1.2}$$

Suppose $\|\Gamma a\| \le \|a\|$ for all a. Expanding $0 \le \|b + \Gamma a\|^2$ gives $2\, b^{\mathsf{T}}\Gamma a \ge -\|b\|^2 - \|\Gamma a\|^2$, so the right side of (1.2) is at least $\|a\|^2 - \|\Gamma a\|^2 \ge 0$. Conversely, if the right side is nonnegative for all a, b, the choice $b = -\Gamma a$ gives $\|a\|^2 - \|\Gamma a\|^2 \ge 0$. Theorem 1.3 completes both directions. □

Remark 1.5. For bosonic covariances the corresponding condition is $\gamma - \mathrm{i}\Omega_n \succeq 0$ with Ω_n the symplectic form on the quadratures; for fermions the identity and the symplectic form exchange roles. The coordinates x_j, y_j of these notes label the two Majorana operators of a mode and are not quadratures.

Theorem 1.6 (Purity). Lean ✓ *Every pure covariance is admissible, satisfies* $\Gamma^{\mathsf{T}}\Gamma = 1$*, and is an isometry:* $\|\Gamma v\| = \|v\|$ *for all* v*.*

Proof. $\Gamma^{\mathsf{T}}\Gamma = -\Gamma^2 = 1$, so $1 - \Gamma^{\mathsf{T}}\Gamma = 0 \succeq 0$, and $\|\Gamma v\|^2 = v^{\mathsf{T}}\Gamma^{\mathsf{T}}\Gamma v = \|v\|^2$. □

1.2 Changes of basis, subsystems and direct sums

A unitary U that maps each Majorana operator to a real linear combination, $U c_j U^* = \sum_k O_{jk} c_k$ with O real, has O orthogonal, because the images again satisfy $c_j c_k + c_k c_j = 2\delta_{jk}$; the state $U^*\rho U$ then has covariance $O\Gamma O^{\mathsf{T}}$ [1, Sec. 2] (U implements O in the terminology of Section 7). Admissibility and purity survive this congruence (Theorem 1.7). Discarding modes restricts Γ to the remaining coordinates; admissibility survives (Theorem 1.8), purity need not (Remark 1.9). Preparing two systems independently gives the direct sum of their covariances, and admissibility and purity are decided blockwise (Theorem 1.10). We build matrix product states from these three operations and a fourth, contraction (Section 2).

Theorem 1.7 (Orthogonal congruence). Lean ✓ *Let* O *be a real* $2n \times 2n$ *matrix with* $OO^{\mathsf{T}} = 1$*. If* Γ *is admissible, then* $O\Gamma O^{\mathsf{T}}$ *is admissible; if* Γ *is pure, then* $O\Gamma O^{\mathsf{T}}$ *is pure.*

Proof. Since O is square, $OO^{\mathsf{T}} = 1$ makes O invertible with inverse O^{T}, so $O^{\mathsf{T}}O = 1$ as well. Antisymmetry: $(O\Gamma O^{\mathsf{T}})^{\mathsf{T}} = O\Gamma^{\mathsf{T}}O^{\mathsf{T}} = -O\Gamma O^{\mathsf{T}}$. Inserting $O^{\mathsf{T}}O = 1$, $(O\Gamma O^{\mathsf{T}})^{\mathsf{T}}(O\Gamma O^{\mathsf{T}}) = O\Gamma^{\mathsf{T}}\Gamma O^{\mathsf{T}}$, hence $1 - (O\Gamma O^{\mathsf{T}})^{\mathsf{T}}(O\Gamma O^{\mathsf{T}}) = O(1 - \Gamma^{\mathsf{T}}\Gamma)O^{\mathsf{T}} \succeq 0$ by the congruence fact above. For purity, $(O\Gamma O^{\mathsf{T}})^2 = O\Gamma^2 O^{\mathsf{T}} = -OO^{\mathsf{T}} = -1$. □

To select a subsystem, we take both Majorana coordinates of each chosen mode together. An injective map $e : \{1, \dots, m\} \to \{1, \dots, n\}$ lists m distinct modes; write $\bar{e}(x_a) = x_{e(a)}$ and $\bar{e}(y_a) = y_{e(a)}$ for the induced map on coordinates.

Theorem 1.8 (Mode selection). Lean ✓ *Let* e *be injective and let* Γ *be admissible. Then the* $2m \times 2m$ *matrix* $\Gamma'_{\alpha\beta} = \Gamma_{\bar{e}(\alpha)\bar{e}(\beta)}$ *is admissible. The empty selection* $m = 0$ *is included.*

Proof. Antisymmetry passes to the restriction, since $\Gamma'_{\beta\alpha} = \Gamma_{\bar{e}(\beta)\bar{e}(\alpha)} = -\Gamma_{\bar{e}(\alpha)\bar{e}(\beta)}$. For $v \in \mathbb{R}^{2m}$ let $\tilde{v} \in \mathbb{R}^{2n}$ have $\tilde{v}_{\bar{e}(\alpha)} = v_\alpha$ and all other entries zero; this is well defined because $\bar{e}$ is injective, and $\|\tilde{v}\| = \|v\|$. For each α, $(\Gamma' v)_\alpha = \sum_\beta \Gamma_{\bar{e}(\alpha)\bar{e}(\beta)} v_\beta = (\Gamma\tilde{v})_{\bar{e}(\alpha)}$. A sum of squares over the coordinates $\bar{e}(\alpha)$ is at most the sum over all coordinates, so $\|\Gamma' v\|^2 \le \|\Gamma\tilde{v}\|^2 \le \|\tilde{v}\|^2 = \|v\|^2$ by Theorem 1.3, which then also gives admissibility of Γ'. □

Remark 1.9. Purity does not pass to subsystems: the reduced covariance of an entangled pure state is not pure.

For Γ_1 of size $2n_1$ and Γ_2 of size $2n_2$, the *direct sum* $\Gamma_1 \oplus \Gamma_2$ (the covariance of two uncorrelated systems) is the $2(n_1+n_2)$ matrix with entries (Γ_1) on the coordinates of the first n_1 modes, (Γ_2) on those of the last n_2 modes, and zero cross entries, in the (x, y) ordering of the composite system.

Theorem 1.10 (Independent sums)**.** Lean ✓ *$\Gamma_1 \oplus \Gamma_2$ is admissible if and only if Γ_1 and Γ_2 are admissible, and pure if and only if both are pure.*

Proof. Conjugation by the permutation matrix that groups the coordinates of each subsystem together carries $\Gamma_1 \oplus \Gamma_2$ to $\operatorname{diag}(\Gamma_1, \Gamma_2)$. This conjugation is a ring isomorphism of matrices that commutes with transposition and fixes 1; it therefore carries antisymmetry, $1-\Gamma^{\mathsf{T}}\Gamma$ and Γ^2 to the corresponding block-diagonal quantities, and it preserves $\succeq 0$ by the congruence fact with O the permutation matrix or its transpose. It remains to decide each property blockwise. Transposition and products of block-diagonal matrices are blockwise, so $\operatorname{diag}(\Gamma_1, \Gamma_2)$ is antisymmetric, or squares to -1, exactly when both blocks do. For $v = (v_1, v_2)$,

$$v^{\mathsf{T}} \operatorname{diag}(A_1, A_2) v = v_1^{\mathsf{T}} A_1 v_1 + v_2^{\mathsf{T}} A_2 v_2,$$

so $\operatorname{diag}(A_1, A_2) \succeq 0$ when both blocks are, and conversely taking $v_2 = 0$ or $v_1 = 0$ isolates each block. Apply this with $A_i = 1 - \Gamma_i^{\mathsf{T}}\Gamma_i$. □

1.3 The standard covariance and one mode

Let

$$\Omega_n = \begin{pmatrix} 0 & 1_n \\ -1_n & 0 \end{pmatrix} \tag{1.3}$$

in the coordinates (x, y). It is antisymmetric and $\Omega_n^2 = -1$, so Ω_n is pure. On Fock space, Ω_n is the covariance of the state with every mode occupied and $-\Omega_n$ that of the vacuum (Corollary 5.9). The covariance of every state with a definite occupation in each mode is Ω_n with the sign of each mode's 2×2 block chosen independently (Theorem 5.8); Proposition 1.11 shows that these signed blocks are the only pure one-mode covariances.

Proposition 1.11 (One mode)**.** Lean ✓ *Every real antisymmetric 2×2 matrix has the form $\Gamma_\lambda = \lambda\Omega_1$ with $\lambda \in \mathbb{R}$. Γ_λ is admissible if and only if $|\lambda| \le 1$, and pure if and only if $|\lambda| = 1$.*

Proof. Antisymmetry forces zero diagonal entries and opposite off-diagonal entries, which is the stated form with λ the (x_1, y_1) entry. Then $\Gamma_\lambda^{\mathsf{T}}\Gamma_\lambda = \lambda^2 \cdot 1$ and $\Gamma_\lambda^2 = -\lambda^2 \cdot 1$. For a real number c, the matrix $c \cdot 1$ satisfies $c \cdot 1 \succeq 0$ exactly when $c \ge 0$, since $v^{\mathsf{T}}(c \cdot 1)v = c\|v\|^2$; with $c = 1 - \lambda^2$ this gives admissibility exactly when $\lambda^2 \le 1$. Purity is $-\lambda^2 \cdot 1 = -1$, that is $\lambda^2 = 1$. □

2 Contraction of covariances

To build a matrix product state on a chain, we join the virtual indices of on-site tensors by maximally entangled pairs and then sum over them. For Gaussian states the on-site tensors are covariances, the pairs are the bond covariance of Definition 2.7, and the summation is one operation on matrices, the contraction of Definition 2.1: it eliminates the labels of the joined modes and returns a covariance on the labels that remain. The result is admissible whenever the inputs are (Theorem 2.4), so that a chain built by repeated contraction is a covariance at every stage, and contracting two bonds through a bond returns a bond (Theorem 2.9), which fixes the sign of the bond. That the contraction is the covariance of the state obtained by projecting the eliminated modes onto the pair is the fermionic linear-optics calculus [1, Sec. 4], [4, Sec. II]; Section 10 proves it on Fock space (Theorem 10.10).

In this section we call a real matrix Γ indexed by an arbitrary finite set I *admissible* if $\Gamma^{\mathsf{T}} = -\Gamma$ and $1 - \Gamma^{\mathsf{T}}\Gamma \succeq 0$, and *pure* if $\Gamma^{\mathsf{T}} = -\Gamma$ and $\Gamma^2 = -1$; on the Majorana labels of n modes these

are the conditions of Section 1, and the norm form of Theorem 1.3 holds for every finite I with the same proof. "M invertible" means $\det M \neq 0$, and M^{-1} is then the inverse matrix. For disjoint index sets I_1, I_2, $I_1 \sqcup I_2$ is their union, and we write a matrix on $I_1 \sqcup I_2$ in blocks with the I_1 labels first. The *direct sum* of Γ_1 on I_1 and Γ_2 on I_2 is $\mathrm{diag}(\Gamma_1, \Gamma_2)$ on $I_1 \sqcup I_2$, the matrix $\Gamma_1 \oplus \Gamma_2$ of Theorem 1.10 after grouping each system's labels together. A set of modes L carries the Majorana labels x_a, y_a for $a \in L$, written x_L, y_L collectively; 1_L is the identity matrix indexed by L.

2.1 The contraction and its admissibility

Let $I = R \sqcup E$ be split into *retained* and *eliminated* labels (in the chains of Section 3: the physical and the virtual Majorana labels), and write a matrix on I in blocks

$$\Gamma = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad A \in \mathbb{R}^{R\times R},\ B \in \mathbb{R}^{R\times E},\ C \in \mathbb{R}^{E\times R},\ D \in \mathbb{R}^{E\times E}.$$

Let Γ_p be a matrix on E, the *pair covariance*.

Definition 2.1 (Contraction)**.** If $1 - D\Gamma_p$ is invertible,

$$\Gamma \star \Gamma_p = A + B\,\Gamma_p\,(1 - D\Gamma_p)^{-1} C,$$

a matrix on R.

The formula is the Schur complement of the eliminated block of a 2×2 block system: if $(x, w) \mapsto (u, v)$ is the linear map Γ and we impose $w = \Gamma_p v$ on the eliminated coordinates, then $u = (\Gamma \star \Gamma_p)x$ (displayed in the proof of Theorem 2.4). The contraction preserves admissibility because Γ and Γ_p are norm-nonincreasing (Theorem 1.3), hence so is $x \mapsto u$. For a pure Γ_p the formula becomes $A + B\,(D + \Gamma_p)^{-1} B^{\mathsf{T}}$ (Theorem 2.6), the form of [4, Sec. II]; the sign agrees with that reference once the pair is oriented as in Definition 2.7, with $+1$ from the first copy to the second, and reversing the orientation replaces Γ_p by $-\Gamma_p$.

Lemma 2.2 (Push-through)**.** Lean ✓ *For square X, Y of the same size,* $\det(1 - XY) = \det(1 - YX)$*, and if $1 - XY$ is invertible then* $Y(1 - XY)^{-1} = (1 - YX)^{-1} Y$.

Proof. The determinant identity is Sylvester's: both sides equal the determinant of the block matrix $\begin{pmatrix} 1 & X \\ Y & 1 \end{pmatrix}$, obtained by eliminating either diagonal block. From $Y(1-XY) = Y - YXY = (1 - YX)Y$, multiply by $(1 - YX)^{-1}$ on the left and $(1 - XY)^{-1}$ on the right. □

Theorem 2.3 (Antisymmetry)**.** Lean ✓ *Let $\Gamma^{\mathsf{T}} = -\Gamma$, $\Gamma_p^{\mathsf{T}} = -\Gamma_p$, and $1 - D\Gamma_p$ invertible. Then* $(\Gamma \star \Gamma_p)^{\mathsf{T}} = -\Gamma \star \Gamma_p$.

Proof. Antisymmetry of Γ gives $A^{\mathsf{T}} = -A$, $D^{\mathsf{T}} = -D$ and $C = -B^{\mathsf{T}}$, so $C^{\mathsf{T}} = -B$; antisymmetry of Γ_p and D gives $(1 - D\Gamma_p)^{\mathsf{T}} = 1 - \Gamma_p^{\mathsf{T}} D^{\mathsf{T}} = 1 - \Gamma_p D$. Then

$$\begin{aligned} \big(B\Gamma_p(1 - D\Gamma_p)^{-1}C\big)^{\mathsf{T}} &= C^{\mathsf{T}}\big((1 - D\Gamma_p)^{\mathsf{T}}\big)^{-1}\Gamma_p^{\mathsf{T}} B^{\mathsf{T}} = (-B)(1 - \Gamma_p D)^{-1}(-\Gamma_p)B^{\mathsf{T}} \\ &= B\,(1 - \Gamma_p D)^{-1}\Gamma_p\,B^{\mathsf{T}} = B\,\Gamma_p(1 - D\Gamma_p)^{-1}B^{\mathsf{T}} = -B\,\Gamma_p(1 - D\Gamma_p)^{-1}C, \end{aligned}$$

using Lemma 2.2 with $X = D$, $Y = \Gamma_p$ in the second line and then $B^{\mathsf{T}} = -C$. So the second term changes sign under transposition, as does A. □

Theorem 2.4 (Admissibility of the contraction)**.** Lean ✓ *Let Γ and Γ_p be admissible and $1 - D\Gamma_p$ invertible. Then $\Gamma \star \Gamma_p$ is admissible.*

Proof. Antisymmetry is Theorem 2.3; by the norm form it remains to show $\|(\Gamma \star \Gamma_p)x\| \le \|x\|$ for real $x \in \mathbb{R}^R$.

(a) Put
$$v = (1 - D\Gamma_p)^{-1} C x \in \mathbb{R}^E, \qquad w = \Gamma_p v \in \mathbb{R}^E, \qquad u = Ax + Bw \in \mathbb{R}^R.$$
Then $u = (\Gamma \star \Gamma_p)x$ by definition, and $(1 - D\Gamma_p)v = Cx$ gives $v = Cx + Dw$. Hence
$$\Gamma \begin{pmatrix} x \\ w \end{pmatrix} = \begin{pmatrix} Ax + Bw \\ Cx + Dw \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix}.$$

(b) Admissibility of Γ in norm form gives $\|u\|^2 + \|v\|^2 \le \|x\|^2 + \|w\|^2$, and admissibility of Γ_p gives $\|w\|^2 = \|\Gamma_p v\|^2 \le \|v\|^2$. Adding, $\|u\|^2 \le \|x\|^2$.

□

Remark 2.5. Admissibility of Γ and Γ_p does not imply that $1 - D\Gamma_p$ is invertible: for $\Gamma = \begin{pmatrix} 0 & 0 \\ 0 & D \end{pmatrix}$ with D pure and $\Gamma_p = -D$, $1 - D\Gamma_p = 1 - 1 = 0$. For a pure pair covariance, Theorem 2.6 turns the hypothesis into the invertibility of $D + \Gamma_p$, which we check directly in each example.

Theorem 2.6 (Pure pair covariances)**.** Lean ✓ *Let $\Gamma^{\mathsf{T}} = -\Gamma$ and let Γ_p be pure. Then $1 - D\Gamma_p$ is invertible if and only if $D + \Gamma_p$ is, and in that case*
$$\Gamma \star \Gamma_p = A + B\,(D + \Gamma_p)^{-1} B^{\mathsf{T}}.$$

Proof. (a) Since $\Gamma_p^2 = -1$, $\Gamma_p(-\Gamma_p) = 1$, so $-\Gamma_p$ is invertible with $(-\Gamma_p)^{-1} = \Gamma_p$, and $(D + \Gamma_p)(-\Gamma_p) = -D\Gamma_p + 1 = 1 - D\Gamma_p$. Taking determinants, $\det(1 - D\Gamma_p)$ and $\det(D + \Gamma_p)$ vanish together.

(b) When they do not vanish, $(1 - D\Gamma_p)^{-1} = (-\Gamma_p)^{-1}(D + \Gamma_p)^{-1} = \Gamma_p(D + \Gamma_p)^{-1}$, hence $\Gamma_p(1 - D\Gamma_p)^{-1} = \Gamma_p^2(D + \Gamma_p)^{-1} = -(D + \Gamma_p)^{-1}$ and, with $C = -B^{\mathsf{T}}$, $B\Gamma_p(1 - D\Gamma_p)^{-1}C = B(D + \Gamma_p)^{-1}B^{\mathsf{T}}$.

□

For a pair covariance that is not pure the two expressions differ in general.

2.2 The bond covariance and teleportation

Let L and L' be two sets of modes matched by a bijection $a \mapsto a'$, with Majorana labels $(x_L, x_{L'}, y_L, y_{L'})$. In the chain of Section 3 the first copy L is the left virtual set of a site and the second copy L' the right virtual set of the previous site; we orient the bond from the first copy to the second, $\beta_{x_a x_{a'}} = +1$.

Definition 2.7 (Bond covariance)**.** The bond covariance on $L \sqcup L'$ is
$$\beta = \begin{pmatrix} J & 0 \\ 0 & -J \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & 1_L \\ -1_L & 0 \end{pmatrix},$$
where the outer blocks are the x and y labels and the inner blocks of J are L and L': $\beta_{x_a x_{a'}} = +1$, $\beta_{y_a y_{a'}} = -1$, all other entries zero.

For one pair of modes, β has $\mathrm{i}c_{x_a}c_{x_{a'}}$ and $-\mathrm{i}c_{y_a}c_{y_{a'}}$ both of expectation 1; in the vacuum of Corollary 5.9 the nonzero expectations are instead $\mathrm{i}c_{x_a}c_{y_a}$ within one mode. The two-mode state with covariance β is Theorem 5.10.

Theorem 2.8 (Purity of the bond)**.** Lean ✓ *β is pure.*

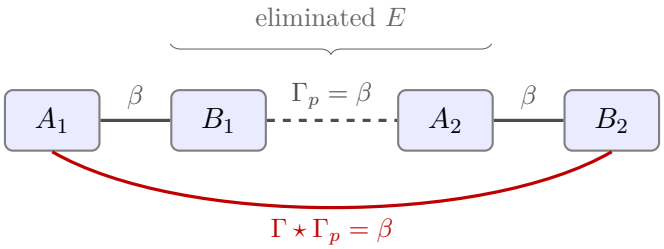


Figure 2: Teleportation (Theorem 2.9). Each block is a copy of the virtual label set L; solid lines are the bonds β on (A_1, B_1) and (A_2, B_2), the dashed line is the pair covariance $\Gamma_p = \beta$ on the eliminated labels $E = B_1 \sqcup A_2$. Contracting returns the bond β on the retained pair (A_1, B_2), with the same sign and the same x, y roles.

Proof. $J^{\mathsf{T}} = -J$ and $J^2 = -1$, so $\beta^{\mathsf{T}} = -\beta$ and $\beta^2 = \mathrm{diag}(J^2, J^2) = -1$. □

Theorem 2.9 (Teleportation). Lean ✓ *Let two systems $I_1 = A_1 \sqcup B_1$ and $I_2 = A_2 \sqcup B_2$, each a matched pair of copies of L, carry the bond covariance β on (A_1, B_1) and on (A_2, B_2), and let Γ be their direct sum on $I_1 \sqcup I_2$. Retain $R = A_1 \sqcup B_2$, eliminate $E = B_1 \sqcup A_2$, and take $\Gamma_p = \beta$ on (B_1, A_2). Then $1 - D\Gamma_p$ is invertible and $\Gamma \star \Gamma_p = \beta$ on (A_1, B_2).*

Proof. Neither bond has an entry within $A_1 \sqcup B_2$ or within $B_1 \sqcup A_2$, so $A = 0$ and $D = 0$; hence $1 - D\Gamma_p = 1$ is invertible and $\Gamma \star \Gamma_p = B\Gamma_p C = -B\Gamma_p B^{\mathsf{T}}$. All matrices are block diagonal in the x and y labels. In the x sector the two bonds contribute J on (A_1, B_1) and on (A_2, B_2), so the retained–eliminated block is

$$B = \begin{array}{c|cc} & x_{B_1} & x_{A_2} \\ \hline x_{A_1} & 1_L & 0 \\ x_{B_2} & 0 & -1_L \end{array}$$

and $\Gamma_p = J$ on (B_1, A_2). Then

$$-BJB^{\mathsf{T}} = -\begin{pmatrix} 1_L & 0 \\ 0 & -1_L \end{pmatrix}\begin{pmatrix} 0 & 1_L \\ -1_L & 0 \end{pmatrix}\begin{pmatrix} 1_L & 0 \\ 0 & -1_L \end{pmatrix} = -\begin{pmatrix} 0 & -1_L \\ 1_L & 0 \end{pmatrix} = J$$

on (x_{A_1}, x_{B_2}). In the y sector B and Γ_p both change sign, so $-B\Gamma_p B^{\mathsf{T}}$ changes sign and equals $-J$ on (y_{A_1}, y_{B_2}). Together, $\Gamma \star \Gamma_p = \beta$ on (A_1, B_2). □

Theorem 2.9 (Figure 2) fixes the convention: with the bond of Definition 2.7 used both as the on-site resource and as the contracted pair, contraction reproduces the bond without a change of sign or of the x, y roles. Here we orient the contracted pair from B_1 to A_2. The chain of Section 3 orients its inter-site pairs the other way, from A_{i+1} to B_i; since $A = 0$ and $D = 0$ here, the contraction is linear in Γ_p, and with the chain orientation the same computation returns $-\beta$ on (A_1, B_2), that is β on (B_2, A_1).

3 Periodic chains

A Gaussian fermionic matrix product state on a ring of N sites has, at each site, P physical modes and two sets of M virtual modes; neighbouring sites share a maximally entangled pair of virtual sets, and projecting every pair gives the state of the physical modes (Figure 3). On covariances this is one contraction against the bond covariance β (Definition 3.2). The result is admissible whenever the on-site covariances are, since the bond is pure (Theorem 3.4); we compute a three-site example in closed form (Proposition 3.6).

The sites of the ring are $i \in \mathbb{Z}/N$, identified with $\{0, \ldots, N-1\}$ (the empty set for $N = 0$), with successor $i+1$ read modulo N, and the numbers $N, M, P \geq 0$ are arbitrary. As in Figure 3,

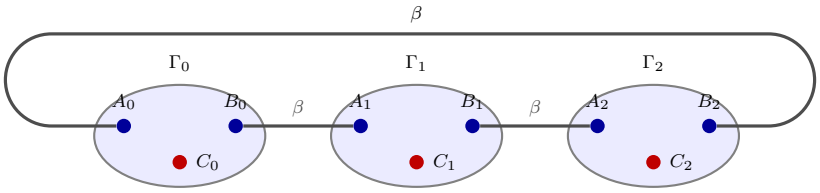


Figure 3: The periodic chain for $N = 3$, $M = P = 1$, drawn as in [4, Fig. 1]. Dots are modes: the virtual modes A_i, B_i (blue) and the physical mode C_i (red). A line joining B_i to A_{i+1} is the maximally entangled pair of Theorem 5.10, whose covariance is β on the ordered pair $(L, L') = (A_{i+1}, B_i)$, $\beta_{x_{A_{i+1}} x_{B_i}} = 1$. The shaded region around site i is the on-site covariance Γ_i on A_i, B_i, C_i; projecting every pair leaves a state of the physical modes, and on covariances this is the contraction of Definition 3.2.

site i carries M left virtual modes A_i, M right virtual modes B_i and P physical modes C_i, with the Majorana labels $I_{\text{site}} = (x_A, x_B, x_C, y_A, y_B, y_C)$; a family of labels indexed by (site, site label) is ordered *site by site* when all labels of site 0 come first, then those of site 1, and so on. The pairs (site, mode) form the sets $Q = \{(i, c) : i \in \mathbb{Z}/N,\ c \in \{1, \dots, P\}\}$ for the physical modes and $L = \{(i, a) : i \in \mathbb{Z}/N,\ a \in \{1, \dots, M\}\}$ for the virtual ones; the bond of Definition 2.7 will be placed on $L \sqcup L'$ with L' a second copy of L, and $x^L_{(i,a)}$, $x^{L'}_{(i,a)}$ denote the x-labels of the two copies.

3.1 The covariance of a periodic chain

Collecting the labels of all sites into the split $R \sqcup E$ of Section 2 reorders them, and one notation records the reordering. For an index bijection $e : I \to I'$ and a matrix Γ on I', the relabelled matrix Γ^e on I has entries $(\Gamma^e)_{ab} = \Gamma_{e(a)e(b)}$.

Definition 3.1 (On-site covariances and the input matrix)**.** The on-site covariances are matrices Γ_i on I_{site}, one per site. The input matrix is obtained by stacking them block-diagonally, site by site, and then permuting rows and columns so that the physical labels come first and the virtual labels follow, grouped as the pairs (A_i, B_{i-1}) that the bonds join. In symbols, let $R = (x_Q, y_Q)$ and $E = (x_L, x_{L'}, y_L, y_{L'})$, and define $e : R \sqcup E \to \bigsqcup_i I_i$ by

$$e(x_{(i,c)}) = x_{C_i,c}, \quad e(x^L_{(i,a)}) = x_{A_i,a}, \quad e(x^{L'}_{(i,a)}) = x_{B_{i-1},a},$$

and the same with y in place of x; e is a bijection. The *input matrix* is $\Gamma_{\text{in}} = \big(\operatorname{diag}(\Gamma_i)_i\big)^e$, a matrix on $R \sqcup E$.

$\operatorname{diag}(\Gamma_i)$ is the direct sum over sites, ordered site by site; the shift $i - 1$ in the L' labels makes (A_i, B_{i-1}) the matched pair (L, L') of Definition 2.7, L the left virtual set of site i and L' the right virtual set of site $i - 1$.

Definition 3.2 (Bonds of the chain and its covariance)**.** The bonds of the chain form the matrix $\beta_N = \beta$ on E, the bond covariance of $L \sqcup L'$: $(\beta_N)_{x_{A_i,a}\, x_{B_{i-1},a}} = 1$, $(\beta_N)_{y_{A_i,a}\, y_{B_{i-1},a}} = -1$, all other entries zero. If $1 - D\beta_N$ is invertible, where D is the $E \times E$ block of Γ_{in}, the *covariance of the chain* with on-site covariances $\Gamma_\bullet$ is

$$\Gamma_N(\Gamma_\bullet) = \Gamma_{\text{in}} \star \beta_N,$$

a matrix on R.

Lemma 3.3 (Relabelling and direct sums)**.** Lean ✓ *Let $e : I \to I'$ be a bijection. Γ on I' is admissible (pure) if and only if Γ^e is. If Γ_i are matrices on I_i, then $\operatorname{diag}(\Gamma_i)_i$ on $\bigsqcup_i I_i$ is admissible if and only if every Γ_i is.*

Proof. (a) $(\Gamma^e)^{\mathsf{T}} = (\Gamma^{\mathsf{T}})^e$, $\Gamma^e\Delta^e = (\Gamma\Delta)^e$, $1^e = 1$, and $(\Gamma^e v)\cdot(\Gamma^e v) = (\Gamma\tilde{v})\cdot(\Gamma\tilde{v})$, $v\cdot v = \tilde{v}\cdot\tilde{v}$ for $\tilde{v} = v\circ e^{-1}$. Antisymmetry and $\Gamma^2 = -1$ transfer by the first three identities; the norm form of Theorem 1.3 transfers by the last two.

(b) Transposition of a block-diagonal matrix is blockwise, and $(\mathrm{diag}(\Gamma_i)v)\cdot(\mathrm{diag}(\Gamma_i)v) = \sum_i(\Gamma_i v_i)\cdot(\Gamma_i v_i)$, $v\cdot v = \sum_i v_i\cdot v_i$ with v_i the restriction of v to I_i. The norm inequality for $\mathrm{diag}(\Gamma_i)$ follows from those of the blocks by summing; conversely v supported on one block isolates that block.

□

Theorem 3.4 (The chain covariance is admissible)**.** Lean ✓ *Let every Γ_i be admissible and $1 - D\beta_N$ invertible. Then $\Gamma_N(\Gamma_\bullet)$ is admissible.*

Proof. β_N is pure (Theorem 2.8), hence admissible. Γ_{in} is admissible by Lemma 3.3. Apply Theorem 2.4. □

Remark 3.5. For $N = 1$ the bond joins B_0 to A_0 within the site.

3.2 Three sites

Take $N = 3$ and $M = P = 1$. Then $I_{\mathrm{site}} = (x_A, x_B, x_C, y_A, y_B, y_C)$, R consists of the six labels $x_{C_0}, x_{C_1}, x_{C_2}, y_{C_0}, y_{C_1}, y_{C_2}$, and E of the twelve labels x_i^L, $x_i^{L'}$, y_i^L, $y_i^{L'}$ ($i \in \mathbb{Z}/3$, in this order of groups), with $x_i^L = x_{A_i}$ and $x_i^{L'} = x_{B_{i-1}}$. Let Γ_{seed} be the direct sum of the bond covariance β on (A, B) (Definition 2.7) and $-\Omega_1$ on C, that is $(\Gamma_{\mathrm{seed}})_{x_C y_C} = -1$ (Corollary 5.9), and let O be the rotation with $O_{x_A x_A} = O_{y_C y_C} = \frac{3}{5}$, $O_{x_A y_C} = -\frac{4}{5}$, $O_{y_C x_A} = \frac{4}{5}$ and $O = 1$ on the other four labels. The on-site covariance at every site is

$$\Gamma_{\mathrm{site}} = O\,\Gamma_{\mathrm{seed}}\,O^{\mathsf{T}} = \frac{1}{5}\begin{pmatrix} 0 & 3 & -4 & 0 & 0 & 0 \\ -3 & 0 & 0 & 0 & 0 & -4 \\ 4 & 0 & 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 5 & 0 & 0 \\ 0 & 4 & 3 & 0 & 0 & 0 \end{pmatrix},$$

pure by Theorem 1.7.

Proposition 3.6 (Three-site chain)**.** Lean ✓ *For the on-site covariances $\Gamma_i = \Gamma_{\mathrm{site}}$, $i \in \mathbb{Z}/3$: $\det(1 - D\beta_3) = (304/125)^2 \neq 0$, and in the labels $(x_{C_0}, x_{C_1}, x_{C_2}, y_{C_0}, y_{C_1}, y_{C_2})$*

$$\Gamma_3(\Gamma_\bullet) = \begin{pmatrix} 0 & -\Theta \\ \Theta^{\mathsf{T}} & 0 \end{pmatrix}, \qquad \Theta = \frac{1}{19}\begin{pmatrix} 15 & -6 & 10 \\ 10 & 15 & -6 \\ -6 & 10 & 15 \end{pmatrix},$$

which is pure ($\Theta\Theta^{\mathsf{T}} = 1$) and translation invariant (Θ is circulant).

Proof. (a) The blocks of Γ_{in}: $A = -\frac{3}{5}\Omega_3$, with Ω_3 as in (1.3), from the (x_C, y_C) entry of Γ_{site} at each site; B has the entries $B_{x_{C_i}x_i^L} = \frac{4}{5}$ and $B_{y_{C_i}x_{i+1}^{L'}} = \frac{4}{5}$ (the entries (x_C, x_A) and (y_C, x_B) of Γ_{site}; the label x_{B_i} is $x_{i+1}^{L'}$) and no others; $C = -B^{\mathsf{T}}$; D is the (x_A, x_B, y_A, y_B) block of Γ_{site} at each site, relabelled by e. Let S be the cyclic shift on $\mathbb{Z}/3$, $\mathsf{S}_{i,i+1} = 1$, so that $\mathsf{S}^{\mathsf{T}} = \mathsf{S}^{-1} = \mathsf{S}^2$. With the eliminated labels in the groups $(x^L, x^{L'}, y^L, y^{L'})$ and the retained labels in the groups (x_C, y_C),

$$B = \tfrac{4}{5}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \mathsf{S} & 0 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 0 & \frac{3}{5}\mathsf{S} & 0 & 0 \\ -\frac{3}{5}\mathsf{S}^{\mathsf{T}} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\mathsf{S} \\ 0 & 0 & \mathsf{S}^{\mathsf{T}} & 0 \end{pmatrix}, \qquad \beta_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

(b) D couples $x_i^L = x_{A_i}$ to $x_{i+1}^{L'} = x_{B_i}$ and β_3 couples $x_{i+1}^{L'}$ back to x_{i+1}^L, and likewise for y, so $D\beta_3$ maps each of the four label groups $x^L, x^{L'}, y^L, y^{L'}$ to itself by a circulant matrix. Hence $1 - D\beta_3$ and its inverse Y are block diagonal in these four groups with circulant blocks, and $\det(1 - D\beta_3) = (304/125)^2$. Explicitly, $1 - D\beta_3 = \mathrm{diag}(1 + \frac{3}{5}\mathsf{S},\ 1 + \frac{3}{5}\mathsf{S}^\mathsf{T},\ 1 + \mathsf{S},\ 1 + \mathsf{S}^\mathsf{T})$. Since $\mathsf{S}^3 = 1$ and the eigenvalues of S are the cube roots of unity, $\det(1 + \frac{3}{5}\mathsf{S}) = 1 + (\frac{3}{5})^3 = \frac{152}{125}$ and $\det(1 + \mathsf{S}) = 2$, the same for S^T, and $(1 + \frac{3}{5}\mathsf{S})^{-1} = \frac{125}{152}(1 - \frac{3}{5}\mathsf{S} + \frac{9}{25}\mathsf{S}^2)$, $(1 + \mathsf{S})^{-1} = \frac{1}{2}(1 - \mathsf{S} + \mathsf{S}^2)$. Hence $\det(1 - D\beta_3) = (152/125)^2 \cdot 2^2$ and

$$152\, Y = \mathrm{diag}(125 - 75\mathsf{S} + 45\mathsf{S}^2,\ 125 - 75\mathsf{S}^\mathsf{T} + 45\mathsf{S},\ 76(1 - \mathsf{S} + \mathsf{S}^2),\ 76(1 - \mathsf{S}^\mathsf{T} + \mathsf{S})).$$

(c) $\Gamma_3 = A + B\beta_3 YC$ is the stated matrix. The blocks of $152Y$ are integer circulant matrices, and we evaluate the products blockwise. Only the x^L and $x^{L'}$ blocks $Y_L = (1 + \frac{3}{5}\mathsf{S})^{-1}$ and $Y_{L'} = (1 + \frac{3}{5}\mathsf{S}^\mathsf{T})^{-1}$ of Y enter, and

$$B\beta_3 YC = -\tfrac{16}{25}\begin{pmatrix} 0 & Y_{L'}\mathsf{S}^\mathsf{T} \\ -\mathsf{S}Y_L & 0 \end{pmatrix} = \frac{1}{95}\begin{pmatrix} 0 & -(18 - 30\mathsf{S} + 50\mathsf{S}^\mathsf{T}) \\ (18 - 30\mathsf{S} + 50\mathsf{S}^\mathsf{T})^\mathsf{T} & 0 \end{pmatrix}.$$

Adding $A = -\frac{3}{5}\Omega_3$ gives $\Theta = \frac{3}{5} + \frac{1}{95}(18 - 30\mathsf{S} + 50\mathsf{S}^\mathsf{T}) = \frac{1}{19}(15 - 6\mathsf{S} + 10\mathsf{S}^\mathsf{T})$, the stated circulant matrix, and $\Theta\Theta^\mathsf{T} = \frac{1}{361}(361 + 60(\mathsf{S} + \mathsf{S}^\mathsf{T}) - 60(\mathsf{S}^2 + (\mathsf{S}^\mathsf{T})^2)) = 1$ since $\mathsf{S}^2 = \mathsf{S}^\mathsf{T}$.

In the formalization, $5\Gamma_{\mathrm{site}}$, $152Y$ and 19Θ are integer matrices; the identities $(1 - D\beta_3)(152Y) = 152 \cdot 1$ and $72200\,(A + B\beta_3 YC) = 72200\,\Gamma_3$ are decided over $\mathbb{Z}$ by evaluation and then divided. formalization

□

The physical modes are correlated: $\Gamma_{x_{C_i} y_{C_{i+1}}} = 6/19$ and $\Gamma_{x_{C_i} y_{C_{i-1}}} = -10/19$. For pure on-site covariances Γ_N is pure (Corollary 11.5; like Corollary 11.4, a fact about the matrices of Part I whose proof in these notes goes through Fock space), and Section 10 constructs the state of the physical modes whose covariance is Γ_N.

Checks. Every quantity in the proof of Proposition 3.6, the determinant included, is recomputed in exact rational arithmetic by `checks/three_site_exact.py`, which also prints the full matrices $1 - D\beta_3$, Y and $B\beta_3 YC$.

4 Homogeneous chains in momentum space

When every site of the periodic chain of Section 3 carries the same on-site covariance Γ, the chain covariance commutes with the shift of sites, and the discrete Fourier transform over sites splits it into N blocks of size $2P \times 2P$, one per momentum k. Each block is again a contraction (Definition 2.1), now of a single site against a single bond, with the shift of the right virtual modes turned into a phase, and the inverse transform recovers Γ_N from its blocks; antisymmetry and admissibility of Γ pass to every block.

Throughout this section $N \geq 1$ and $M, P \geq 0$; the on-site covariances are $\Gamma_i = \Gamma$ for every site, a fixed real matrix on $I_{\mathrm{site}} = (x_A, x_B, x_C, y_A, y_B, y_C)$, and $\Gamma_N = \Gamma_N(\Gamma, \ldots, \Gamma)$ is the covariance of the chain (Definition 3.2), with D the $E \times E$ block of Γ_{in}. We write retained labels as (i, a) with $i \in \mathbb{Z}/N$ the site and $a \in R_0 = (x_C, y_C)$ one of the $2P$ physical labels of a site. The site labels split as $I_{\mathrm{site}} = R_0 \sqcup E_0$ with $E_0 = (x_A, x_B, y_A, y_B)$; β_M is the real bond covariance of E_0 (Definition 2.7 with $L = A$, $L' = B$ and the index set $\{1, \ldots, M\}$). For a complex matrix X, X^* is the conjugate transpose, $\overline{X}$ the entrywise conjugate, and X^{-1} the inverse of an invertible X.

4.1 Twisted on-site covariances

Definition 4.1 (Characters and the twisted on-site covariance)**.** For $k, i \in \mathbb{Z}/N$ let $\chi_k(i) = e^{-2\pi \mathrm{i}\, ki/N}$ and $\omega_k = \chi_k(1)$. The *twist* $t_k : I_{\mathrm{site}} \to \mathbb{C}$ is $t_k(u) = \omega_k$ for $u \in \{x_B, y_B\}$ and $t_k(u) = 1$ otherwise, and the *twisted on-site covariance* at momentum k (the on-site covariance conjugated by the momentum phase on the right virtual modes) is

$$\Gamma^{(k)} = \mathrm{diag}(t_k)\, \Gamma\, \mathrm{diag}(t_k)^*, \qquad \Gamma^{(k)}_{uv} = t_k(u)\, \Gamma_{uv}\, \overline{t_k(v)}.$$

With this sign of χ_k, Theorem 4.8 expands Γ_N in $e^{+2\pi \mathrm{i}\, k(i-j)/N}\, \Sigma(k)$. The opposite convention expands in $e^{-2\pi \mathrm{i}\, k(i-j)/N}$, and in that convention the block at momentum k is $\Sigma(-k) = \overline{\Sigma(k)}$ (Theorem 4.10(a)).

Definition 4.2 (Momentum-space covariance)**.** Write $\Gamma^{(k)} = \begin{pmatrix} A_k & B_k \\ C_k & D_k \end{pmatrix}$ in the split $I_{\mathrm{site}} = R_0 \sqcup E_0$. The *momentum-space covariance* at k (the symbol of the block-circulant matrix Γ_N) is the complex $R_0 \times R_0$ matrix

$$\Sigma(k) = A_k + B_k\, \beta_M\, (1 - D_k \beta_M)^{-1}\, C_k,$$

the contraction formula of Definition 2.1 applied over $\mathbb{C}$ to $\Gamma^{(k)}$ and the bond of one site.

When $1 - D_k\beta_M$ is singular, we read $\Sigma(k)$ with $(1 - D_k\beta_M)^{-1} = 0$, the value of the formalization's inverse on singular matrices; only Theorem 4.10(a) uses $\Sigma(k)$ in that case. formalization

Definition 4.3 (Fourier coefficients of the chain)**.** For $k, l \in \mathbb{Z}/N$ and $a, b \in R_0$,

$$\widehat{\Gamma}_N(k,l)_{ab} = \frac{1}{N} \sum_{i,j \in \mathbb{Z}/N} \chi_k(i)\, (\Gamma_N)_{(i,a)(j,b)}\, \overline{\chi_l(j)}.$$

4.2 The Fourier transform of the chain

Let Φ_N be the $N \times N$ matrix with entries $(\Phi_N)_{ki} = \chi_k(i)/\sqrt{N}$; by the orthogonality of characters, $\sum_i \chi_k(i)\overline{\chi_l(i)} = N\delta_{kl}$, it is unitary. Identify the retained labels with pairs, $R \cong (\mathbb{Z}/N) \times R_0$ by (i, a), and the eliminated labels likewise, $E \cong (\mathbb{Z}/N) \times E_0$ by $x^L_{(i,a)} \mapsto (i, x_{A,a})$, $x^{L'}_{(i,a)} \mapsto (i, x_{B,a})$ and the same for y; a label (i, u) with $u \in \{x_B, y_B\}$ is thus the right virtual mode of site $i - 1$ (Definition 3.1). Let $\Phi_R = \Phi_N \otimes 1_{R_0}$, $\Phi_E = \Phi_N \otimes 1_{E_0}$ and $\Phi = \Phi_R \oplus \Phi_E$ on $R \sqcup E$.

Lemma 4.4 (Block diagonalization of the input)**.** Lean ✓ *$\Phi_E \beta_N \Phi_E^* = \beta_N$, and $\Phi\Gamma_{\mathrm{in}}\Phi^* = \bigoplus_k \Gamma^{(k)}$ in the split $R \sqcup E$, that is, the four blocks of $\Phi\Gamma_{\mathrm{in}}\Phi^*$ are $\bigoplus_k A_k$, $\bigoplus_k B_k$, $\bigoplus_k C_k$, $\bigoplus_k D_k$.*

Proof. (a) In the pair identification, $\beta_N = 1_N \otimes \beta_M$, because the bond pairs $x^L_{(i,a)}$ with $x^{L'}_{(i,a)}$ at the same pair (i, a) (Definition 3.2); it commutes with $\Phi_N \otimes 1$.

(b) Let $s(u) = 1$ for $u \in \{x_B, y_B\}$ and $s(u) = 0$ otherwise. The label (i, u) with $u \in \{x_B, y_B\}$ names a mode of site $i - 1$, and $\mathrm{diag}(\Gamma_\bullet)$ is site diagonal, so $(\Gamma_{\mathrm{in}})_{(i,u)(j,v)} = \Gamma_{uv}\, [\, i - s(u) = j - s(v)\,]$. Hence

$$(\Phi\Gamma_{\mathrm{in}}\Phi^*)_{(k,u)(l,v)} = \frac{\Gamma_{uv}}{N} \sum_i \chi_k(i)\, \overline{\chi_l(i - s(u) + s(v))} = \delta_{kl}\, \Gamma_{uv}\, \omega_k^{s(u)}\, \overline{\omega_k}^{\,s(v)} = \delta_{kl}\, \Gamma^{(k)}_{uv},$$

by $\chi_l(i+m) = \chi_l(i)\chi_l(m)$, $\chi_l(1) = \omega_l$, and orthogonality; the last equality is Definition 4.1. □

Lemma 4.5 (Contraction and relabelling). Lean ✓ *Let Γ be a matrix on $R \sqcup E$, Γ_p a matrix on E with $1 - D\Gamma_p$ invertible, and Y_R, Y_E invertible matrices on the retained and on the eliminated labels. Then, with the contraction formula taken over $\mathbb{C}$,*

$$\begin{pmatrix} Y_R A Y_R^{-1} & Y_R B Y_E^{-1} \\ Y_E C Y_R^{-1} & Y_E D Y_E^{-1} \end{pmatrix} \star (Y_E \Gamma_p Y_E^{-1}) = Y_R \, (\Gamma \star \Gamma_p) \, Y_R^{-1}.$$

If $\Gamma = \bigoplus_k \Gamma_k$ and $\Gamma_p = \bigoplus_k \Gamma_{p,k}$ are block diagonal in a common index k, then $\det(1 - D\Gamma_p) = \prod_k \det(1 - D_k \Gamma_{p,k})$, and when this is nonzero, $\Gamma \star \Gamma_p = \bigoplus_k \Gamma_k \star \Gamma_{p,k}$.

Proof. $1 - Y_E D Y_E^{-1} \, Y_E \Gamma_p Y_E^{-1} = Y_E (1 - D\Gamma_p) Y_E^{-1}$, whose inverse is $Y_E (1 - D\Gamma_p)^{-1} Y_E^{-1}$; the four products then telescope. For block diagonal data, determinants multiply over blocks, the inverse of an invertible block diagonal matrix is block diagonal with the inverse blocks, and sums and products are blockwise. □

Theorem 4.6 (Determinant factorization). Lean ✓ $\det(1 - D\beta_N) = \prod_{k \in \mathbb{Z}/N} \det(1 - D_k \beta_M)$. *In particular, if $1 - D\beta_N$ is invertible then $1 - D_k \beta_M$ is invertible for every k.*

Proof. $\Phi_E (1 - D\beta_N) \Phi_E^* = 1 - \bigoplus_k D_k \beta_M$ by Lemma 4.4; Φ_E is unitary, so the determinants agree, and the block determinant is the product (Lemma 4.5). A nonzero product has nonzero factors. □

Theorem 4.7 (Translation invariance and diagonalization). Lean ✓ *Let $1 - D\beta_N$ be invertible. Then for all $i, j \in \mathbb{Z}/N$, $k, l \in \mathbb{Z}/N$ and $a, b \in R_0$,*

$$(\Gamma_N)_{(i+1,a)(j+1,b)} = (\Gamma_N)_{(i,a)(j,b)}, \qquad \widehat{\Gamma}_N(k,l) = \begin{cases} \Sigma(k) & k = l, \\ 0 & k \neq l. \end{cases}$$

Theorem 4.8 (Reconstruction). Lean ✓ *Let $1 - D\beta_N$ be invertible. Then for all $i, j \in \mathbb{Z}/N$ and $a, b \in R_0$,*

$$(\Gamma_N)_{(i,a)(j,b)} = \frac{1}{N} \sum_{k \in \mathbb{Z}/N} \overline{\chi_k(i)} \, \Sigma(k)_{ab} \, \chi_k(j).$$

Proof of Theorems 4.7 and 4.8. (a) By Lemma 4.4, Lemma 4.5 with $Y = \Phi$ (so $Y^{-1} = \Phi^*$), and Theorem 4.6,

$$\Phi_R \, \Gamma_N \, \Phi_R^* = (\Phi \Gamma_{\text{in}} \Phi^*) \star (\Phi_E \beta_N \Phi_E^*) = \Big(\bigoplus_k \Gamma^{(k)} \Big) \star \Big(\bigoplus_k \beta_M \Big) = \bigoplus_k \Sigma(k).$$

(b) The $((k,a),(l,b))$ entry of the left side is $\widehat{\Gamma}_N(k,l)_{ab}$ by the definition of Φ_R; this is the second part of Theorem 4.7.

(c) Multiplying (a) by Φ_R^* on the left and Φ_R on the right gives $\Gamma_N = \Phi_R^* \big(\bigoplus_k \Sigma(k) \big) \Phi_R$, whose $((i,a),(j,b))$ entry is the sum of Theorem 4.8.

(d) $\overline{\chi_k(i+1)} \, \chi_k(j+1) = \overline{\chi_k(i)} \, \overline{\omega_k} \, \chi_k(j) \, \omega_k = \overline{\chi_k(i)} \, \chi_k(j)$, so replacing (i,j) by $(i+1, j+1)$ leaves the sum in (c) unchanged; this is the first part of Theorem 4.7. □

Remark 4.9. The invertibility hypothesis enters translation invariance only through this route; the chain construction is shift symmetric, and we make no claim here about the singular case.

4.3 Admissibility in momentum space

Theorem 4.10 (Symmetries and admissibility of the momentum-space covariance)**.** Lean ✓

(a) If $1 - D_k\beta_M$ is invertible then $\Sigma(-k) = \overline{\Sigma(k)}$; the formalized statement holds for every k, with the inverse of a singular matrix read as 0. formalization

(b) If $\Gamma^{\mathsf{T}} = -\Gamma$ and $1 - D\beta_N$ is invertible, then $\Sigma(k)^* = -\Sigma(k)$ for every k.

(c) If Γ is admissible (Definition 1.1) and $1 - D\beta_N$ is invertible, then $1 - \Sigma(k)^*\Sigma(k) \succeq 0$ for every k, that is, $\|\Sigma(k)z\| \le \|z\|$ for every $z \in \mathbb{C}^{R_0}$.

Proof. (a) $\omega_{-k} = \overline{\omega_k}$, so $t_{-k} = \overline{t_k}$ and, Γ being real, $\Gamma^{(-k)} = \overline{\Gamma^{(k)}}$; β_M is real, and the inverse, read as 0 on singular matrices, commutes with entrywise conjugation, so the formula of Definition 4.2 commutes with conjugation.

(b) $\Gamma_N^{\mathsf{T}} = -\Gamma_N$ (Theorem 2.3, applied to Γ_{in}, which is antisymmetric with Γ, and to β_N), and Γ_N is real, so $\Gamma_N^* = -\Gamma_N$ and $\Phi_R\Gamma_N\Phi_R^*$ is skew-Hermitian; by step (a) of the proof of Theorem 4.7 its blocks $\Sigma(k)$ are skew-Hermitian.

(c) Γ_N is admissible (Theorem 3.4), so $1 - \Gamma_N^{\mathsf{T}}\Gamma_N \succeq 0$ over $\mathbb{R}$, hence over $\mathbb{C}$: for $z = a + \mathrm{i}b$ with a, b real, $z^*Xz = a^{\mathsf{T}}Xa + b^{\mathsf{T}}Xb$ when X is real symmetric. Conjugating by Φ_R and using step (a) of the proof of Theorem 4.7, $\Phi_R(1 - \Gamma_N^{\mathsf{T}}\Gamma_N)\Phi_R^* = 1 - \bigoplus_k \Sigma(k)^*\Sigma(k) \succeq 0$, and a block diagonal matrix is positive semidefinite exactly when each block is (test with vectors supported on one block). □

Part (c) is the momentum-space form of admissibility. For pure Γ the chain covariance Γ_N is pure (Corollary 11.5, a matrix fact whose proof in these notes goes through Fock space), and the argument of (c) with $1 - \Gamma_N^{\mathsf{T}}\Gamma_N = 0$ gives $\Sigma(k)^*\Sigma(k) = 1$ for every k.

Part II
States on Fock space

5 Majorana operators on Fock space

A *configuration* of n modes is a map $s : \{1, \dots, n\} \to \{0, 1\}$; s_j is the occupation of mode j. Fock space is $\mathcal{F}_n = \mathbb{C}^{\{0,1\}^n}$ with the standard basis $\{|s\rangle\}$, and operators on it are complex matrices indexed by configurations. For a matrix M, M^* is its conjugate transpose, and 1 denotes the identity of whatever size the context requires. For a configuration s let $s^{(j)}$ be s with the occupation of mode j replaced by $s_j + 1$ modulo 2; then $(s^{(j)})^{(j)} = s$ and $(s^{(j)})^{(k)} = (s^{(k)})^{(j)}$. Write $\varsigma(b) = (-1)^b$ for $b \in \{0, 1\}$, so $\varsigma(s_j + 1) = -\varsigma(s_j)$, and $[\,\cdot\,]$ for the number 1 if the statement in the bracket holds and 0 otherwise.

Definition 5.1 (Pauli matrices and the string)**.** For each mode j define

$$(X_j)_{st} = [t = s^{(j)}], \qquad Z_j = \operatorname{diag}\big(\varsigma(s_j)\big)_s, \qquad Z_{<j} = \operatorname{diag}\Big(\prod_{k<j} \varsigma(s_k)\Big)_s.$$

X_j is the permutation matrix of the flip of mode j; Z_j and $Z_{<j}$ are real diagonal, and $Z_{<1} = 1$. The matrices X_j and X_k of different modes commute (Lemma 5.2(b)), and so do the diagonal matrices Z_j and Z_k; operators built from these alone would commute across modes, whereas Theorem 5.4 requires $c_\alpha c_\beta = -c_\beta c_\alpha$ for labels of different modes. $Z_{<j}$ records the parity of the modes before j, and Lemma 5.2(e) says that X_j reverses that parity exactly when j lies before the string's mode; this is the sign rule Theorem 5.4 uses. The ordering of modes is a convention; a different ordering gives operators related to these by a unitary conjugation, which we do not prove here.

Lemma 5.2 (Commutation rules). Lean ✓ *For all modes j, k:*

(a) $X_j^2 = Z_j^2 = Z_{<j}^2 = 1$, *and* $X_j, Z_j, Z_{<j}$ *are real symmetric.*

(b) $(X_jX_k)_{st} = [t = (s^{(j)})^{(k)}]$; *hence* $X_jX_k = X_kX_j$, *and* $(X_jX_k)_{ss} = [j = k]$.

(c) For a diagonal matrix $D = \mathrm{diag}(d_s)$, $X_jD = D'X_j$ *with* $D' = \mathrm{diag}(d_{s^{(j)}})$.

(d) $X_jZ_j = -Z_jX_j$, *and* $X_jZ_k = Z_kX_j$ *for* $k \neq j$.

(e) $X_jZ_{<k} = \eta_{jk}Z_{<k}X_j$, *where* $\eta_{jk} = -1$ *if* $j < k$ *and* $\eta_{jk} = 1$ *otherwise.*

Proof. (a) $(X_j^2)_{st} = \sum_u [u = s^{(j)}][t = u^{(j)}] = [t = s]$; Z_j^2 and $Z_{<j}^2$ are diagonal with entries $\varsigma(\cdot)^2 = 1$. Symmetry of X_j: $t = s^{(j)}$ if and only if $s = t^{(j)}$.

(b) $(X_jX_k)_{st} = \sum_u [u = s^{(j)}][t = u^{(k)}] = [t = (s^{(j)})^{(k)}]$. The two flips commute, and $(s^{(j)})^{(k)} = s$ exactly when $j = k$, since for $j \neq k$ the configuration $(s^{(j)})^{(k)}$ differs from s at mode j.

(c) $(X_jD)_{st} = \sum_u [u = s^{(j)}]\, d_u [u = t] = d_{s^{(j)}}[t = s^{(j)}] = (D'X_j)_{st}$.

(d) By (c) with $d_s = \varsigma(s_j)$, $d_{s^{(j)}} = -\varsigma(s_j)$; for $k \neq j$, $d_s = \varsigma(s_k)$ is unchanged by flipping mode j.

(e) By (c) with $d_s = \prod_{k'<k} \varsigma(s_{k'})$: flipping mode j changes the sign of the factor $k' = j$ if $j < k$ and of no factor otherwise.

□

Definition 5.3 (Majorana operators). With Majorana labels $x_1, \ldots, x_n, y_1, \ldots, y_n$ as in Section 1, the Jordan–Wigner operators are

$$c_{x_j} = Z_{<j}X_j, \qquad c_{y_j} = \mathrm{i}\, Z_{<j}X_jZ_j.$$

X_j, Z_j and $\mathrm{i}X_jZ_j = \sigma_j^y$ are the Pauli matrices $\sigma_j^x, \sigma_j^z, \sigma_j^y$ of mode j, so $c_{x_j} = Z_{<j}\sigma_j^x$ and $c_{y_j} = Z_{<j}\sigma_j^y$ with $Z_{<j}$ the Jordan–Wigner string. For one mode, in the basis $(|0\rangle, |1\rangle)$,

$$c_{x_1} = X_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad c_{y_1} = \mathrm{i}X_1Z_1 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}.$$

The annihilation operator of mode j is $a_j = \frac{1}{2}(c_{x_j} + \mathrm{i}c_{y_j}) = \frac{1}{2}Z_{<j}X_j(1 - Z_j)$; since $Z_{<j}$ and X_j commute (Lemma 5.2(e)),

$$a_j^*a_j = \tfrac{1}{4}(1 - Z_j)X_jZ_{<j}Z_{<j}X_j(1 - Z_j) = \tfrac{1}{4}(1 - Z_j)^2 = \tfrac{1}{2}(1 - Z_j) = \mathrm{diag}(s_j)_s,$$

the number operator of mode j, and $\mathrm{i}\, c_{x_j}c_{y_j} = \mathrm{i} \cdot \mathrm{i}\, Z_{<j}X_jZ_{<j}X_jZ_j = -Z_j = 2a_j^*a_j - 1$. For one mode $a_1 = |0\rangle\langle 1|$, and $c_{x_1} = a_1 + a_1^*$, $c_{y_1} = \mathrm{i}(a_1^* - a_1)$.

Every Majorana operator is a *signed flip*, a permutation matrix of one bit flip with signs on the entries. By Lemma 5.2(c),

$$c_{x_j} = \mathrm{diag}(f_{x_j})\, X_j, \qquad c_{y_j} = \mathrm{diag}(f_{y_j})\, X_j, \qquad f_{x_j}(s) = \prod_{k<j} \varsigma(s_k), \quad f_{y_j}(s) = -\mathrm{i}\,\varsigma(s_j) \prod_{k<j} \varsigma(s_k), \tag{5.1}$$

where for c_{y_j} we move the factor $Z_j = \mathrm{diag}(\varsigma(s_j))$ through X_j, which turns it into $\mathrm{diag}(\varsigma(s_j{+}1)) = \mathrm{diag}(-\varsigma(s_j))$. For labels α, β with modes j, k, Lemma 5.2(c) gives the product formula

$$c_\alpha c_\beta = \mathrm{diag}\big(f_\alpha(s)\, f_\beta(s^{(j)})\big)\, X_jX_k. \tag{5.2}$$

Within one mode, σ_j^x and σ_j^y are Pauli matrices: they square to 1 and anticommute. For $j < k$ the string $Z_{<k}$ contains the factor Z_j, which anticommutes with the X_j in c_{x_j} and c_{y_j}, while every other factor of one operator commutes with every factor of the other; the string supplies the sign between different modes.

Theorem 5.4 (Canonical anticommutation relations). Lean ✓ *For all Majorana labels α, β,*

$$c_\alpha c_\beta + c_\beta c_\alpha = 2\delta_{\alpha\beta}\, 1, \qquad c_\alpha^* = c_\alpha.$$

Proof. By (5.2) and Lemma 5.2(b), the anticommutator equals $\operatorname{diag}(g_{\alpha\beta})\, X_j X_k$ with

$$g_{\alpha\beta}(s) = f_\alpha(s) f_\beta(s^{(j)}) + f_\beta(s) f_\alpha(s^{(k)}).$$

Flipping mode j multiplies $\prod_{k'<k} \varsigma(s_{k'})$ by η_{jk} and $\varsigma(s_j)$ by -1; for $j \neq k$, $\eta_{jk} + \eta_{kj} = 0$. Write $p_j = \prod_{k'<j} \varsigma(s_{k'})$ and $\sigma_j = \varsigma(s_j)$. Then $g_{\alpha\beta}(s)$ is:

(i) $j \neq k$, $\alpha = x_j$, $\beta = x_k$: $p_j p_k (\eta_{jk} + \eta_{kj}) = 0$.

(ii) $j \neq k$, $\alpha = x_j$, $\beta = y_k$: $-\mathrm{i}\, p_j p_k \sigma_k (\eta_{jk} + \eta_{kj}) = 0$; the case $\alpha = y_j$, $\beta = x_k$ is the same with σ_j.

(iii) $j \neq k$, $\alpha = y_j$, $\beta = y_k$: $(-\mathrm{i})^2 p_j p_k \sigma_j \sigma_k (\eta_{jk} + \eta_{kj}) = 0$.

(iv) $j = k$, $\{\alpha, \beta\} = \{x_j, y_j\}$: $-\mathrm{i}\, p_j^2 (\varsigma(s_j + 1) + \sigma_j) = 0$.

(v) $\alpha = \beta = x_j$: $2p_j^2 = 2$; $\alpha = \beta = y_j$: $2(-\mathrm{i})^2 p_j^2 \sigma_j \varsigma(s_j + 1) = 2$.

For $\alpha \neq \beta$ the anticommutator is therefore 0; for $\alpha = \beta$ it is $\operatorname{diag}(2)\, X_j^2 = 2 \cdot 1$ by (a).

Hermiticity: by (a) and (c), $c_\alpha^* = X_j \operatorname{diag}(\overline{f_\alpha}) = \operatorname{diag}(\overline{f_\alpha(s^{(j)})}) X_j$, and $\overline{f_\alpha(s^{(j)})} = f_\alpha(s)$: for x_j the product p_j does not involve mode j; for y_j, $\overline{-\mathrm{i}\,\varsigma(s_j+1) p_j} = \mathrm{i}\,(-\sigma_j) p_j = f_{y_j}(s)$. □

5.1 Covariance of a density operator

The covariance collects the second moments $\operatorname{Tr}(\rho\, c_\alpha c_\beta)$; by Theorem 5.4 their symmetric part is fixed, $\operatorname{Tr}(\rho\, c_\alpha c_\beta) + \operatorname{Tr}(\rho\, c_\beta c_\alpha) = 2\delta_{\alpha\beta}$, so only the antisymmetric part carries information about ρ, and the factor $\frac{\mathrm{i}}{2}$ makes it real. A state is Gaussian when Wick's rule (Definition 9.1) determines all its higher moments from these; in closed form, $\operatorname{Tr}(\rho\, c_{\alpha_1} \cdots c_{\alpha_{2k}})$ is then the Pfaffian of the $2k \times 2k$ submatrix of $-\mathrm{i}\Gamma$ on the distinct labels $\alpha_1, \ldots, \alpha_{2k}$, and odd moments vanish [1, Sec. 2].

Definition 5.5 (Density operator and covariance). A *density operator* is a matrix ρ on $\mathcal{F}_n$ with $\rho \succeq 0$ and $\operatorname{Tr}\rho = 1$. Its *Majorana covariance* is the real $2n \times 2n$ matrix

$$\Gamma(\rho)_{\alpha\beta} = \Re\Big(\tfrac{\mathrm{i}}{2} \operatorname{Tr}(\rho\, [c_\alpha, c_\beta])\Big), \qquad [c_\alpha, c_\beta] = c_\alpha c_\beta - c_\beta c_\alpha. \tag{5.3}$$

A density operator is a *pure state* when $\rho = |\psi\rangle\langle\psi|$ for a unit vector ψ; equivalently $\rho^2 = \rho$, since a positive semidefinite matrix with $\rho^2 = \rho$ and $\operatorname{Tr}\rho = 1$ has eigenvalues in $\{0, 1\}$ summing to 1.

Theorem 5.6 (Reality and antisymmetry). Lean ✓ *Let $\rho = \rho^*$. Then $\frac{\mathrm{i}}{2} \operatorname{Tr}(\rho[c_\alpha, c_\beta])$ is real, and $\Gamma(\rho)^{\mathsf{T}} = -\Gamma(\rho)$, $\Gamma(\rho)_{\alpha\alpha} = 0$, and $\Gamma(\rho)_{\alpha\beta} = \Re(\mathrm{i} \operatorname{Tr}(\rho\, c_\alpha c_\beta))$ for $\alpha \neq \beta$.*

Proof. Put $T = \operatorname{Tr}(\rho[c_\alpha, c_\beta])$. Since $\operatorname{Tr}(M^*) = \overline{\operatorname{Tr} M}$ and $[c_\alpha, c_\beta]^* = c_\beta c_\alpha - c_\alpha c_\beta$ by Theorem 5.4, $\overline{T} = \operatorname{Tr}([c_\alpha, c_\beta]^* \rho) = -\operatorname{Tr}([c_\alpha, c_\beta]\rho) = -T$ using $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$. Hence T is purely imaginary and $\frac{\mathrm{i}}{2} T$ is real. Antisymmetry is $[c_\beta, c_\alpha] = -[c_\alpha, c_\beta]$; the diagonal vanishes since $[c_\alpha, c_\alpha] = 0$; for $\alpha \neq \beta$, Theorem 5.4 gives $[c_\alpha, c_\beta] = 2 c_\alpha c_\beta$. The last three statements hold for every ρ. □

Restricted to a single mode, the inequality of Theorem 5.7 reads as follows. With z supported on the labels x_j, y_j and $z_{x_j} = \frac{1}{2}$, $z_{y_j} = -\frac{\mathrm{i}}{2}$, the operator $A = \sum_\alpha \overline{z_\alpha} c_\alpha$ is $a_j = \frac{1}{2}(c_{x_j} + \mathrm{i} c_{y_j})$ and $z^*(1 + \mathrm{i}\Gamma)z = \frac{1}{2}(1 + \Gamma_{x_j y_j}) = \operatorname{Tr}(\rho\, a_j^* a_j)$, the occupation of mode j. Theorem 5.7 is this positivity for every $A = \sum_\alpha \overline{z_\alpha} c_\alpha$ at once, and Theorem 5.8 shows that the bound is attained.

Theorem 5.7 (Admissibility). Lean ✓ *For every density operator ρ, the covariance $\Gamma(\rho)$ is admissible.*

Proof. Write $\Gamma = \Gamma(\rho)$; it is antisymmetric by Theorem 5.6, so by Theorem 1.4 it suffices to show $1 + \mathrm{i}\Gamma \succeq 0$. Fix $z \in \mathbb{C}^{2n}$ and put $A = \sum_\alpha \overline{z_\alpha}\, c_\alpha$, so $A^* = \sum_\alpha z_\alpha c_\alpha$. Theorem 5.4 gives $c_\alpha c_\beta = \delta_{\alpha\beta}\, 1 + \frac{1}{2}[c_\alpha, c_\beta]$, hence

$$A^*A = \sum_{\alpha,\beta} z_\alpha \overline{z_\beta}\, c_\alpha c_\beta = \|z\|^2\, 1 + \tfrac{1}{2} \sum_{\alpha,\beta} z_\alpha \overline{z_\beta}\, [c_\alpha, c_\beta].$$

Taking the trace against ρ, with $\operatorname{Tr}\rho = 1$ and $\frac{1}{2}\operatorname{Tr}(\rho[c_\alpha, c_\beta]) = -\mathrm{i}\,\Gamma_{\alpha\beta}$ by (5.3) and Theorem 5.6,

$$\operatorname{Tr}(\rho A^*A) = \|z\|^2 - \mathrm{i} \sum_{\alpha,\beta} \overline{z_\beta}\, \Gamma_{\alpha\beta}\, z_\alpha = \|z\|^2 - \mathrm{i}\, z^* \Gamma^{\mathsf{T}} z = \|z\|^2 + \mathrm{i}\, z^* \Gamma z = z^*(1 + \mathrm{i}\Gamma) z.$$

Finally $\operatorname{Tr}(\rho A^*A) = \operatorname{Tr}(A \rho A^*) \geq 0$: for every vector w, $w^*(A\rho A^*)w = (A^*w)^* \rho\, (A^*w) \geq 0$, so $A\rho A^* \succeq 0$, and a positive semidefinite matrix has nonnegative diagonal entries, hence nonnegative trace. □

Conversely, every admissible matrix is the covariance of some density operator (Theorem 7.19).

5.2 Occupation-number states

For a configuration s let $\rho_s = |s\rangle\langle s|$, the matrix with a single entry 1 at position (s, s). It is a pure state, and $\operatorname{Tr}(\rho_s M) = M_{ss}$; for $s \equiv 0$ it is the vacuum ρ_{vac}. These are product states: each mode is definitely empty or definitely occupied, and Theorem 5.8 shows that the covariance is a direct sum of the one-mode blocks $\pm\Omega_1$ of Proposition 1.11. Every pure Gaussian state has covariance $O\,\mathsf{D}(\eta)\,O^{\mathsf{T}}$ with $\eta_j = \pm 1$ and O orthogonal, a rotated occupation covariance (Corollary 9.16(b)).

Theorem 5.8 (Covariance of occupation states)**.** Lean ✓ *Write $\nu(b) = -\varsigma(b) = 2b - 1$ for the* occupation sign*, -1 for an empty mode and $+1$ for an occupied one, and let $\Lambda_s = \operatorname{diag}(\nu(s_j))_{j=1}^{n}$. In the (x, y) block ordering,*

$$\Gamma(\rho_s) = \begin{pmatrix} 0 & \Lambda_s \\ -\Lambda_s & 0 \end{pmatrix}:$$

$\Gamma(\rho_s)_{x_j y_j} = \nu(s_j) = -\Gamma(\rho_s)_{y_j x_j}$, and all other entries vanish.

Proof. By Theorem 5.6 the diagonal entries vanish and for $\alpha \neq \beta$ the entry is $(c_\alpha c_\beta)_{ss}$. By (5.2) and Lemma 5.2(b) this is $f_\alpha(s) f_\beta(s^{(j)})\,[j = k]$, so all cross-mode entries vanish. For the same mode, $f_{x_j}(s) f_{y_j}(s^{(j)}) = -\mathrm{i}\,\varsigma(s_j + 1) p_j^2 = \mathrm{i}\,\varsigma(s_j)$ and $f_{y_j}(s) f_{x_j}(s^{(j)}) = -\mathrm{i}\,\varsigma(s_j)$. Then $\Gamma_{x_j y_j} = \Re(\mathrm{i} \cdot \mathrm{i}\,\varsigma(s_j)) = -\varsigma(s_j)$ and $\Gamma_{y_j x_j} = \varsigma(s_j)$. □

Corollary 5.9 (Vacuum and purity)**.** Lean ✓ *The vacuum $s \equiv 0$ has covariance $\Gamma(\rho_{\mathrm{vac}}) = -\Omega_n$, with Ω_n the standard covariance* (1.3)*. Every occupation state has a pure covariance.*

Proof. For $s \equiv 0$, $\Lambda_s = -1$. In general $\Lambda_s^2 = 1$, so $\Gamma(\rho_s)^2 = \operatorname{diag}(-\Lambda_s^2, -\Lambda_s^2) = -1$; antisymmetry is Theorem 5.6. □

We therefore work with the sign convention $\Gamma_{x_j y_j} = -1$ for an empty mode and $+1$ for an occupied one: $\Gamma_{x_j y_j}(\rho) = \operatorname{Tr}(\rho\, \mathrm{i} c_{x_j} c_{y_j}) = \operatorname{Tr}\big(\rho\,(2a_j^* a_j - 1)\big)$ is twice the occupation of mode j minus one, and the sign of the vacuum covariance is that of the factor i in $c_{y_j} = \mathrm{i}(a_j^* - a_j)$.

5.3 The bond as a state of two modes

The bond covariance β of Definition 2.7 is the covariance of a pure state of two modes, $\frac{1}{\sqrt{2}}(|00\rangle + \mathrm{i}|11\rangle)$. The phase i is what pairs x with x and y with y: for $|00\rangle + |11\rangle$ the nonzero expectations would be those of $c_{x_1} c_{y_2}$ and $c_{y_1} c_{x_2}$. With the Jordan–Wigner operators of Definition 5.3: $c_{x_1} = X_1$, $c_{y_1} = \mathrm{i} X_1 Z_1$, $c_{x_2} = Z_1 X_2$, $c_{y_2} = \mathrm{i} Z_1 X_2 Z_2$.

Theorem 5.10 (Fock-space realization of the bond). Lean ✓ *Let* $\psi = \frac{1}{\sqrt{2}}(|00\rangle + \mathrm{i}\,|11\rangle)$ *and* $\rho = |\psi\rangle\langle\psi|$. *Then* ρ *is a density operator and, in the labels* (x_1, x_2, y_1, y_2),

$$\Gamma(\rho) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \beta \quad \textit{for } L = \{1\},\ L' = \{2\}.$$

Proof. $\rho \succeq 0$ and $\operatorname{Tr}\rho = \|\psi\|^2 = 1$. Since the c_α anticommute for $\alpha \neq \beta$, $\Gamma(\rho)_{\alpha\beta} = \Re(\mathrm{i}\,\langle\psi|c_\alpha c_\beta|\psi\rangle)$ for $\alpha \neq \beta$, and the diagonal vanishes. With $XZ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$:

(a) $c_{x_1}c_{x_2} = X_1Z_1X_2$. Since $XZ|0\rangle = |1\rangle$ and $XZ|1\rangle = -|0\rangle$, $X_1Z_1X_2|00\rangle = |11\rangle$ and $X_1Z_1X_2|11\rangle = -|00\rangle$, so $X_1Z_1X_2\psi = \frac{1}{\sqrt{2}}(|11\rangle - \mathrm{i}|00\rangle) = -\mathrm{i}\psi$ and $\Gamma_{x_1x_2} = \Re(\mathrm{i}\cdot(-\mathrm{i})) = 1$.

(b) $c_{y_1}c_{y_2} = (\mathrm{i}X_1Z_1)(\mathrm{i}Z_1X_2Z_2) = -X_1X_2Z_2$; $X_1X_2Z_2|00\rangle = |11\rangle$, $X_1X_2Z_2|11\rangle = -|00\rangle$, so $X_1X_2Z_2\psi = -\mathrm{i}\psi$, $\langle\psi|c_{y_1}c_{y_2}|\psi\rangle = \mathrm{i}$, and $\Gamma_{y_1y_2} = \Re(\mathrm{i}\cdot\mathrm{i}) = -1$.

(c) $c_{x_1}c_{y_1} = \mathrm{i}Z_1$ and $c_{x_2}c_{y_2} = \mathrm{i}Z_2$; $\langle\psi|Z_j|\psi\rangle = \frac{1}{2}(1-1) = 0$.

(d) $c_{x_1}c_{y_2} = \mathrm{i}X_1Z_1X_2Z_2$ and $c_{x_2}c_{y_1} = \mathrm{i}Z_1X_2X_1Z_1 = -\mathrm{i}X_1X_2$ (using $Z_1X_1 = -X_1Z_1$ and $Z_1^2 = 1$). Both $X_1Z_1X_2Z_2\psi$ and $X_1X_2\psi$ equal $\frac{1}{\sqrt{2}}(|11\rangle + \mathrm{i}|00\rangle)$, which is orthogonal to ψ: $\langle\psi|(|11\rangle + \mathrm{i}|00\rangle) = \frac{1}{\sqrt{2}}(-\mathrm{i}+\mathrm{i}) = 0$.

Antisymmetry gives the remaining entries. □

6 Examples

Two families of states are chains of Section 3: the occupation states of Section 5, as chains without virtual modes (Theorem 6.2), and the Kitaev chain at its solvable point, as a chain with one virtual mode per bond (Theorem 6.5). The third example is the hopping gate on $\mathcal{F}_n$, a Gaussian unitary (Theorem 6.8); such unitaries act on covariances by orthogonal congruence (Theorem 6.9).

6.1 Product states

A chain with $M = 0$ virtual modes has no bonds.

Proposition 6.1 (Chains without virtual modes). Lean ✓ *Let* $M = 0$ *and* $P \geq 0$. *For any on-site covariances* $\Gamma_\bullet$ *(matrices on* $I_{\text{site}} = (x_C, y_C)$*),* $1 - D\beta_N = 1$ *is invertible and*

$$\Gamma_N(\Gamma_\bullet)_{(i,a)(j,b)} = \begin{cases} (\Gamma_i)_{ab} & i = j \\ 0 & i \neq j, \end{cases}$$

the direct sum of the on-site covariances. It is admissible (pure) if and only if every Γ_i *is.*

Proof. E is empty, so D, β_N and the term $B\beta_N(1-D\beta_N)^{-1}C$ are empty matrices and $\Gamma_N = A = \Gamma_{\text{in}}$, which is $\operatorname{diag}(\Gamma_i)$ relabelled by e restricted to R (Definition 3.1). Admissibility and purity transfer by Lemma 3.3. □

Theorem 6.2 (Occupation states as chains). Lean ✓ *Let* $s : \mathbb{Z}/N \times \{1, \ldots, P\} \to \{0, 1\}$ *and let* s_i *be its restriction to site* i. *Identify the* NP *modes with the pairs* (i, c) *site by site,* $(i, c) \mapsto c + Pi$. *Then the covariance of the occupation state* $|s\rangle\langle s|$ *on* NP *modes equals* $\Gamma_N(\Gamma_\bullet)$ *with* $M = 0$ *and* $\Gamma_i = \begin{pmatrix} 0 & \Lambda_{s_i} \\ -\Lambda_{s_i} & 0 \end{pmatrix}$, *the occupation covariance of the site (Theorem 5.8). In particular the vacuum on* NP *modes is the chain with* $\Gamma_i = -\Omega_P$ *at every site.*

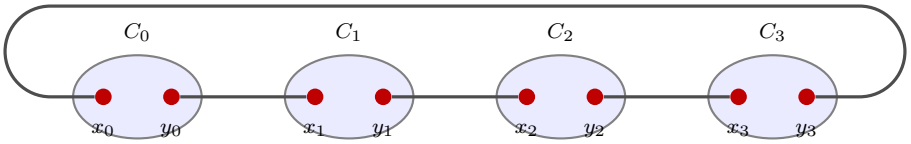


Figure 4: The Kitaev chain covariance K_N for $N = 4$. Dots are the Majorana labels x_i, y_i of the physical mode C_i, $i \in \mathbb{Z}/N$; a line joining y_i to x_{i+1} is the pair $(K_N)_{y_i x_{i+1}} = 1$. Every label lies on exactly one line, which is the purity $K_N^2 = -1$.

Proof. Theorem 5.8 gives $\Gamma(|s\rangle\langle s|) = \begin{pmatrix} 0 & \Lambda_s \\ -\Lambda_s & 0 \end{pmatrix}$ on the NP modes; Λ_s is diagonal, so the (x, y) block couples only equal modes, and after the site-by-site relabelling this is the right side of Proposition 6.1. The vacuum is Corollary 5.9. □

6.2 The Kitaev chain

The Kitaev chain at the point where the chemical potential vanishes and the hopping equals the pairing amplitude has a ground state in which y_i is paired with x_{i+1} around the ring [5, Sec. 2]. With sites $i \in \mathbb{Z}/N$ and the labels $(x_0, \dots, x_{N-1}, y_0, \dots, y_{N-1})$, this pairing is the matrix K_N with

$$(K_N)_{y_i\, x_{i+1}} = 1, \qquad (K_N)_{x_{i+1}\, y_i} = -1, \qquad \text{all other entries } 0$$

(Figure 4); for $N = 1$, $K_1 = -\Omega_1$ is the vacuum covariance. K_N is the ground covariance of the quadratic Hamiltonian $H_{-K_N} = -\frac{\mathrm{i}}{2}\sum_i c_{y_i} c_{x_{i+1}}$ (Lemma 11.10 and Theorem 11.12); Kitaev's $H = \mathrm{i}t\sum_j c_{2j} c_{2j+1}$ with $t > 0$, which in the dictionary $c_{2j-1} = x_j$, $c_{2j} = y_j$ reads $\mathrm{i}t\sum_i c_{y_i} c_{x_{i+1}} = 2t\, H_{+K_N}$, has ground covariance $-K_N$; we remove the sign of t by reflecting the y Majoranas (Remark 11.11). The chain construction of Section 3 produces K_N from one virtual mode per bond and an on-site covariance that is admissible but not pure.

Definition 6.3 (Kitaev on-site covariance)**.** Let $M = P = 1$. Γ_K is the matrix on the labels $I_{\text{site}} = (x_A, x_B, x_C, y_A, y_B, y_C)$ with

$$(\Gamma_K)_{x_C\, x_A} = 1, \qquad (\Gamma_K)_{y_C\, x_B} = 1,$$

their antisymmetric partners -1, and every other entry 0; the rows and columns of y_A and y_B are zero.

Lemma 6.4. Lean ✓ *Γ_K is admissible and not pure: $\Gamma_K^{\mathsf{T}} = -\Gamma_K$, $1 - \Gamma_K^{\mathsf{T}}\Gamma_K$ is the projector onto (y_A, y_B), and $\Gamma_K^2 \neq -1$.*

Proof. Antisymmetry is by construction. Each nonzero column of Γ_K has a single entry ± 1, and the four nonzero columns x_A, x_C, x_B, y_C are supported on the rows x_C, x_A, y_C, x_B respectively, so $\Gamma_K^{\mathsf{T}}\Gamma_K = \mathrm{diag}(1, 1, 1, 0, 0, 1)$ on $(x_A, x_B, x_C, y_A, y_B, y_C)$ and $1 - \Gamma_K^{\mathsf{T}}\Gamma_K$ is the projector onto (y_A, y_B), positive semidefinite. $\Gamma_K^2 = -\Gamma_K^{\mathsf{T}}\Gamma_K \neq -1$. □

Theorem 6.5 (The Kitaev chain as a chain with one virtual mode)**.** Lean ✓ *For every $N \geq 1$, with $\Gamma_i = \Gamma_K$ at every site: $\det(1 - D\beta_N) = 1$ and*

$$\Gamma_N(\Gamma_K, \dots, \Gamma_K) = K_N.$$

K_N is pure and translation invariant, and its momentum-space covariance (Definition 4.2) is

$$\Sigma(k) = \begin{pmatrix} 0 & -\omega_k \\ \overline{\omega_k} & 0 \end{pmatrix} \quad \text{on } (x_C, y_C), \qquad k \in \mathbb{Z}/N.$$

Proof. (a) Γ_K has no entry between two virtual labels and none between x_C and y_C, so the retained block A and the eliminated block D of Γ_{in} vanish: $\det(1 - D\beta_N) = \det 1 = 1$ and $\Gamma_N = B\beta_N C$. In the labels of Definition 3.1, the only entries of B are $B_{y_{C_i} x_{B_i}} = (\Gamma_K)_{y_C x_B} = 1$ and $B_{x_{C_i} x_{A_i}} = (\Gamma_K)_{x_C x_A} = 1$, $C = -B^{\mathsf{T}}$, and β_N pairs $x_{A_{i+1}}$ with x_{B_i}: $(\beta_N)_{x_{A_{i+1}} x_{B_i}} = 1$ (Definition 3.2). Each row of $B\beta_N C$ therefore has a single nonzero entry,

$$(\Gamma_N)_{y_{C_i}\, x_{C_{i+1}}} = B_{y_{C_i} x_{B_i}}\, (\beta_N)_{x_{B_i} x_{A_{i+1}}}\, C_{x_{A_{i+1}} x_{C_{i+1}}} = 1 \cdot (-1) \cdot (-1) = 1,$$
$$(\Gamma_N)_{x_{C_{i+1}}\, y_{C_i}} = B_{x_{C_{i+1}} x_{A_{i+1}}}\, (\beta_N)_{x_{A_{i+1}} x_{B_i}}\, C_{x_{B_i} y_{C_i}} = 1 \cdot 1 \cdot (-1) = -1,$$

which is K_N; for $N = 1$ the same computation pairs y_{C_0} with x_{C_0} and gives $-\Omega_1$.

(b) $K_N^2 = -1$: every label is paired with exactly one other label, with product of entries -1. The entries depend on i, j only through $j - i$, which is translation invariance.

(c) The twisted on-site covariance $\Gamma_K^{(k)}$ of Definition 4.1 has $A_k = 0$ and $D_k = 0$ as in (a), so $\Sigma(k) = B_k \beta_1 C_k$. B_k has the entries $(B_k)_{x_C\, x_A} = 1$ and $(B_k)_{y_C\, x_B} = \overline{\omega_k}$ (the twist puts $\overline{\omega_k}$ on the x_B column), $C_k = -B_k^*$, and β_1 pairs x_A with x_B with $(\beta_1)_{x_A x_B} = 1$. Hence

$$\Sigma(k)_{y_C x_C} = (B_k)_{y_C x_B} (\beta_1)_{x_B x_A} (C_k)_{x_A x_C} = \overline{\omega_k} \cdot (-1) \cdot (-1) = \overline{\omega_k},$$
$$\Sigma(k)_{x_C y_C} = (B_k)_{x_C x_A} (\beta_1)_{x_A x_B} (C_k)_{x_B y_C} = 1 \cdot 1 \cdot (-\omega_k) = -\omega_k,$$

and the diagonal entries vanish. □

Proposition 6.6 (A pure on-site choice closes a virtual loop)**.** Lean ✓ *Let Γ'_K, the pure completion of Γ_K, be Γ_K with the additional entries $(\Gamma'_K)_{y_A y_B} = -1$, $(\Gamma'_K)_{y_B y_A} = 1$, which is pure. Then, for the on-site covariances $\Gamma_i = \Gamma'_K$,*

$$\det(1 - D\beta_N) = 2\big(1 - (-1)^N\big) = \begin{cases} 4 & N \text{ odd} \\ 0 & N \text{ even,} \end{cases}$$

so the chain covariance is defined only for odd N, where it equals K_N.

Proof. (a) The twisted virtual block of Γ'_K has the single pair $(D_k)_{y_A y_B} = -\overline{\omega_k}$, $(D_k)_{y_B y_A} = \omega_k$, and β_1 pairs y_A with y_B with entry -1; so $1 - D_k\beta_1 = \operatorname{diag}(1, 1, 1 + \overline{\omega_k}, 1 + \omega_k)$ on (x_A, x_B, y_A, y_B) and $\det(1 - D_k\beta_1) = |1 + \omega_k|^2$. By Theorem 4.6, $\det(1 - D\beta_N) = \big|\prod_k (1 + \omega_k)\big|^2$.

(b) The ω_k, $k \in \mathbb{Z}/N$, are the N distinct N-th roots of unity (they satisfy $\omega_k^N = 1$, and $\omega_k = \omega_l$ forces $k = l$ since $|k - l| < N$), so $z^N - 1 = \prod_k (z - \omega_k)$; at $z = -1$, $(-1)^N - 1 = (-1)^N \prod_k (1 + \omega_k)$, that is $\prod_k (1 + \omega_k) = 1 - (-1)^N$. Hence $\det(1 - D\beta_N) = (1 - (-1)^N)^2 = 2(1 - (-1)^N)$, as $1 - (-1)^N \in \{0, 2\}$.

(c) For odd N every $1 - D_k\beta_1$ is invertible, and $(1 - D_k\beta_1)^{-1} = \operatorname{diag}(1, 1, (1 + \overline{\omega_k})^{-1}, (1 + \omega_k)^{-1})$ acts as the identity on the labels x_A, x_B, the only virtual labels coupled to C; so $\Sigma(k) = B_k\beta_1(1 - D_k\beta_1)^{-1} C_k$ is the matrix of Theorem 6.5(c). By Theorem 4.8, $(\Gamma_N)_{(i,y)(j,x)} = \frac{1}{N} \sum_k \overline{\chi_k(i)}\, \overline{\omega_k}\, \chi_k(j) = \frac{1}{N} \sum_k \overline{\chi_k(i+1)}\, \chi_k(j) = [\, j = i + 1\,]$ by $\chi_k(i + 1) = \chi_k(i)\omega_k$ and orthogonality, likewise $(\Gamma_N)_{(i,x)(j,y)} = -[\, i = j + 1\,]$, and the (x, x) and (y, y) blocks vanish: this is K_N. □

The pairs (y_{A_i}, y_{B_i}) of Γ'_K and the bonds $(y_{B_i}, y_{A_{i+1}})$ form a closed loop of Majorana pairs around the ring with no physical label on it, and the determinant is the product over momenta of the loop's factors $|1 + \omega_k|^2$. The three-site family of Section 3 contains such a loop as well: the rotation of Γ_{seed} leaves the pair (y_A, y_B) untouched, so $D_k\beta_1$ is block diagonal with the same (y_A, y_B) block as in (a), $\det(1 - D_k\beta_1)$ carries the factor $|1 + \omega_k|^2$, and the chain determinant of that family vanishes for even N.

6.3 Hopping gates

Fix n modes and two distinct modes $j \neq k$, with the Jordan–Wigner operators c_a of Definition 5.3, and $\varphi \in \mathbb{R}$.

Definition 6.7 (Hopping gate)**.**

$$U_{jk}(\varphi) = \big(\cos\varphi + \sin\varphi\, c_{x_j} c_{x_k}\big)\big(\cos\varphi + \sin\varphi\, c_{y_j} c_{y_k}\big),$$

a matrix on $\mathcal{F}_n$. The rotation $O_{jk}(\varphi) \in \mathbb{R}^{2n\times 2n}$ acts on the labels (x_j, x_k) and on (y_j, y_k) by

$$\mathsf{R} = \begin{pmatrix} \cos 2\varphi & -\sin 2\varphi \\ \sin 2\varphi & \cos 2\varphi \end{pmatrix}$$

and as the identity on all other labels; in (x, y) blocks, $O_{jk}(\varphi) = \mathrm{diag}(\mathsf{R}_n, \mathsf{R}_n)$ with R_n the rotation by 2φ in the plane (j, k) of $\mathbb{R}^n$.

$U_{jk}(\varphi) = \exp\big(\varphi\,(c_{x_j}c_{x_k} + c_{y_j}c_{y_k})\big) = \exp\big(2\varphi\,(a_j^* a_k - a_k^* a_j)\big)$ with $a_j = \frac{1}{2}(c_{x_j} + \mathrm{i}c_{y_j})$, the number-conserving hopping gate of angle 2φ; on covariances the hopping gate is the Givens rotation by 2φ of the modes j, k (Theorem 6.9). We do not prove the exponential form in these notes.

Theorem 6.8 (The hopping gate is a Gaussian unitary)**.** Lean ✓ *$U_{jk}(\varphi)$ is unitary, and for every label a,*

$$U_{jk}(\varphi)\, c_a\, U_{jk}(\varphi)^* = \sum_b \big(O_{jk}(\varphi)\big)_{ab}\, c_b,$$

that is, $U c_{x_j} U^* = \cos 2\varphi\, c_{x_j} - \sin 2\varphi\, c_{x_k}$, $U c_{x_k} U^* = \sin 2\varphi\, c_{x_j} + \cos 2\varphi\, c_{x_k}$, the same with y, and $U c_a U^* = c_a$ for the other labels.

Proof. Let $g = c_{x_j} c_{x_k}$. By Theorem 5.4, $g^2 = -1$, $g^* = -g$, g anticommutes with c_{x_j} and c_{x_k} and commutes with every other c_a, including c_{y_j}, c_{y_k}; moreover $g c_{x_j} = -c_{x_k}$, $c_{x_j} g = c_{x_k}$, $g c_{x_k} = c_{x_j}$, $c_{x_k} g = -c_{x_j}$. Hence $V = \cos\varphi + \sin\varphi\, g$ satisfies $VV^* = (\cos\varphi + \sin\varphi\, g)(\cos\varphi - \sin\varphi\, g) = \cos^2\varphi - \sin^2\varphi\, g^2 = 1$, likewise $V^*V = 1$, and

$$\begin{aligned} V c_{x_j} V^* &= (\cos\varphi\, c_{x_j} + \sin\varphi\, g c_{x_j})(\cos\varphi - \sin\varphi\, g) = (\cos\varphi\, c_{x_j} - \sin\varphi\, c_{x_k})(\cos\varphi - \sin\varphi\, g) \\ &= (\cos^2\varphi - \sin^2\varphi)\, c_{x_j} - 2\sin\varphi\cos\varphi\, c_{x_k} = \cos 2\varphi\, c_{x_j} - \sin 2\varphi\, c_{x_k}; \end{aligned}$$

similarly $V c_{x_k} V^* = \sin 2\varphi\, c_{x_j} + \cos 2\varphi\, c_{x_k}$, and $V c_a V^* = c_a V V^* = c_a$ for the other labels. The same computation applies to the y factor $W = \cos\varphi + \sin\varphi\, c_{y_j} c_{y_k}$, which commutes with c_{x_j}, c_{x_k}; $U = VW$ is unitary as a product of unitaries, and $U c_a U^* = V(W c_a W^*)V^*$ gives the formulas label by label. □

Theorem 6.9 (Gaussian unitaries act by orthogonal congruence)**.** Lean ✓ *Let U be unitary and $O \in \mathbb{R}^{2n\times 2n}$ with $U c_a U^* = \sum_b O_{ab} c_b$ for all a. Then $OO^{\mathsf{T}} = 1$, and for every density operator ρ, $U\rho U^*$ is a density operator with*

$$\Gamma(U\rho U^*) = O^{\mathsf{T}}\, \Gamma(\rho)\, O.$$

Proof. (a) $\mathrm{Tr}(c_a c_b) = 2^n \delta_{ab}$: $\mathrm{Tr}(c_a c_b) = \mathrm{Tr}(c_b c_a)$, so $2\,\mathrm{Tr}(c_a c_b) = \mathrm{Tr}(c_a c_b + c_b c_a) = 2\delta_{ab}\,\mathrm{Tr}\,1$ (Theorem 5.4).

(b) $OO^{\mathsf{T}} = 1$: $\mathrm{Tr}\big((U c_a U^*)(U c_b U^*)\big) = \mathrm{Tr}(U c_a c_b U^*) = \mathrm{Tr}(c_a c_b) = 2^n \delta_{ab}$, and by the hypothesis the same trace is $\sum_{a',b'} O_{aa'} O_{bb'}\,\mathrm{Tr}(c_{a'} c_{b'}) = 2^n \sum_{a'} O_{aa'} O_{ba'} = 2^n (OO^{\mathsf{T}})_{ab}$.

(c) $U^* c_a U = \sum_b O_{ba} c_b$: conjugating the hypothesis by U^* gives $c_b = \sum_{b'} O_{bb'}\, U^* c_{b'} U$, and $O^{\mathsf{T}} O = 1$ inverts this relation.

(d) Density: $U\rho U^* \succeq 0$ and $\operatorname{Tr}(U\rho U^*) = \operatorname{Tr}\rho = 1$. Covariance: $\Gamma(U\rho U^*)_{ab} = \frac{\mathrm{i}}{2}\operatorname{Tr}(U\rho U^*[c_a, c_b]) = \frac{\mathrm{i}}{2}\operatorname{Tr}(\rho\, U^*[c_a, c_b]U)$, and by (c)

$$U^*[c_a, c_b]U = [U^*c_aU, U^*c_bU] = \sum_{a'b'} O_{a'a}O_{b'b}[c_{a'}, c_{b'}];$$

taking the trace gives $\sum_{a'b'} O_{a'a}\Gamma(\rho)_{a'b'}O_{b'b} = (O^{\mathsf{T}}\Gamma(\rho)O)_{ab}$, both sides being real (Theorem 5.6). □

Corollary 6.10 (One gate on an occupation state)**.** Lean ✓ *For $n = 2$, $j = 1$, $k = 2$ and $\rho = |10\rangle\langle 10|$,*

$$\Gamma(U_{12}(\varphi)\rho U_{12}(\varphi)^*) = \begin{pmatrix} 0 & X \\ -X^{\mathsf{T}} & 0 \end{pmatrix}, \qquad X = \begin{pmatrix} \cos 4\varphi & -\sin 4\varphi \\ -\sin 4\varphi & -\cos 4\varphi \end{pmatrix},$$

in the labels (x_1, x_2, y_1, y_2). The vacuum is fixed by every hopping gate: $\Gamma(U_{jk}(\varphi)\,|0\rangle\langle 0|\,U_{jk}(\varphi)^*) = -\Omega_n$.

Proof. Theorem 6.9 with $\Gamma(\rho) = \begin{pmatrix} 0 & \Lambda_{10} \\ -\Lambda_{10} & 0 \end{pmatrix}$, $\Lambda_{10} = \operatorname{diag}(1, -1)$ (Theorem 5.8), and $O = \operatorname{diag}(\mathsf{R}, \mathsf{R})$ with R the rotation by 2φ: the (x, y) block is $\mathsf{R}^{\mathsf{T}}\Lambda_{10}\mathsf{R} = X$, with $\cos^2 2\varphi - \sin^2 2\varphi = \cos 4\varphi$ and $2\sin 2\varphi\cos 2\varphi = \sin 4\varphi$. For the vacuum, $\Lambda = -1_n$ and $\mathsf{R}_n^{\mathsf{T}}(-1)\mathsf{R}_n = -1$ since $\mathsf{R}_n^{\mathsf{T}}\mathsf{R}_n = 1$, which follows from $O^{\mathsf{T}}O = 1$. □

A composition of gates acts by the product of the rotations, by Theorem 6.9 applied to each factor.

7 Quadratic Hamiltonians and the realization of covariances

Every admissible matrix is the covariance of a density operator (Theorem 7.19), the converse of Theorem 5.7, and every quadratic Hamiltonian has a Gaussian ground state (Theorem 7.20 and Corollary 9.11(c)). Both rest on one matrix fact, the normal form $\Gamma = O\,\mathsf{D}(\varepsilon)\,O^{\mathsf{T}}$ of a real antisymmetric matrix (Theorem 7.13), and on one Fock-space fact, that every rotation O is implemented by a unitary (Theorem 7.17).

Throughout, n modes carry the labels $x_1, \dots, x_n, y_1, \dots, y_n$, vectors of $\mathbb{R}^{2n}$ are indexed by the labels, e_a is the unit vector of the label a, $u\cdot v$ is the Euclidean inner product, and $\langle u, v\rangle = \sum_a \overline{u_a}v_a$ is the inner product of $\mathbb{C}^{2n}$, conjugate-linear in the first argument. For a function d on the configurations $s \in \{0,1\}^n$, $\operatorname{diag}(d)$ is the matrix on $\mathcal{F}_n$ with $d(s)$ at the position (s, s) and zero elsewhere. The Majorana operators c_a are those of Definition 5.3, ρ is a density operator and $\Gamma(\rho)$ its covariance (Definition 5.5). A unitary U *implements* a real matrix O when $Uc_aU^* = \sum_b O_{ab}c_b$ for every label a (U is the Gaussian unitary, or Bogoliubov transformation, of the rotation O; Theorem 6.9); then $OO^{\mathsf{T}} = 1$, $U^*c_aU = \sum_b O_{ba}c_b$, U^* implements O^{T}, and $\Gamma(U\rho U^*) = O^{\mathsf{T}}\Gamma(\rho)O$, $\Gamma(U^*\rho U) = O\,\Gamma(\rho)\,O^{\mathsf{T}}$.

7.1 The canonical form and the energy of a quadratic Hamiltonian

Definition 7.1 (Canonical form)**.** For $\varepsilon \in \mathbb{R}^n$, $\mathsf{D}(\varepsilon)$ is the matrix on the labels with

$$\mathsf{D}(\varepsilon)_{x_jy_j} = \varepsilon_j, \qquad \mathsf{D}(\varepsilon)_{y_jx_j} = -\varepsilon_j,$$

and every other entry 0; in (x, y) blocks, $\mathsf{D}(\varepsilon) = \begin{pmatrix} 0 & \operatorname{diag}(\varepsilon) \\ -\operatorname{diag}(\varepsilon) & 0 \end{pmatrix}$.

The covariances of Section 5 are of this form: $\Omega_n = \mathsf{D}(1,\dots,1)$ (equation (1.3)), the vacuum has $\mathsf{D}(-1,\dots,-1)$, and the occupation state $|s\rangle\langle s|$ has $\mathsf{D}(\nu(s))$ with $\nu_j(s) = -1$ for $s_j = 0$ and $+1$ for $s_j = 1$ (Theorem 5.8). Since $\mathsf{D}(\varepsilon)^{\mathsf{T}}\mathsf{D}(\varepsilon) = \operatorname{diag}(\varepsilon^2, \varepsilon^2)$, $\mathsf{D}(\varepsilon)$ is admissible if and only if $|\varepsilon_j| \le 1$ for all j and pure if and only if $|\varepsilon_j| = 1$ for all j.

Definition 7.2 (Quadratic Hamiltonian)**.** For a real matrix h on the labels,

$$H_h = \frac{\mathrm{i}}{4} \sum_{a,b} h_{ab}\, c_a c_b,$$

a matrix on $\mathcal{F}_n$. The results below assume $h^{\mathsf{T}} = -h$.

For one mode and $h = \mathsf{D}(\varepsilon)$, $H_h = \frac{\mathrm{i}}{4}\varepsilon(c_x c_y - c_y c_x) = \frac{\mathrm{i}}{2}\varepsilon\, c_x c_y$; with $c_x = X$ and $c_y = \mathrm{i}XZ$ (Definition 5.3, $n = 1$) this is $-\frac{1}{2}\varepsilon Z = \operatorname{diag}(-\frac{\varepsilon}{2}, \frac{\varepsilon}{2})$ on $(|0\rangle, |1\rangle)$: the empty mode has energy $-\varepsilon/2$, the filled mode $+\varepsilon/2$. Lemma 7.4 is this computation for n modes.

Lemma 7.3 (Hermiticity, energy, conjugation)**.** Lean ✓ *Let $h^{\mathsf{T}} = -h$.*

(a) $H_h^* = H_h$.

(b) For every Hermitian ρ, $\operatorname{Tr}(\rho H_h) = \frac{1}{4}\sum_{a,b} h_{ab}\,\Gamma(\rho)_{ab}$, a real number.

(c) If the unitary U implements O, then $U H_h U^* = H_{O^{\mathsf{T}} h O}$.

Proof. (a) $(c_a c_b)^* = c_b c_a$, so $H_h^* = -\frac{\mathrm{i}}{4}\sum_{ab} h_{ab} c_b c_a = -\frac{\mathrm{i}}{4}\sum_{ab} h_{ba} c_a c_b = H_h$. (b) By Theorem 5.4, $c_a c_b = \delta_{ab} + \frac{1}{2}[c_a, c_b]$; the diagonal terms carry $h_{aa} = 0$, so $\operatorname{Tr}(\rho H_h) = \frac{\mathrm{i}}{8}\sum_{ab} h_{ab} \operatorname{Tr}(\rho[c_a, c_b]) = \frac{1}{4}\sum_{ab} h_{ab}\Gamma(\rho)_{ab}$, real by Theorem 5.6. (c) $U c_a c_b U^* = (U c_a U^*)(U c_b U^*) = \sum_{c,d} O_{ac} O_{bd}\, c_c c_d$, hence $U H_h U^* = \frac{\mathrm{i}}{4}\sum_{c,d}(\sum_{a,b} O_{ac} h_{ab} O_{bd}) c_c c_d = H_{O^{\mathsf{T}} h O}$. □

Lemma 7.4 (The Hamiltonian of a canonical form is diagonal)**.** Lean ✓ *Let $n_j = \operatorname{diag}(s \mapsto s_j)$ be the number operator of mode j. Then $\mathrm{i}\, c_{x_j} c_{y_j} = 2n_j - 1$, and for every $\varepsilon \in \mathbb{R}^n$*

$$H_{\mathsf{D}(\varepsilon)} = \sum_j \varepsilon_j (n_j - \tfrac{1}{2}) = \operatorname{diag}\Big(s \mapsto \sum_j \varepsilon_j (s_j - \tfrac{1}{2})\Big).$$

Proof. $Z_{<j}$ commutes with X_j (Lemma 5.2), so $c_{x_j} c_{y_j} = \mathrm{i}\, Z_{<j} X_j Z_{<j} X_j Z_j = \mathrm{i} Z_j$ and $\mathrm{i} c_{x_j} c_{y_j} = -Z_j = \operatorname{diag}(-(-1)^{s_j}) = 2n_j - 1$. The only nonzero entries of $\mathsf{D}(\varepsilon)$ are ε_j at (x_j, y_j) and $-\varepsilon_j$ at (y_j, x_j), so $H_{\mathsf{D}(\varepsilon)} = \frac{\mathrm{i}}{4}\sum_j \varepsilon_j (c_{x_j} c_{y_j} - c_{y_j} c_{x_j}) = \frac{\mathrm{i}}{2}\sum_j \varepsilon_j c_{x_j} c_{y_j} = \frac{1}{2}\sum_j \varepsilon_j (2n_j - 1)$. □

Write $d_\varepsilon(s) = \sum_j \varepsilon_j (s_j - \frac{1}{2})$ for the diagonal entry of $H_{\mathsf{D}(\varepsilon)}$ at the configuration s.

7.2 Ground states of a canonical form

Lemma 7.5 (Ground state of a diagonal operator)**.** Lean ✓ *Let $d : \{0,1\}^n \to \mathbb{R}$, $D = \operatorname{diag}(d)$, and let $s^\star$ satisfy $d(s) \ge d(s^\star)$ for all s. For every density operator ρ, $\operatorname{Tr}(\rho D) = \sum_s d(s)\rho_{ss} \ge d(s^\star)$. If moreover $d(s) > d(s^\star)$ for every $s \ne s^\star$, then $\operatorname{Tr}(\rho D) = d(s^\star)$ implies $\rho = |s^\star\rangle\langle s^\star|$.*

Proof. $\rho_{ss} \ge 0$ and $\sum_s \rho_{ss} = 1$, so $\operatorname{Tr}(\rho D) - d(s^\star) = \sum_s (d(s) - d(s^\star))\rho_{ss} \ge 0$. Under the strict gaps, equality forces $\rho_{ss} = 0$ for $s \ne s^\star$. For $\rho \succeq 0$ a vanishing diagonal entry ρ_{ss} forces the row and column s to vanish: the 2×2 principal submatrix on $\{s, t\}$ is positive semidefinite, so $\rho_{ss}\rho_{tt} - |\rho_{st}|^2 \ge 0$, that is $\rho_{st} = 0$. Hence $\rho = \rho_{s^\star s^\star} |s^\star\rangle\langle s^\star|$, and $\operatorname{Tr}\rho = 1$ gives $\rho_{s^\star s^\star} = 1$. □

Definition 7.6 (Ground configuration and energy of a canonical form)**.** For $\varepsilon \in \mathbb{R}^n$ let $s^\varepsilon \in \{0,1\}^n$ have $s_j^\varepsilon = 1$ if $\varepsilon_j < 0$ and $s_j^\varepsilon = 0$ otherwise, and let $\mathcal{E}_{\mathrm{gs}}(\varepsilon) = -\frac{1}{2}\sum_j |\varepsilon_j|$. With the convention $\operatorname{sgn} 0 = 1$, $\nu(s^\varepsilon) = -\operatorname{sgn}\varepsilon$ entrywise.

Theorem 7.7 (Ground states of a canonical form)**.** Lean ✓ *Let $\varepsilon \in \mathbb{R}^n$.*

(a) For every density operator ρ, $\mathrm{Tr}(\rho H_{\mathsf{D}(\varepsilon)}) \geq \mathcal{E}_{\mathrm{gs}}(\varepsilon)$, with equality for $\rho = |s^\varepsilon\rangle\langle s^\varepsilon|$, whose covariance is $\mathsf{D}(-\operatorname{sgn}\varepsilon)$.

(b) If $\varepsilon_j \neq 0$ for all j, then $\mathrm{Tr}(\rho H_{\mathsf{D}(\varepsilon)}) = \mathcal{E}_{\mathrm{gs}}(\varepsilon)$ implies $\rho = |s^\varepsilon\rangle\langle s^\varepsilon|$.

Proof. By Lemma 7.4, $H_{\mathsf{D}(\varepsilon)} = \mathrm{diag}(d)$ with $d(s) = \sum_j \varepsilon_j(s_j - \frac{1}{2})$. For each j, $\varepsilon_j(s_j - s_j^\varepsilon)$ equals $|\varepsilon_j|$ if $s_j \neq s_j^\varepsilon$ and 0 otherwise. Hence $d(s) - d(s^\varepsilon) = \sum_{j:\, s_j \neq s_j^\varepsilon} |\varepsilon_j| \geq 0$ and $d(s^\varepsilon) = -\frac{1}{2}\sum_j |\varepsilon_j|$. (a) is Lemma 7.5 with $s^\star = s^\varepsilon$, and the covariance is Theorem 5.8. (b) When every $\varepsilon_j \neq 0$ the gap $d(s) - d(s^\varepsilon)$ is positive for $s \neq s^\varepsilon$. □

Theorem 7.8 (Ground states of a rotated canonical form)**.** Lean ✓ *Let $h = O\,\mathsf{D}(\varepsilon)\,O^{\mathsf{T}}$ with $OO^{\mathsf{T}} = 1$, and let the unitary U implement O.*

(a) For every density operator ρ, $\mathrm{Tr}(\rho H_h) \geq \mathcal{E}_{\mathrm{gs}}(\varepsilon)$.

(b) Equality holds for the pure state $\rho_{\mathrm{gs}} = U^*|s^\varepsilon\rangle\langle s^\varepsilon|U$, and $\Gamma(\rho_{\mathrm{gs}}) = O\,\mathsf{D}(-\operatorname{sgn}\varepsilon)\,O^{\mathsf{T}}$.

(c) If $\varepsilon_j \neq 0$ for all j, then $\mathrm{Tr}(\rho H_h) = \mathcal{E}_{\mathrm{gs}}(\varepsilon)$ implies $\rho = \rho_{\mathrm{gs}}$.

Proof. By Lemma 7.3(c) and $O^{\mathsf{T}}O = 1$, $UH_hU^* = H_{\mathsf{D}(\varepsilon)}$, so $\mathrm{Tr}(\rho H_h) = \mathrm{Tr}\big((U\rho U^*)H_{\mathsf{D}(\varepsilon)}\big)$ with $U\rho U^*$ a density operator (Theorem 6.9). (a) is Theorem 7.7(a) for $U\rho U^*$. (b) $U\rho_{\mathrm{gs}}U^* = |s^\varepsilon\rangle\langle s^\varepsilon|$ attains $\mathcal{E}_{\mathrm{gs}}(\varepsilon)$, and $\Gamma(U^*\rho U) = O\,\Gamma(\rho)\,O^{\mathsf{T}}$ with $\Gamma(|s^\varepsilon\rangle\langle s^\varepsilon|) = \mathsf{D}(-\operatorname{sgn}\varepsilon)$. (c) Theorem 7.7(b) gives $U\rho U^* = |s^\varepsilon\rangle\langle s^\varepsilon|$. □

The energy $\frac{1}{4}\sum_{ab} h_{ab}\Gamma_{ab}$ of Lemma 7.3(b) is a function of the covariance alone, and in the frame of O it is a weighted sum of the (x_j, y_j) entries.

Lemma 7.9 (Energy in the rotated frame)**.** Lean ✓ *Let $h = O\,\mathsf{D}(\varepsilon)\,O^{\mathsf{T}}$ with $OO^{\mathsf{T}} = 1$, let $\Gamma^{\mathsf{T}} = -\Gamma$, and put $\Gamma' = O^{\mathsf{T}}\Gamma O$.*

(a) $(\Gamma')^{\mathsf{T}} = -\Gamma'$, and Γ' is admissible (pure) if and only if Γ is.

(b) $\frac{1}{4}\sum_{a,b} h_{ab}\Gamma_{ab} = \frac{1}{2}\sum_j \varepsilon_j \Gamma'_{x_j y_j}$.

(c) If $\Gamma = O\,\mathsf{D}(\lambda)\,O^{\mathsf{T}}$, then $\Gamma' = \mathsf{D}(\lambda)$ and $\frac{1}{4}\sum_{a,b} h_{ab}\Gamma_{ab} = \frac{1}{2}\sum_j \varepsilon_j\lambda_j$.

Proof. (a) $(O^{\mathsf{T}}\Gamma O)^{\mathsf{T}} = O^{\mathsf{T}}\Gamma^{\mathsf{T}}O = -\Gamma'$; admissibility and purity transfer by Theorem 1.7 in both directions, since $\Gamma = O\Gamma' O^{\mathsf{T}}$. (b) $\sum_{ab} h_{ab}\Gamma_{ab} = \mathrm{Tr}(h^{\mathsf{T}}\Gamma) = \mathrm{Tr}(O\mathsf{D}(\varepsilon)^{\mathsf{T}}O^{\mathsf{T}}\Gamma) = \mathrm{Tr}(\mathsf{D}(\varepsilon)^{\mathsf{T}}\Gamma') = \sum_{cd}\mathsf{D}(\varepsilon)_{cd}\Gamma'_{cd} = \sum_j \varepsilon_j(\Gamma'_{x_jy_j} - \Gamma'_{y_jx_j}) = 2\sum_j \varepsilon_j\Gamma'_{x_jy_j}$. (c) $O^{\mathsf{T}}O = 1$ gives $\Gamma' = \mathsf{D}(\lambda)$, whose (x_j, y_j) entry is λ_j. □

Theorem 7.10 (The variational bound on covariances)**.** Lean ✓ *Let $h = O\,\mathsf{D}(\varepsilon)\,O^{\mathsf{T}}$ with $OO^{\mathsf{T}} = 1$, and let Γ be admissible. Then*

$$\tfrac{1}{4}\sum_{a,b} h_{ab}\Gamma_{ab} \;\geq\; \mathcal{E}_{\mathrm{gs}}(\varepsilon).$$

If $\varepsilon_j \neq 0$ for all j, equality holds if and only if $\Gamma = O\,\mathsf{D}(-\operatorname{sgn}\varepsilon)\,O^{\mathsf{T}}$.

Proof. Put $\Gamma' = O^{\mathsf{T}}\Gamma O$; by Lemma 7.9, Γ' is admissible and $\frac{1}{4}\sum h_{ab}\Gamma_{ab} = \frac{1}{2}\sum_j \varepsilon_j\Gamma'_{x_jy_j}$. Every entry of an admissible matrix has modulus at most 1, since $|\Gamma'_{ab}| \leq \|\Gamma' e_b\| \leq 1$ (Theorem 1.3). Hence $\frac{1}{2}\sum_j \varepsilon_j\Gamma'_{x_jy_j} \geq -\frac{1}{2}\sum_j |\varepsilon_j|$, with equality if and only if $\Gamma'_{x_jy_j} = -\operatorname{sgn}\varepsilon_j$ for every j with $\varepsilon_j \neq 0$. Suppose all $\varepsilon_j \neq 0$ and equality holds. Fix j. The column $\Gamma' e_{y_j}$ has norm at most 1 and the entry $-\operatorname{sgn}\varepsilon_j$ at x_j, so all its other entries vanish; by antisymmetry $\Gamma'_{y_jx_j} = \operatorname{sgn}\varepsilon_j$, and the same argument on the column $\Gamma' e_{x_j}$ leaves only that entry. Doing this for every j pins every column of Γ', including the entries between different modes, and gives $\Gamma' = \mathsf{D}(-\operatorname{sgn}\varepsilon)$, that is $\Gamma = O\,\mathsf{D}(-\operatorname{sgn}\varepsilon)\,O^{\mathsf{T}}$. Conversely this matrix is admissible and has $\Gamma'_{x_jy_j} = -\operatorname{sgn}\varepsilon_j$ (Lemma 7.9(a),(c)). □

The covariance is linear in the state, so a mode that is occupied with probability $\frac{1}{2}(1+\lambda_j)$ and empty otherwise contributes the entry λ_j; a product of such one-mode mixtures realizes every admissible canonical form.

Definition 7.11 (Product mixture)**.** For $\lambda \in [-1,1]^n$ put $p_j(0) = \frac{1}{2}(1-\lambda_j)$, $p_j(1) = \frac{1}{2}(1+\lambda_j)$, and $\rho_\lambda = \sum_{s\in\{0,1\}^n} \big(\prod_j p_j(s_j)\big)|s\rangle\langle s|$.

Theorem 7.12 (Canonical forms are realized by product mixtures)**.** Lean ✓ *For $\lambda \in [-1,1]^n$, ρ_λ is a density operator with $\Gamma(\rho_\lambda) = \mathsf{D}(\lambda)$. If $|\lambda_j| = 1$ for all j, then $\rho_\lambda = |s\rangle\langle s|$ with $s_j = 1$ if $\lambda_j = 1$ and $s_j = 0$ if $\lambda_j = -1$.*

Proof. The weights are nonnegative and $\sum_s \prod_j p_j(s_j) = \prod_j (p_j(0) + p_j(1)) = 1$; a convex combination of density operators is a density operator. The covariance is linear in ρ, so $\Gamma(\rho_\lambda) = \mathsf{D}\big(\sum_s p(s)\nu(s)\big)$ with j-th entry $\big(\sum_b p_j(b)\nu(b)\big)\prod_{k\neq j}\big(\sum_b p_k(b)\big) = -p_j(0) + p_j(1) = \lambda_j$. If $|\lambda_j| = 1$ then p_j is concentrated on one bit and the sum has one term. □

7.3 The normal form

Theorems 7.8 and 7.10 assume h is a rotation of a canonical form. Every real antisymmetric matrix is one (Theorem 7.13): the Hermitian matrix $\mathrm{i}\Gamma$ has an orthonormal eigenbasis whose real and imaginary parts pair the labels.

Theorem 7.13 (Real antisymmetric normal form)**.** Lean ✓ *Let $\Gamma \in \mathbb{R}^{2n\times 2n}$ with $\Gamma^{\mathsf{T}} = -\Gamma$. There exist O with $OO^{\mathsf{T}} = 1$ and $\varepsilon \in \mathbb{R}^n$ with $\varepsilon_j \geq 0$ such that $\Gamma = O\,\mathsf{D}(\varepsilon)\,O^{\mathsf{T}}$.*

(a) There also exist such O, ε with $\det O = 1$, one entry of ε possibly negative.

(b) For every decomposition $\Gamma = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$ with $OO^{\mathsf{T}} = 1$: if Γ is admissible then $|\varepsilon_j| \leq 1$ for all j, and if Γ is pure then $|\varepsilon_j| = 1$ for all j.

Proof. $M = \mathrm{i}\Gamma$ is Hermitian with $\overline{M} = -M$. Let $v_1, \ldots, v_{2n}$ be an orthonormal eigenbasis of $\mathbb{C}^{2n}$, $Mv_k = \mu_k v_k$, $\mu_k \in \mathbb{R}$. Conjugating, $M\overline{v_k} = -\mu_k\overline{v_k}$, so $\overline{v_k}$ is an eigenvector for $-\mu_k$, and eigenvectors for different eigenvalues are orthogonal. Let $\mathcal{P} = \{k : \mu_k > 0\}$, and for $k \in \mathcal{P}$ let $p_k = \sqrt{2}\,\mathrm{Re}\,v_k$ and $q_k = -\sqrt{2}\,\mathrm{Im}\,v_k$, so that $v_k = (p_k - \mathrm{i}q_k)/\sqrt{2}$.

(i) Pairing. The real and imaginary parts of $\mathrm{i}\Gamma(p_k - \mathrm{i}q_k) = \mu_k(p_k - \mathrm{i}q_k)$ give $\Gamma q_k = \mu_k p_k$ and $\Gamma p_k = -\mu_k q_k$.

(ii) Orthonormality. For $j, k \in \mathcal{P}$, $\langle v_j, v_k\rangle = \delta_{jk}$ and $\langle\overline{v_j}, v_k\rangle = 0$. In real terms, $2\langle v_j, v_k\rangle = (p_j \cdot p_k + q_j \cdot q_k) + \mathrm{i}(q_j \cdot p_k - p_j \cdot q_k)$ and $2\langle\overline{v_j}, v_k\rangle = (p_j \cdot p_k - q_j \cdot q_k) - \mathrm{i}(p_j \cdot q_k + q_j \cdot p_k)$; adding and subtracting, $p_j \cdot p_k = q_j \cdot q_k = \delta_{jk}$ and $p_j \cdot q_k = 0$. So $\{p_k, q_k\}_{k\in\mathcal{P}}$ is orthonormal in $\mathbb{R}^{2n}$ and $|\mathcal{P}| \leq n$.

(iii) Completion. Choose an injection $\phi : \mathcal{P} \to \{1, \ldots, n\}$, place p_k at the label $x_{\phi(k)}$ and q_k at $y_{\phi(k)}$, and extend to an orthonormal basis $(b_\ell)_\ell$ of $\mathbb{R}^{2n}$ indexed by the labels.

(iv) Kernel. Let $x \in \mathbb{R}^{2n}$ be orthogonal to every p_k, q_k, $k \in \mathcal{P}$. Then $\langle v_k, x\rangle = 0$ and $\langle\overline{v_k}, x\rangle = 0$ for $k \in \mathcal{P}$. For $\mu_l < 0$, expanding $\overline{v_l}$ in the eigenbasis gives $\langle\overline{v_l}, x\rangle = \sum_k \langle\overline{v_l}, v_k\rangle\langle v_k, x\rangle$ with only $k \in \mathcal{P}$ contributing, so $\langle\overline{v_l}, x\rangle = 0$ and $\langle v_l, x\rangle = \overline{\langle\overline{v_l}, x\rangle} = 0$ (x is real). Hence $Mx = \sum_l \mu_l\langle v_l, x\rangle v_l = 0$, that is $\Gamma x = 0$; this applies to every b_ℓ at a label outside the image of ϕ.

(v) The matrix. Let O have the columns b_ℓ, so $OO^{\mathsf{T}} = 1$, and let $\varepsilon_j = \mu_k$ if $j = \phi(k)$ and $\varepsilon_j = 0$ otherwise. The entry $(O^{\mathsf{T}}\Gamma O)_{\ell\ell'} = b_\ell \cdot \Gamma b_{\ell'}$ is determined by the column ℓ': for a kernel label it vanishes by (iv); for $\ell' = y_{\phi(k)}$, $\Gamma b_{\ell'} = \mu_k p_k$ by (i), and $b_\ell \cdot p_k = [\ell = x_{\phi(k)}]$ by the orthonormality of the basis; for $\ell' = x_{\phi(k)}$, $\Gamma b_{\ell'} = -\mu_k q_k$ and $b_\ell \cdot q_k = [\ell = y_{\phi(k)}]$. Every label ℓ' is of one of the three kinds, so $O^{\mathsf{T}}\Gamma O = \mathsf{D}(\varepsilon)$ entry by entry.

(a) For $n = 0$, $O = 1$. For $n \geq 1$ and $\det O = -1$, let $\mathcal{R}$ be the reflection of the label y_1; then $O\mathcal{R}$ is orthogonal with determinant 1, $\mathcal{R}\mathsf{D}(\varepsilon)\mathcal{R} = \mathsf{D}(\varepsilon')$ with $\varepsilon'_1 = -\varepsilon_1$ and $\varepsilon'_j = \varepsilon_j$ otherwise, and $\Gamma = (O\mathcal{R})\mathsf{D}(\varepsilon')(O\mathcal{R})^{\mathsf{T}}$. (b) $\mathsf{D}(\varepsilon) = O^{\mathsf{T}}\Gamma O$ is admissible (pure) with Γ (Theorem 1.7), and the criterion after Definition 7.1 gives the bounds. □

For one mode, $\Gamma = \begin{pmatrix} 0 & \gamma \\ -\gamma & 0 \end{pmatrix}$ and $\mathrm{i}\Gamma$ has the eigenvectors $v = (1, -\mathrm{i})/\sqrt{2}$ with eigenvalue γ and $\bar{v}$ with eigenvalue $-\gamma$; for $\gamma > 0$ the vectors $p = \sqrt{2}\,\mathrm{Re}\, v = (1, 0)$ and $q = -\sqrt{2}\,\mathrm{Im}\, v = (0, 1)$ give $O = 1$, $\varepsilon = \gamma$; for $\gamma < 0$ the eigenvector for the positive eigenvalue $-\gamma$ is $\bar{v} = (1, \mathrm{i})/\sqrt{2}$, so $p = (1, 0)$, $q = (0, -1)$, $O = \mathrm{diag}(1, -1)$ and $\varepsilon = -\gamma$; this O has determinant -1, and (a) restores $\det O = 1$ by reflecting y_1, which returns $O = 1$, $\varepsilon = \gamma$.

7.4 Rotations are implemented by Gaussian unitaries

Theorem 6.8 implemented the rotations of two hopping gates. Every rotation in $SO(2n)$ is a product of plane rotations (Theorem 7.15), and each plane rotation is implemented by one factor $\cos\varphi + \sin\varphi\, c_a c_b$ (Lemma 7.16); hence every rotation is implemented (Theorem 7.17).

Definition 7.14 (Plane rotations)**.** For labels $a \neq b$ and $(p, q) \in \mathbb{R}^2$ with $p^2 + q^2 = 1$, $\mathsf{R}_{ab}(p, q)$ is the matrix on the labels equal to the identity except for the rows a and b, which are $(p, -q)$ and (q, p) on the columns (a, b) (a Givens rotation). A *rotation product* is a finite product of plane rotations, the empty product being 1.

$\mathsf{R}_{ab}(p, q)$ is orthogonal with determinant 1 and $\mathsf{R}_{ab}(p, q)^{\mathsf{T}} = \mathsf{R}_{ab}(p, -q)$, so rotation products form a group; a plane rotation of the labels other than i, extended by the identity on i, is a plane rotation of all labels. We state the definition and Theorem 7.15 for an arbitrary finite set of labels, of size m.

Theorem 7.15 ($SO(m)$ is generated by plane rotations)**.** Lean ✓ *Let O be a real matrix on a finite label set of size m with $OO^{\mathsf{T}} = 1$ and $\det O = 1$. Then O is a rotation product.*

Proof. Induction on m. For $m = 0$, $O = 1$; for $m = 1$, $O = (\det O) = (1)$. Let $m \geq 2$ and fix a label i.

Sweep. For every unit vector v there is a rotation product W of plane rotations in planes (a, i), $a \neq i$, with $Wv = e_i$. For a set S of labels not containing i, we build a rotation product W_S with $(W_S v)_a = 0$ for all $a \in S$ and, if $S \neq \emptyset$, $(W_S v)_i \geq 0$, one label at a time: $W_\emptyset = 1$; given W_S and a label $a \notin S \cup \{i\}$, put $w = W_S v$ and $r = \sqrt{w_a^2 + w_i^2}$; if $r = 0$ keep W_S, otherwise $W_{S\cup\{a\}} = \mathsf{R}_{ai}(w_i/r,\, w_a/r)\, W_S$, since $\mathsf{R}_{ai}(w_i/r, w_a/r)$ sends (w_a, w_i) to $(0, r)$ and fixes the other coordinates. With S the set of all labels other than i, $W_S v$ has zeros outside i, norm 1 and $(W_S v)_i \geq 0$, so $W_S v = e_i$.

Reduction. Apply the sweep to $v = Oe_i$. Then $\tilde{O} = WO$ is orthogonal with $\tilde{O}e_i = e_i$, so its column i is e_i; the rows of an orthogonal matrix are orthonormal too, and the row i has $\tilde{O}_{ii} = 1$, so $\tilde{O}_{ia} = 0$ for $a \neq i$. Hence $\tilde{O} = O' \oplus 1$ with O' orthogonal on the labels other than i and $\det O' = \det \tilde{O} = \det W \, \det O = 1$.

Induction. By the induction hypothesis O' is a rotation product on $m - 1$ labels, hence so is $\tilde{O} = O' \oplus 1$, and $O = W^{\mathsf{T}}\tilde{O}$ is a rotation product, W^{T} being the product of the transposed plane rotations in reverse order. □

Lemma 7.16 (One factor implements one plane rotation)**.** Lean ✓ *Let $a \neq b$ and $\varphi \in \mathbb{R}$. The unitary $V = \cos\varphi + \sin\varphi\, c_a c_b$ implements $\mathsf{R}_{ab}(\cos 2\varphi, \sin 2\varphi)$. For every (p, q) with $p^2 + q^2 = 1$ there is φ with $(\cos 2\varphi, \sin 2\varphi) = (p, q)$. If U implements O and V implements O', then UV implements $O'O$; the identity implements 1.*

Proof. The conjugation formulas of Theorem 6.8's proof, which use only the anticommutation relations of two distinct labels, are the rows of $\mathsf{R}_{ab}(\cos 2\varphi, \sin 2\varphi)$: row a is $(\cos 2\varphi, -\sin 2\varphi)$ on (a, b), row b is $(\sin 2\varphi, \cos 2\varphi)$, every other row is a unit vector. For the angle take $\theta = \arccos p$ if $q \geq 0$ and $\theta = -\arccos p$ otherwise, so that $\cos\theta = p$, $\sin\theta = \pm\sqrt{1-p^2} = q$, and $\varphi = \theta/2$. Product rule: $(UV)c_a(UV)^* = U\big(\sum_b O'_{ab}c_b\big)U^* = \sum_d (O'O)_{ad}c_d$. □

Theorem 7.17 (Every rotation is implemented by a Gaussian unitary). Lean ✓ *For every real matrix O on the Majorana labels with $OO^{\mathsf{T}} = 1$ and $\det O = 1$ there is a unitary U, a product of factors $\cos\varphi + \sin\varphi\, c_a c_b$, that implements O.*

Proof. By Theorem 7.15, $O = \mathsf{R}_1\mathsf{R}_2\cdots\mathsf{R}_T$ with plane rotations R_t; by Lemma 7.16 choose U_t implementing R_t; then $U_T\cdots U_2U_1$ implements $\mathsf{R}_1\mathsf{R}_2\cdots\mathsf{R}_T$ by the product rule. □

Remark 7.18. We do not implement reflections ($\det O = -1$) here; Theorem 7.13(a) supplies a decomposition with $\det O = 1$.

7.5 Realization and ground states

Theorem 7.19 (Realization of admissible covariances). Lean ✓ *Let Γ be admissible. Write $\Gamma = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$ with $\det O = 1$ and $|\varepsilon_j| \leq 1$ (Theorem 7.13), and let U implement O (Theorem 7.17). Then $\rho = U^*\rho_\varepsilon U$ (Definition 7.11) is a density operator with $\Gamma(\rho) = \Gamma$. If Γ is pure then $\rho = U^*|s\rangle\langle s|U$ with $s_j = 1$ if $\varepsilon_j = 1$ and $s_j = 0$ if $\varepsilon_j = -1$, a pure state.*

Proof. By Theorem 7.12, ρ_ε is a density operator with covariance $\mathsf{D}(\varepsilon)$; U^* implements O^{T}, so $U^*\rho_\varepsilon U$ is a density operator with covariance $O\,\mathsf{D}(\varepsilon)\,O^{\mathsf{T}} = \Gamma$ (Theorem 6.9). If Γ is pure then $|\varepsilon_j| = 1$ and $\rho_\varepsilon = |s\rangle\langle s|$. □

With Theorem 5.7: the covariances of density operators on n modes are exactly the admissible matrices on $2n$ labels, and every pure covariance is the covariance of a pure state.

Theorem 7.20 (Ground state of a quadratic Hamiltonian). Lean ✓ *Let $h^{\mathsf{T}} = -h$. There exist O with $OO^{\mathsf{T}} = 1$, $\det O = 1$, $\varepsilon \in \mathbb{R}^n$ with $h = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$, and a unitary U implementing O; for every such choice Theorems 7.8 and 7.10 hold. In particular:*

(a) $\min\{\mathrm{Tr}(\rho H_h) : \rho$ a density operator$\}$ exists, equals $-\frac{1}{2}\sum_j |\varepsilon_j|$, and is attained by the pure state $\rho_{\mathrm{gs}} = U^*|s^\varepsilon\rangle\langle s^\varepsilon|U$, whose covariance $O\mathsf{D}(-\operatorname{sgn}\varepsilon)O^{\mathsf{T}}$ is pure.

(b) If $\varepsilon_j \neq 0$ for all j, the minimizer is unique, and among admissible matrices the minimizer of $\Gamma \mapsto \frac{1}{4}\sum_{ab} h_{ab}\Gamma_{ab}$ is unique and equals $\Gamma(\rho_{\mathrm{gs}})$.

Proof. Theorem 7.13(a) and Theorem 7.17 supply O, ε, U; Theorems 7.8 and 7.10 give (a) and (b). □

For a Hermitian matrix M write $\operatorname{sgn}(M)$ for the Hermitian matrix with the eigenvectors of M and the signs of its eigenvalues. When all $\varepsilon_j \neq 0$, the ground covariance is $\Gamma(\rho_{\mathrm{gs}}) = \mathrm{i}\,\operatorname{sgn}(\mathrm{i}h)$: $\mathrm{i}h = O\,(\mathrm{i}\mathsf{D}(\varepsilon))\,O^{\mathsf{T}}$ with O real orthogonal, so $\operatorname{sgn}(\mathrm{i}h) = O\,\operatorname{sgn}(\mathrm{i}\mathsf{D}(\varepsilon))\,O^{\mathsf{T}}$; on the labels of mode j, $(\mathrm{i}\mathsf{D}(\varepsilon))^2 = \varepsilon_j^2\,1$, so $\operatorname{sgn}(\mathrm{i}\mathsf{D}(\varepsilon)) = \mathrm{i}\mathsf{D}(\operatorname{sgn}\varepsilon) = -\mathrm{i}\mathsf{D}(-\operatorname{sgn}\varepsilon)$, and $\operatorname{sgn}(\mathrm{i}h) = -\mathrm{i}\,O\mathsf{D}(-\operatorname{sgn}\varepsilon)O^{\mathsf{T}} = -\mathrm{i}\,\Gamma(\rho_{\mathrm{gs}})$. The sign in this formula is that of the covariance convention of Definition 5.5; with Γ defined with the opposite sign, the ground covariance is $-\mathrm{i}\,\operatorname{sgn}(\mathrm{i}h)$.

Corollary 7.21 (A pure covariance determines the state). Lean ✓ *Let ρ be a density operator whose covariance is pure. Write $\Gamma(\rho) = O\mathsf{D}(\eta)O^{\mathsf{T}}$ with $\det O = 1$, $\eta_j = \pm 1$ (Theorem 7.13(a),(b)), and let U implement O (Theorem 7.17). Then $\rho = U^*|s\rangle\langle s|U$ with $s_j = 1$ if $\eta_j = 1$ and $s_j = 0$ otherwise. In particular two density operators with the same pure covariance are equal.*

Proof. Let $h = O\mathsf{D}(-\eta)O^{\mathsf{T}}$, so $\varepsilon = -\eta$ has no zero entry, $s^{\varepsilon} = s$ and $-\operatorname{sgn}\varepsilon = \eta$. By Lemma 7.3(b) and Lemma 7.9(c) with $\Gamma(\rho) = O\mathsf{D}(\eta)O^{\mathsf{T}}$, $\operatorname{Tr}(\rho H_h) = \frac{1}{2}\sum_j(-\eta_j)\eta_j = -\frac{n}{2} = \mathcal{E}_{\mathrm{gs}}(\varepsilon)$, and Theorem 7.8(c) gives $\rho = U^*|s\rangle\langle s|U$. □

Remark 7.22. A covariance that is not pure does not determine the state: for $n = 1$, every $\rho = \frac{1}{2}(1 + tX)$ with $|t| \le 1$ has $\Gamma(\rho) = 0$, since $c_x c_y = \mathrm{i}Z$ and $\operatorname{Tr} Z = \operatorname{Tr}(XZ) = 0$.

8 Thermal states and unitary evolution

For $h = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$, the Gibbs state $e^{-H_h}/\operatorname{Tr} e^{-H_h}$ of the quadratic Hamiltonian H_h has covariance $O\,\mathsf{D}(-\tanh(\varepsilon/2))\,O^{\mathsf{T}}$ (Theorem 8.3), and the unitary $e^{-\mathrm{i}tH_h}$ implements the rotation e^{-th} of the Majorana operators (Theorem 8.13). Both reduce to the canonical form by the conjugation $U^*H_{\mathsf{D}(\varepsilon)}U = H_h$ of Section 7, since $H_{\mathsf{D}(\varepsilon)}$ is diagonal and we compute its exponentials entrywise.

The conventions of Section 7 are in force. For a square matrix A, real or complex, $e^A = \sum_{k\ge 0} A^k/k!$; we use the facts $e^{\operatorname{diag}(v)} = \operatorname{diag}(e^v)$ entrywise, $e^{WAW^{-1}} = We^AW^{-1}$ for invertible W, $e^{A+B} = e^Ae^B$ when $AB = BA$, $(e^A)^* = e^{A^*}$, $(e^A)^{\mathsf{T}} = e^{A^{\mathsf{T}}}$, and $\operatorname{Tr}(WAW^{-1}) = \operatorname{Tr} A$.

8.1 The Gibbs family

Definition 8.1 (Partition function, Gibbs state, thermal state)**.** For a real matrix h on the labels, the *partition function* and the *Gibbs state* of H_h are

$$\mathcal{Z}(h) = \operatorname{Tr} e^{-H_h}, \qquad \rho_h = \mathcal{Z}(h)^{-1}e^{-H_h},$$

and for $\tau \in \mathbb{R}$ (the inverse temperature, written β in the literature; here β is the bond covariance) the *thermal state* of H_h at inverse temperature τ is $\rho_{\tau h} = e^{-\tau H_h}/\operatorname{Tr} e^{-\tau H_h}$. The results below assume $h^{\mathsf{T}} = -h$.

For one mode and $h = \mathsf{D}(\varepsilon)$, $H_h = \operatorname{diag}(-\frac{\varepsilon}{2}, \frac{\varepsilon}{2})$ on $(|0\rangle, |1\rangle)$ (Lemma 7.4 for $n = 1$), so $e^{-H_h} = \operatorname{diag}(e^{\varepsilon/2}, e^{-\varepsilon/2})$ and $\mathcal{Z} = 2\cosh(\varepsilon/2)$. Hence $\rho_h = \operatorname{diag}(\frac{1}{2}(1 + \tanh\frac{\varepsilon}{2}), \frac{1}{2}(1 - \tanh\frac{\varepsilon}{2}))$, and for $\varepsilon > 0$ the empty mode has the larger weight, as in the ground state of Theorem 7.7 ($s_j^{\varepsilon} = 0$ for $\varepsilon_j > 0$). ρ_h is the thermal state at $\tau = 1$; Corollary 8.4 passes to $\rho_{\tau h}$ by the substitution $\varepsilon \mapsto \tau\varepsilon$.

Lemma 8.2 (The Gibbs state of a canonical form)**.** Lean ✓ *Let $\varepsilon \in \mathbb{R}^n$ and $\lambda_j = -\tanh(\varepsilon_j/2)$. Then*

(a) $e^{-H_{\mathsf{D}(\varepsilon)}} = \operatorname{diag}\big(s \mapsto e^{-d_\varepsilon(s)}\big)$*;*

(b) $\mathcal{Z}(\mathsf{D}(\varepsilon)) = \prod_j 2\cosh(\varepsilon_j/2)$*, a positive real number;*

(c) $\rho_{\mathsf{D}(\varepsilon)} = \rho_\lambda$*, the product mixture with parameter λ; hence $\rho_{\mathsf{D}(\varepsilon)}$ is a density operator with covariance $\mathsf{D}(\lambda)$.*

Proof. (a) $-H_{\mathsf{D}(\varepsilon)} = \operatorname{diag}(-d_\varepsilon)$ (Lemma 7.4) and the exponential of a diagonal matrix is the diagonal matrix of the exponentials of its entries. (b) $e^{-d_\varepsilon(s)} = \prod_j e^{-\varepsilon_j(s_j - 1/2)}$, so

$$\operatorname{Tr} e^{-H_{\mathsf{D}(\varepsilon)}} = \sum_s \prod_j e^{-\varepsilon_j(s_j-1/2)} = \prod_j (e^{\varepsilon_j/2} + e^{-\varepsilon_j/2}) = \prod_j 2\cosh(\varepsilon_j/2),$$

the sum over $s \in \{0,1\}^n$ of a product factorizing over j. (c) By (a) and (b), $\rho_{\mathsf{D}(\varepsilon)} = \sum_s p(s)|s\rangle\langle s|$ with $p(s) = \prod_j e^{-\varepsilon_j(s_j-1/2)}/(2\cosh(\varepsilon_j/2))$. For one factor and $b \in \{0,1\}$,

$$\frac{e^{-\varepsilon(b-1/2)}}{2\cosh(\varepsilon/2)} = \frac{\cosh(\varepsilon/2) \mp \sinh(\varepsilon/2)}{2\cosh(\varepsilon/2)} = \frac{1 \mp \tanh(\varepsilon/2)}{2} = p_j(b) \quad (\text{sign } - \text{ for } b = 1,\ + \text{ for } b = 0),$$

with p_j the Bernoulli weight of parameter $\lambda_j = -\tanh(\varepsilon_j/2)$ of Definition 7.11. Hence $p(s) = \prod_j p_j(s_j)$ and $\rho_{\mathsf{D}(\varepsilon)} = \rho_\lambda$. Theorem 7.12 gives the rest. □

Theorem 8.3 (The Gibbs family). Lean ✓ *Let $h^{\mathsf{T}} = -h$.*

(a) Let $h = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$ with $OO^{\mathsf{T}} = 1$ and let the unitary U implement O. Then $\mathcal{Z}(h) = \prod_j 2\cosh(\varepsilon_j/2)$ and

$$\rho_h = U^*\rho_{\mathsf{D}(\varepsilon)}U = U^*\rho_\lambda U, \qquad \lambda_j = -\tanh(\varepsilon_j/2),$$

so ρ_h is a density operator with $\Gamma(\rho_h) = O\,\mathsf{D}(-\tanh(\varepsilon/2))\,O^{\mathsf{T}}$ and energy $\operatorname{Tr}(\rho_h H_h) = -\frac{1}{2}\sum_j \varepsilon_j \tanh(\varepsilon_j/2)$.

(b) For every real antisymmetric h: $\mathcal{Z}(h)$ is a positive real number, ρ_h is a density operator, and there are O with $OO^{\mathsf{T}} = 1$, $\det O = 1$, and $\varepsilon \in \mathbb{R}^n$ with $h = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$ and $\Gamma(\rho_h) = O\,\mathsf{D}(-\tanh(\varepsilon/2))\,O^{\mathsf{T}}$.

Proof. (a) $U^*H_{\mathsf{D}(\varepsilon)}U = H_h$ (proof of Theorem 7.8) and $U^* = U^{-1}$, so $e^{-H_h} = U^*e^{-H_{\mathsf{D}(\varepsilon)}}U$ and $\mathcal{Z}(h) = \operatorname{Tr} e^{-H_{\mathsf{D}(\varepsilon)}} = \mathcal{Z}(\mathsf{D}(\varepsilon))$ by Lemma 8.2(b). Hence $\rho_h = U^*\rho_{\mathsf{D}(\varepsilon)}U = U^*\rho_\lambda U$ by Lemma 8.2(c), which is the state built in the proof of Theorem 7.19 (the proof uses only that U implements O, not $\det O = 1$) for the admissible matrix $O\mathsf{D}(\lambda)O^{\mathsf{T}}$: a density operator with covariance $O\mathsf{D}(\lambda)O^{\mathsf{T}}$. For the energy, Lemma 7.3(b) and Lemma 7.9(c) with $\Gamma(\rho_h) = O\mathsf{D}(\lambda)O^{\mathsf{T}}$ give $\operatorname{Tr}(\rho_h H_h) = \frac{1}{4}\sum_{ab} h_{ab}\Gamma(\rho_h)_{ab} = \frac{1}{2}\sum_j \varepsilon_j\lambda_j = -\frac{1}{2}\sum_j \varepsilon_j \tanh(\varepsilon_j/2)$. (b) Theorem 7.13(a) and Theorem 7.17 supply O, ε, U; apply (a). □

Corollary 8.4 (Thermal states). Lean ✓ *Let $h = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$ with $OO^{\mathsf{T}} = 1$, U implementing O, and $\tau \in \mathbb{R}$. The thermal state $\rho_{\tau h}$ is a density operator with*

$$\Gamma(\rho_{\tau h}) = O\,\mathsf{D}(-\tanh(\tau\varepsilon/2))\,O^{\mathsf{T}} \qquad \text{and} \qquad \operatorname{Tr}(\rho_{\tau h}H_h) = -\tfrac{1}{2}\sum_j \varepsilon_j \tanh(\tau\varepsilon_j/2).$$

For $h = 0$ (or $\tau = 0$) the thermal state is $2^{-n}1$, with covariance $\Gamma(2^{-n}1) = 0$.

Proof. $\tau h = O\mathsf{D}(\tau\varepsilon)O^{\mathsf{T}}$, and Theorem 8.3(a) applies. The energy is

$$\tfrac{1}{4}\sum_{ab} h_{ab}\Gamma_{ab} = \tfrac{1}{2}\sum_j \varepsilon_j(-\tanh(\tau\varepsilon_j/2))$$

by Lemma 7.9(c) with $\Gamma(\rho_{\tau h}) = O\mathsf{D}(-\tanh(\tau\varepsilon/2))O^{\mathsf{T}}$. For $h = 0$: $H_0 = 0$, $e^0 = 1$, $\mathcal{Z}(0) = 2^n$, and Theorem 8.3(a) with $O = 1$, $U = 1$, $\varepsilon = 0$ gives $\Gamma(2^{-n}1) = \mathsf{D}(0) = 0$. □

Remark 8.5. As $\tau \to \infty$, $-\tanh(\tau\varepsilon_j/2) \to -\operatorname{sgn}\varepsilon_j$ for every $\varepsilon_j \neq 0$, so when every $\varepsilon_j \neq 0$ the thermal covariance tends entrywise to the ground covariance $O\mathsf{D}(-\operatorname{sgn}\varepsilon)O^{\mathsf{T}}$ of Theorem 7.8 and the energy to the ground energy $-\frac{1}{2}\sum_j|\varepsilon_j|$. For $h = \mathsf{D}(\varepsilon)$ with all $\varepsilon_j > 0$ the limit is the vacuum covariance $-\Omega_n$ of Corollary 5.9.

Proposition 8.6 (Strict admissibility). Lean ✓ *Call Γ* strictly admissible *when $\Gamma^{\mathsf{T}} = -\Gamma$ and $1 - \Gamma^{\mathsf{T}}\Gamma \succ 0$ (positive definite). For $h^{\mathsf{T}} = -h$:*

(a) $\Gamma(\rho_h)$ is strictly admissible;

(b) if $n \geq 1$, $\Gamma(\rho_h)$ is not pure;

(c) $O\Gamma O^{\mathsf{T}}$ is strictly admissible with Γ when $OO^{\mathsf{T}} = 1$, and $\mathsf{D}(\lambda)$ is strictly admissible if and only if $|\lambda_j| < 1$ for all j.

Proof. (c) $1 - (O\Gamma O^{\mathsf{T}})^{\mathsf{T}}(O\Gamma O^{\mathsf{T}}) = O(1 - \Gamma^{\mathsf{T}}\Gamma)O^{\mathsf{T}}$, and $M \mapsto OMO^{\mathsf{T}}$ preserves positive definiteness for invertible O; $1 - \mathsf{D}(\lambda)^{\mathsf{T}}\mathsf{D}(\lambda) = \operatorname{diag}(1-\lambda^2, 1-\lambda^2)$ is positive definite if and only if every $1 - \lambda_j^2 > 0$. (a) By Theorem 8.3(b), $\Gamma(\rho_h) = O\mathsf{D}(\lambda)O^{\mathsf{T}}$ with $|\lambda_j| = |-\tanh(\varepsilon_j/2)| < 1$; apply (c). (b) If $\Gamma(\rho_h)$ were pure, $\mathsf{D}(\lambda) = O^{\mathsf{T}}\Gamma(\rho_h)O$ would be pure (Theorem 1.7 with O^{T}, since $OO^{\mathsf{T}} = 1$ gives $O^{\mathsf{T}}O = 1$), forcing $|\lambda_j| = 1$ for every j; with $n \geq 1$ there is a j, and $|\lambda_j| < 1$ by (a), a contradiction. Hence $\Gamma(\rho_h)$ is not pure. □

Remark 8.7. In the normal form, $(\mathrm{i}\mathsf{D}(\varepsilon))^2 = \operatorname{diag}(\varepsilon^2, \varepsilon^2)$, so for an odd function f the matrix function $f(\mathrm{i}\mathsf{D}(\varepsilon))$ of the Hermitian matrix $\mathrm{i}\mathsf{D}(\varepsilon)$ is $\mathrm{i}\mathsf{D}(f(\varepsilon))$. Theorem 8.3 therefore reads $\Gamma(\rho_h) = \mathrm{i}\tanh(\mathrm{i}h/2)$, and Theorem 8.8(a) reads $h = 2\mathrm{i}\operatorname{artanh}(\mathrm{i}\Gamma(\rho_h))$: the coordinate-free form of the two theorems.

Theorem 8.8 (Realization by the Gibbs family)**.** Lean ✓

(a) Let $\Gamma = O\mathsf{D}(\eta)O^{\mathsf{T}}$ with $OO^{\mathsf{T}} = 1$, $|\eta_j| < 1$ for all j, and let U implement O. Put $\varepsilon_j = -2\operatorname{artanh}(\eta_j)$ and $h = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$. Then $h^{\mathsf{T}} = -h$ and $\Gamma(\rho_h) = \Gamma$.

(b) Every strictly admissible Γ is the covariance of a Gibbs state: there is h with $h^{\mathsf{T}} = -h$ and $\Gamma(\rho_h) = \Gamma$.

(c) (Uniqueness of the parameter in a frame.) Let $OO^{\mathsf{T}} = 1$, U implement O, and $\varepsilon, \varepsilon' \in \mathbb{R}^n$. If $\Gamma(\rho_{O\mathsf{D}(\varepsilon)O^{\mathsf{T}}}) = \Gamma(\rho_{O\mathsf{D}(\varepsilon')O^{\mathsf{T}}})$ then $\varepsilon = \varepsilon'$.

Proof. (a) $\mathsf{D}(\varepsilon)^{\mathsf{T}} = -\mathsf{D}(\varepsilon)$ gives $h^{\mathsf{T}} = -h$. By Theorem 8.3(a),

$$\Gamma(\rho_h) = O\mathsf{D}(-\tanh(-\operatorname{artanh}\eta))O^{\mathsf{T}} = O\mathsf{D}(\eta)O^{\mathsf{T}},$$

since tanh is odd and $\tanh(\operatorname{artanh} x) = x$ for $|x| < 1$. (b) Theorem 7.13(a) gives $\Gamma = O\mathsf{D}(\eta)O^{\mathsf{T}}$ with $OO^{\mathsf{T}} = 1$, $\det O = 1$; $\mathsf{D}(\eta) = O^{\mathsf{T}}\Gamma O$ is strictly admissible by Proposition 8.6(c), so $|\eta_j| < 1$; Theorem 7.17 gives U; apply (a). (c) Theorem 8.3(a) gives $O\mathsf{D}(\lambda)O^{\mathsf{T}} = O\mathsf{D}(\lambda')O^{\mathsf{T}}$ with $\lambda_j = -\tanh(\varepsilon_j/2)$, $\lambda'_j = -\tanh(\varepsilon'_j/2)$; multiplying by O^{T} on the left and O on the right ($O^{\mathsf{T}}O = 1$), $\mathsf{D}(\lambda) = \mathsf{D}(\lambda')$, so $\lambda = \lambda'$ (the entries $x_j y_j$), and tanh is injective. □

Uniqueness of the Gibbs state with a given covariance, beyond the fixed frame of (c), is Corollary 9.16(c).

8.2 Unitary evolution

In the canonical form the unitary $e^{-\mathrm{i}tH_{\mathsf{D}(\varepsilon)}}$ is diagonal with the phases $e^{-\mathrm{i}td_\varepsilon(s)}$, a product of the factors $\cos\varphi_j + \sin\varphi_j\, c_{x_j} c_{y_j}$ of Lemma 7.16 with $\varphi_j = t\varepsilon_j/2$, and it rotates each plane (x_j, y_j) by the angle $t\varepsilon_j$ (Lemma 8.12). For $h = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$ the conjugation by U turns this into the rotation e^{-th} (Theorem 8.13), so a covariance evolves as $\Gamma \mapsto e^{th}\Gamma e^{-th}$ and the Gibbs states are stationary (Corollary 8.14).

Definition 8.9 (Evolution operator, rotation of the planes)**.** For a real matrix h on the labels and $t \in \mathbb{R}$, $U_h(t) = e^{-\mathrm{i}tH_h}$. For $\theta \in \mathbb{R}^n$, $\mathsf{R}(\theta)$ is the matrix on the labels written in (x, y) blocks as

$$\mathsf{R}(\theta) = \begin{pmatrix} \operatorname{diag}(\cos\theta) & -\operatorname{diag}(\sin\theta) \\ \operatorname{diag}(\sin\theta) & \operatorname{diag}(\cos\theta) \end{pmatrix},$$

that is, the rows x_j and y_j are $(\cos\theta_j, -\sin\theta_j)$ and $(\sin\theta_j, \cos\theta_j)$ on the columns (x_j, y_j) and zero elsewhere: the product of the plane rotations $\mathsf{R}_{x_j y_j}(\cos\theta_j, \sin\theta_j)$ of Definition 7.14 over j.

Lemma 8.10 (Unitarity and the group law)**.** Lean ✓ *Let $h^{\mathsf{T}} = -h$ and $t, t' \in \mathbb{R}$. Then $U_h(t)^* = U_h(-t)$, $U_h(t)U_h(t)^* = U_h(t)^*U_h(t) = 1$, $U_h(t)U_h(t') = U_h(t+t')$ and $U_h(0) = 1$.*

Proof. $H_h^* = H_h$ (Lemma 7.3(a)), so $(-\mathrm{i}tH_h)^* = \mathrm{i}tH_h$ and $U_h(t)^* = e^{\mathrm{i}tH_h} = U_h(-t)$. The exponents $-\mathrm{i}tH_h$ and $-\mathrm{i}t'H_h$ commute, so $U_h(t)U_h(t') = e^{-\mathrm{i}(t+t')H_h} = U_h(t+t')$; with $t' = -t$ this gives $U_h(t)U_h(t)^* = U_h(0) = e^0 = 1$, and the other order likewise. □

Lemma 8.11 (The exponential of a canonical form is a rotation)**.** Lean ✓ *For $\theta \in \mathbb{R}^n$, $e^{\mathsf{D}(\theta)} = \mathsf{R}(-\theta)$; hence $e^{-t\mathsf{D}(\varepsilon)} = \mathsf{R}(t\varepsilon)$ for $t \in \mathbb{R}$, $\varepsilon \in \mathbb{R}^n$. Moreover $\mathsf{R}(\theta)\mathsf{R}(\theta)^{\mathsf{T}} = 1$ and $\mathsf{R}(\theta)^{\mathsf{T}} = \mathsf{R}(-\theta)$.*

Proof. Let $\Upsilon = \operatorname{diag}(\theta, \theta)$, the block-diagonal matrix with θ_j at x_j and at y_j. Then $\mathsf{D}(\theta)^2 = -\Upsilon^2$ and Υ commutes with $\mathsf{D}(\theta)$, so for $m \geq 0$

$$\mathsf{D}(\theta)^{2m} = (-1)^m \Upsilon^{2m} = (-1)^m \operatorname{diag}(\theta^{2m}, \theta^{2m}), \quad \mathsf{D}(\theta)^{2m+1} = (-1)^m \Upsilon^{2m} \mathsf{D}(\theta) = (-1)^m \mathsf{D}(\theta^{2m+1}),$$

by induction on m, with θ^k entrywise. The exponential series converges absolutely, so we may split it into its even and odd terms:

$$e^{\mathsf{D}(\theta)} = \sum_{m \geq 0} \frac{(-1)^m}{(2m)!} \operatorname{diag}(\theta^{2m}, \theta^{2m}) + \sum_{m \geq 0} \frac{(-1)^m}{(2m+1)!} \mathsf{D}(\theta^{2m+1}).$$

Both sums are entrywise: the first is $\operatorname{diag}(\cos\theta, \cos\theta)$ and the second is $\mathsf{D}(\sin\theta)$, by the series $\cos x = \sum_m (-1)^m x^{2m}/(2m)!$ and $\sin x = \sum_m (-1)^m x^{2m+1}/(2m+1)!$ applied to each θ_j (the maps $v \mapsto \operatorname{diag}(v, v)$ and $v \mapsto \mathsf{D}(v)$ are linear and continuous, so they commute with the sums). Hence $e^{\mathsf{D}(\theta)}$ has the blocks $\operatorname{diag}(\cos\theta)$, $\operatorname{diag}(\sin\theta)$, $-\operatorname{diag}(\sin\theta)$, $\operatorname{diag}(\cos\theta)$, which is $\mathsf{R}(-\theta)$; and $e^{-t\mathsf{D}(\varepsilon)} = e^{\mathsf{D}(-t\varepsilon)} = \mathsf{R}(t\varepsilon)$. The block products $\mathsf{R}(\theta)\mathsf{R}(\theta)^{\mathsf{T}}$ have the diagonal blocks $\operatorname{diag}(\cos^2\theta + \sin^2\theta) = 1$ and the off-diagonal blocks $\operatorname{diag}(-\cos\theta\sin\theta + \sin\theta\cos\theta) = 0$; $\mathsf{R}(\theta)^{\mathsf{T}} = \mathsf{R}(-\theta)$ by inspection of the blocks. □

For one mode, $\mathsf{D}(\theta) = \begin{pmatrix} 0 & \theta \\ -\theta & 0 \end{pmatrix}$ and $e^{\mathsf{D}(\theta)} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$, the rotation of the plane (x, y) by the angle $-\theta$.

Lemma 8.12 (Evolution in the canonical form)**.** Lean ✓ *Let $\varepsilon \in \mathbb{R}^n$, $t \in \mathbb{R}$, and $\varphi_j = t\varepsilon_j/2$. Then*

$$U_{\mathsf{D}(\varepsilon)}(t) = \operatorname{diag}\big(s \mapsto e^{-\mathrm{i}t\, d_\varepsilon(s)}\big) = \operatorname{diag}\Big(s \mapsto \prod_j e^{\mathrm{i}\varphi_j (-1)^{s_j}}\Big),$$

and $U_{\mathsf{D}(\varepsilon)}(t)$ implements $\mathsf{R}(t\varepsilon)$: for every j,

$$U\, c_{x_j} U^* = \cos(t\varepsilon_j)\, c_{x_j} - \sin(t\varepsilon_j)\, c_{y_j}, \qquad U\, c_{y_j} U^* = \sin(t\varepsilon_j)\, c_{x_j} + \cos(t\varepsilon_j)\, c_{y_j},$$

with $U = U_{\mathsf{D}(\varepsilon)}(t)$.

Proof. $-\mathrm{i}tH_{\mathsf{D}(\varepsilon)} = \operatorname{diag}(-\mathrm{i}t\, d_\varepsilon)$ (Lemma 7.4), whose exponential is the diagonal of the entrywise exponentials; and $d_\varepsilon(s) = -\frac{1}{2}\sum_j \varepsilon_j (-1)^{s_j}$ since $s_j - \frac{1}{2} = -\frac{1}{2}(-1)^{s_j}$, so $e^{-\mathrm{i}td_\varepsilon(s)} = \prod_j e^{\mathrm{i}\varphi_j(-1)^{s_j}}$. Write $f(s)$ for this entry and $G = \operatorname{diag}(f)$; $|f(s)| = 1$, so $G^* = \operatorname{diag}(\bar f)$. Fix j. The diagonal matrices G, G^*, $Z_{<j}$, Z_j commute with one another, and $GX_j = X_j \operatorname{diag}(f \circ \mathrm{flip}_j)$ (Lemma 5.2(c), read from right to left). Hence

$$\begin{aligned} G c_{x_j} G^* &= Z_{<j}\, G X_j G^* = Z_{<j} X_j\, \operatorname{diag}\big((f \circ \mathrm{flip}_j)\, \bar f\big), \\ G c_{y_j} G^* &= \mathrm{i} Z_{<j}\, G X_j G^*\, Z_j = c_{y_j}\, \operatorname{diag}\big((f \circ \mathrm{flip}_j)\, \bar f\big) \end{aligned}$$

(in the second, the diagonal factor commutes with Z_j). Now $f(\mathrm{flip}_j s)\overline{f(s)} = e^{\mathrm{i}\varphi_j((-1)^{s_j+1} - (-1)^{s_j})} = e^{-2\mathrm{i}\varphi_j(-1)^{s_j}}$, so $\operatorname{diag}((f \circ \mathrm{flip}_j)\bar f) = \cos 2\varphi_j - \mathrm{i}\sin 2\varphi_j\, Z_j$. Therefore

$$G c_{x_j} G^* = \cos 2\varphi_j\, c_{x_j} - \mathrm{i}\sin 2\varphi_j\, c_{x_j} Z_j = \cos 2\varphi_j\, c_{x_j} - \sin 2\varphi_j\, c_{y_j},$$

using $c_{x_j} Z_j = Z_{<j} X_j Z_j = -\mathrm{i} c_{y_j}$, and

$$G c_{y_j} G^* = \cos 2\varphi_j\, c_{y_j} - \mathrm{i}\sin 2\varphi_j\, c_{y_j} Z_j = \cos 2\varphi_j\, c_{y_j} + \sin 2\varphi_j\, c_{x_j},$$

using $c_{y_j} Z_j = \mathrm{i} Z_{<j} X_j Z_j Z_j = \mathrm{i} c_{x_j}$. With $2\varphi_j = t\varepsilon_j$ these are the rows x_j and y_j of $\mathsf{R}(t\varepsilon)$. □

Equivalently, $U_{\mathsf{D}(\varepsilon)}(t) = \prod_j (\cos\varphi_j + \sin\varphi_j\, c_{x_j} c_{y_j})$, since $c_{x_j} c_{y_j} = \mathrm{i} Z_j$ and each factor is $\operatorname{diag}(e^{\mathrm{i}\varphi_j(-1)^{s_j}})$; Lemma 7.16 then gives the same rotation.

Theorem 8.13 (Unitary evolution is a rotation). Lean ✓ *Let $h^{\mathsf{T}} = -h$ and $t \in \mathbb{R}$.*

(a) $U_h(t)$ implements e^{-th}: $U_h(t)\, c_a\, U_h(t)^* = \sum_b (e^{-th})_{ab} c_b$ for every label a. If $h = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$ with $OO^{\mathsf{T}} = 1$ and U implements O, then $U_h(t) = U^* U_{\mathsf{D}(\varepsilon)}(t) U$ and $e^{-th} = O\mathsf{R}(t\varepsilon)O^{\mathsf{T}}$.

(b) (Heisenberg picture.) $U_h(t)^*\, c_a\, U_h(t) = \sum_b (e^{th})_{ab} c_b$.

(c) For every density operator σ, the operator $U_h(t)\sigma U_h(t)^*$ is a density operator with covariance $e^{th}\, \Gamma(\sigma)\, e^{-th}$.

(d) (Generator.) $\mathrm{i}[H_h, c_a] = \sum_b h_{ab} c_b$ for every label a.

Proof. (a) Theorem 7.13(a) and Theorem 7.17 supply $h = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$, $OO^{\mathsf{T}} = 1$, $\det O = 1$, and U implementing O; fix any such data. $U^* H_{\mathsf{D}(\varepsilon)} U = H_h$ gives $-\mathrm{i}tH_h = U^*(-\mathrm{i}tH_{\mathsf{D}(\varepsilon)})U$, so $U_h(t) = U^* U_{\mathsf{D}(\varepsilon)}(t) U$. By Lemma 8.12, $V = U_{\mathsf{D}(\varepsilon)}(t)$ implements $\mathsf{R}(t\varepsilon)$; by the product rule of Lemma 7.16, VU implements $O\mathsf{R}(t\varepsilon)$, and since U^* implements O^{T}, $U^*(VU)$ implements $O\mathsf{R}(t\varepsilon)O^{\mathsf{T}}$. Finally $O\mathsf{R}(t\varepsilon)O^{\mathsf{T}} = Oe^{-t\mathsf{D}(\varepsilon)}O^{\mathsf{T}} = e^{-tO\mathsf{D}(\varepsilon)O^{\mathsf{T}}} = e^{-th}$ by Lemma 8.11 and $O^{\mathsf{T}} = O^{-1}$. (b) For U implementing O, $U^* c_a U = \sum_b O_{ba} c_b$, and $(e^{-th})_{ba} = \left((e^{-th})^{\mathsf{T}}\right)_{ab} = (e^{-th^{\mathsf{T}}})_{ab} = (e^{th})_{ab}$. (c) $U_h(t)$ is unitary (Lemma 8.10) and implements $O = e^{-th}$, so Theorem 6.9 gives a density operator with covariance $O^{\mathsf{T}}\Gamma(\sigma)O = e^{th}\Gamma(\sigma)e^{-th}$. (d) By Theorem 5.4, $[c_b c_c, c_a] = c_b\{c_c, c_a\} - \{c_b, c_a\}c_c = 2\delta_{ca}c_b - 2\delta_{ba}c_c$, so $[H_h, c_a] = \frac{\mathrm{i}}{4}\sum_{b,c} h_{bc}(2\delta_{ca}c_b - 2\delta_{ba}c_c) = \frac{\mathrm{i}}{2}\left(\sum_b h_{ba}c_b - \sum_c h_{ac}c_c\right) = -\mathrm{i}\sum_b h_{ab}c_b$, using $h_{ba} = -h_{ab}$; multiply by i. □

Parts (b) and (d) together are the Heisenberg equation: differentiating the closed form $c_a(t) = U_h(t)^* c_a U_h(t) = \sum_b (e^{th})_{ab} c_b$ of (b) gives $\dot{c}_a(t) = \sum_b h_{ab} c_b(t) = \mathrm{i}[H_h, c_a(t)]$, the commutator of (d) at time t.

Corollary 8.14 (Stationarity of the Gibbs family). Lean ✓ *Let $h^{\mathsf{T}} = -h$ and $t \in \mathbb{R}$. Then $U_h(t)\rho_h U_h(t)^* = \rho_h$, and $e^{th}\, \Gamma(\rho_h)\, e^{-th} = \Gamma(\rho_h)$.*

Proof. $-\mathrm{i}tH_h$ and $-H_h$ commute, so $e^{-\mathrm{i}tH_h} e^{-H_h} = e^{-\mathrm{i}tH_h - H_h} = e^{-H_h} e^{-\mathrm{i}tH_h}$; hence $U_h(t)\rho_h = \rho_h U_h(t)$ and $U_h(t)\rho_h U_h(t)^* = \rho_h$ by Lemma 8.10. Theorem 8.13(c) applied to $\sigma = \rho_h$ (a density operator by Theorem 8.3(b)) gives the covariance identity. □

9 Wick's theorem and the Gaussian class

A state is *Gaussian* when all its Majorana moments are given by Wick's rule from its covariance. The rotated product mixtures $U^*\rho_\lambda U$ of Section 7 are Gaussian (Theorem 9.10), and with them the Gibbs and thermal states and the pure states $U^*|s\rangle\langle s|U$. Since the Majorana monomials form a basis of the operators on $\mathcal{F}_n$, a Gaussian state is determined by its covariance (Theorem 9.15); hence every Gaussian state is a rotated product mixture, and the Gibbs state with a given covariance is unique.

The conventions of Sections 7 and 8 are in force. Vectors $v \in \mathbb{C}^{2n}$ are indexed by the labels $x_1, \dots, x_n, y_1, \dots, y_n$; e_a is the unit vector of the label a, and $v \cdot c = \sum_a v_a c_a$. The annihilation operator of mode j is $a_j = \frac{1}{2}(c_{x_j} + \mathrm{i}c_{y_j})$, so that $a_j = u \cdot c$ with $u = \frac{1}{2}(e_{x_j} + \mathrm{i}e_{y_j})$. For a density operator ρ, $\mathrm{Tr}(\rho c_a c_b) = \delta_{ab} - \mathrm{i}\Gamma(\rho)_{ab}$ (Theorem 5.6 and $c_a c_b = \delta_{ab} + \frac{1}{2}[c_a, c_b]$), which motivates the *two-point matrix*

$$\mathsf{G}(\rho) = 1 - \mathrm{i}\,\Gamma(\rho),$$

a complex matrix on the labels with $\mathsf{G}(\rho)_{ab} = \mathrm{Tr}(\rho c_a c_b)$.

9.1 Wick's rule and the Gaussian class

Definition 9.1 (Wick expansion, Gaussian state)**.** For a complex matrix G on the labels and vectors $v_1, \dots, v_m \in \mathbb{C}^{2n}$, the *Wick expansion* is defined by $\mathsf{W}_{\mathsf{G}}() = 1$ and, for $m \geq 1$,

$$\mathsf{W}_{\mathsf{G}}(v_1, \dots, v_m) = \sum_{j=2}^{m} (-1)^j \, (v_1^{\mathsf{T}} \mathsf{G} v_j) \, \mathsf{W}_{\mathsf{G}}(v_2, \dots, \widehat{v_j}, \dots, v_m),$$

where the hat removes the entry. A density operator ρ is *Gaussian* when

$$\operatorname{Tr}\big(\rho \, (v_1 \cdot c)(v_2 \cdot c) \cdots (v_m \cdot c)\big) = \mathsf{W}_{\mathsf{G}(\rho)}(v_1, \dots, v_m) \qquad \text{for all } m \geq 0 \text{ and all } v_1, \dots, v_m \in \mathbb{C}^{2n}.$$

For $m = 1$ the sum is empty and $\mathsf{W}_{\mathsf{G}}(v_1) = 0$; by induction W_{G} vanishes for every odd m. For $m = 2$, $\mathsf{W}_{\mathsf{G}}(v_1, v_2) = v_1^{\mathsf{T}} \mathsf{G} v_2$, and for $m = 4$,

$$\mathsf{W}_{\mathsf{G}}(v_1, v_2, v_3, v_4) = (v_1^{\mathsf{T}} \mathsf{G} v_2)(v_3^{\mathsf{T}} \mathsf{G} v_4) - (v_1^{\mathsf{T}} \mathsf{G} v_3)(v_2^{\mathsf{T}} \mathsf{G} v_4) + (v_1^{\mathsf{T}} \mathsf{G} v_4)(v_2^{\mathsf{T}} \mathsf{G} v_3),$$

the sum over pairings with the sign of the pairing. For every even m, $\mathsf{W}_{\mathsf{G}}(v_1, \dots, v_m)$ is the Pfaffian of the antisymmetric $m \times m$ matrix whose entries above the diagonal are $v_i^{\mathsf{T}} \mathsf{G} v_j$ [1]: the recursion of Definition 9.1 is the expansion of that Pfaffian along its first row. With $v_i = e_{a_i}$ the rule reads $\operatorname{Tr}(\rho \, c_{a_1} \cdots c_{a_m}) = \mathsf{W}_{\mathsf{G}(\rho)}(e_{a_1}, \dots, e_{a_m})$, Wick's rule in its usual form with the pairing $\operatorname{Tr}(\rho c_a c_b) = \delta_{ab} - \mathrm{i}\Gamma_{ab}$; repeated labels are allowed. We state the rule for vectors rather than labels because a Gaussian unitary acts on the vectors, $U(v \cdot c)U^* = (O^{\mathsf{T}} v) \cdot c$ (Theorem 9.4), and because the relation that drives Theorem 9.10 involves the combinations $c_{x_j} \pm \mathrm{i}\lambda_j c_{y_j}$; the label form is the special case $v_i = e_{a_i}$.

Lemma 9.2 (Multilinearity and the two-point rule)**.** Lean ✓

(a) $\mathsf{W}_{\mathsf{G}}(v_1, \dots, v_m) = 0$ *for odd* m, W_{G} *is linear in* v_1, *and* $\mathsf{W}_{\mathsf{G}}(v, w) = v^{\mathsf{T}} \mathsf{G} w$.

(b) For every density operator ρ *and all* v, w, $\operatorname{Tr}(\rho \, (v \cdot c)(w \cdot c)) = v^{\mathsf{T}} \mathsf{G}(\rho) w$; *in particular Wick's rule holds for* $m = 0$ *and* $m = 2$ *for every density operator.*

(c) $(u \cdot c)(w \cdot c) + (w \cdot c)(u \cdot c) = 2(u^{\mathsf{T}} w)\, 1$ *for all* $u, w \in \mathbb{C}^{2n}$.

Proof. (a) We noted odd vanishing above; linearity in v_1 is linearity of $v_1^{\mathsf{T}} \mathsf{G} v_j$ in each term; $m = 2$ is the definition with one term, $j = 2$. (b) $\operatorname{Tr}(\rho (v \cdot c)(w \cdot c)) = \sum_{ab} v_a w_b \operatorname{Tr}(\rho c_a c_b) = \sum_{ab} v_a \mathsf{G}(\rho)_{ab} w_b$; for $m = 0$ both sides are $\operatorname{Tr}\rho = 1$. (c) By Theorem 5.4, $\sum_{ab} u_a w_b (c_a c_b + c_b c_a) = \sum_{ab} u_a w_b \, 2\delta_{ab} = 2(u^{\mathsf{T}} w)$. □

Remark 9.3. Beyond $m = 2$, Wick's rule is a condition. For $n = 1$, $\rho = \frac{1}{2}(1 + c_{x_1})$ is a density operator (c_{x_1} is Hermitian with $c_{x_1}^2 = 1$) with $\operatorname{Tr}(\rho c_{x_1}) = 1 \neq 0$. For $n = 2$, $M = c_{x_1} c_{y_1} c_{x_2} c_{y_2}$ is Hermitian with $M^2 = 1$, so $\rho = \frac{1}{4}(1 + M)$ is a density operator; $\Gamma(\rho) = 0$ since $c_a c_b M$ is $\pm$ a monomial of two labels for $a \neq b$ (Lemma 9.13), while $\operatorname{Tr}(\rho M) = 1 \neq 0 = \mathsf{W}_1(e_{x_1}, e_{y_1}, e_{x_2}, e_{y_2})$. Theorem 9.10 proves the rule for the product mixtures.

Theorem 9.4 (Transport by a Gaussian unitary)**.** Lean ✓ *Let the unitary* U *implement* O *and let* ρ *be a density operator. Then* $U(v \cdot c)U^* = (O^{\mathsf{T}} v) \cdot c$ *for every* v, $\mathsf{W}_{\mathsf{G}}(O^{\mathsf{T}} v_1, \dots, O^{\mathsf{T}} v_m) = \mathsf{W}_{O\mathsf{G}O^{\mathsf{T}}}(v_1, \dots, v_m)$ *for every* G, *and* $\mathsf{G}(U^* \rho U) = O \, \mathsf{G}(\rho) \, O^{\mathsf{T}}$. *If* ρ *is Gaussian, so is* $U^* \rho U$.

Proof. $U(v \cdot c)U^* = \sum_a v_a \sum_b O_{ab} c_b = \sum_b (O^{\mathsf{T}} v)_b c_b$. The identity for W follows by induction on m from $(O^{\mathsf{T}} v_1)^{\mathsf{T}} \mathsf{G} (O^{\mathsf{T}} v_j) = v_1^{\mathsf{T}} (O \mathsf{G} O^{\mathsf{T}}) v_j$. The two-point matrix transforms with the covariance, $\Gamma(U^* \rho U) = O \Gamma(\rho) O^{\mathsf{T}}$ (Theorem 6.9), and $O \, 1 \, O^{\mathsf{T}} = 1$. Finally, with $X = \prod_i (v_i \cdot c)$, cyclicity of the trace gives

$$\begin{aligned} \operatorname{Tr}(U^* \rho U \, X) = \operatorname{Tr}(\rho \, U X U^*) &= \operatorname{Tr}\Big(\rho \prod_i \big((O^{\mathsf{T}} v_i) \cdot c\big)\Big) \\ &= \mathsf{W}_{\mathsf{G}(\rho)}(O^{\mathsf{T}} v_1, \dots, O^{\mathsf{T}} v_m) = \mathsf{W}_{O\mathsf{G}(\rho)O^{\mathsf{T}}}(v_1, \dots, v_m), \end{aligned}$$

which is $\mathsf{W}_{\mathsf{G}(U^* \rho U)}(v_1, \dots, v_m)$. □

Removing one factor at a time reduces Wick's rule to two identities (Lemma 9.5) and a criterion (Lemma 9.6).

Lemma 9.5 (Moving a factor through a product). Lean ✓ *Let $b = u \cdot c$, $X = (w_1 \cdot c)\cdots(w_k \cdot c)$, and let $X_{\hat\imath}$ be the product with the i-th factor removed. Then*

$$bX = \sum_{i=1}^{k} (-1)^{i-1}\, 2(u^{\mathsf T} w_i)\, X_{\hat\imath} + (-1)^k Xb, \qquad Xb = \sum_{i=1}^{k} (-1)^{k-i}\, 2(u^{\mathsf T} w_i)\, X_{\hat\imath} + (-1)^k bX.$$

Proof. Induction on k; for $k = 0$ both identities read $b = b$. Write $X = (w_1 \cdot c)X'$ with $X' = (w_2 \cdot c)\cdots(w_k \cdot c)$, so that $X_{\hat 1} = X'$ and $X_{\hat\imath} = (w_1 \cdot c)X'_{\widehat{i-1}}$ for $i \geq 2$. By Lemma 9.2(c), $b(w_1 \cdot c) = 2(u^{\mathsf T} w_1) - (w_1 \cdot c)b$, so $bX = 2(u^{\mathsf T} w_1)X' - (w_1 \cdot c)\, bX'$; the induction hypothesis $bX' = \sum_{i=2}^{k}(-1)^{i-2} 2(u^{\mathsf T} w_i) X'_{\widehat{i-1}} + (-1)^{k-1} X'b$, multiplied by $-(w_1 \cdot c)$, turns the signs into $(-1)^{i-1}$ and $(-1)^k$ and the products into $X_{\hat\imath}$ and Xb: the first identity. The second is the first solved for Xb: multiply the first identity by $(-1)^k$ and use $(-1)^k(-1)^{i-1} = -(-1)^{k-i}$. □

Lemma 9.6 (A recursion criterion). Lean ✓ *Let ρ be a density operator such that (i) $\operatorname{Tr}(\rho X) = 0$ for every product X of an odd number of factors $w \cdot c$, and (ii) for every label a and every product $X = (w_1 \cdot c)\cdots(w_k \cdot c)$ with k odd,*

$$\operatorname{Tr}(\rho\, c_a X) = \sum_{i=1}^{k} (-1)^{i-1}\, (e_a^{\mathsf T} \mathsf G(\rho) w_i)\, \operatorname{Tr}(\rho X_{\hat\imath}).$$

Then ρ is Gaussian.

Proof. We prove Wick's rule by induction on m. For m odd both sides vanish, by (i) and Lemma 9.2(a). For m even and $m \geq 2$, write $v_1 \cdot c = \sum_a v_{1a} c_a$ and apply (ii) with $X = (v_2 \cdot c)\cdots(v_m \cdot c)$, $k = m - 1$ odd:

$$\operatorname{Tr}(\rho\,(v_1 \cdot c)X) = \sum_a v_{1a} \sum_{i=1}^{m-1} (-1)^{i-1} (e_a^{\mathsf T} \mathsf G w_i) \operatorname{Tr}(\rho X_{\hat\imath}) = \sum_{j=2}^{m} (-1)^j (v_1^{\mathsf T} \mathsf G v_j)\, \operatorname{Tr}(\rho X_{\widehat{j-1}}),$$

with $w_i = v_{i+1}$, $j = i + 1$, $(-1)^{i-1} = (-1)^j$. By the induction hypothesis $\operatorname{Tr}(\rho X_{\widehat{j-1}}) = \mathsf W_{\mathsf G}(v_2, \ldots, \widehat{v_j}, \ldots, v_m)$, and the right side is $\mathsf W_{\mathsf G}(v_1, \ldots, v_m)$ by Definition 9.1. For $m = 0$ both sides are 1. □

9.2 The product mixtures are Gaussian

Lemma 9.7 (Parity). Lean ✓ *Let $\Pi_n = Z_1 Z_2 \cdots Z_n = \operatorname{diag}\big(\prod_j (-1)^{s_j}\big)$. Then $\Pi_n^2 = 1$, $\Pi_n c_a \Pi_n = -c_a$ for every label a, and $\Pi_n \rho_\lambda \Pi_n = \rho_\lambda$ for every $\lambda \in [-1, 1]^n$. Consequently $\operatorname{Tr}(\rho_\lambda X) = 0$ for every product X of an odd number of factors $w \cdot c$.*

Proof. Π_n is diagonal with entries ± 1, so $\Pi_n^2 = 1$ and Π_n commutes with the diagonal matrix ρ_λ. Writing $c_a = \operatorname{diag}(f_a) X_j$ with j the mode of a (equation (5.1)) and $\pi(s) = \prod_k (-1)^{s_k}$ for the diagonal entry of Π_n, $\Pi_n c_a \Pi_n = \operatorname{diag}(\pi\, f_a) X_j \operatorname{diag}(\pi) = \operatorname{diag}(\pi\, f_a\, (\pi \circ \operatorname{flip}_j)) X_j$ and $\pi(\operatorname{flip}_j s) = -\pi(s)$, so $\Pi_n c_a \Pi_n = -c_a$. For X of odd length, $\Pi_n X \Pi_n = (-1)^{\text{length}} X = -X$, by induction on the number of factors: $\Pi_n (w \cdot c) X' \Pi_n = (\Pi_n (w \cdot c) \Pi_n)(\Pi_n X' \Pi_n) = -(w \cdot c)\, \Pi_n X' \Pi_n$. Hence $\operatorname{Tr}(\rho_\lambda X) = \operatorname{Tr}(\Pi_n \rho_\lambda \Pi_n\, X) = \operatorname{Tr}(\rho_\lambda\, \Pi_n X \Pi_n) = -\operatorname{Tr}(\rho_\lambda X)$. □

Lemma 9.8 (The mode relation). Lean ✓ *Let $\lambda \in [-1, 1]^n$ and fix a mode j. Then*

$$(c_{x_j} - \mathrm{i}\lambda_j c_{y_j})\, \rho_\lambda = \rho_\lambda\, (c_{x_j} + \mathrm{i}\lambda_j c_{y_j}), \qquad (c_{y_j} + \mathrm{i}\lambda_j c_{x_j})\, \rho_\lambda = \rho_\lambda\, (c_{y_j} - \mathrm{i}\lambda_j c_{x_j}).$$

Moreover $\mathsf G(\rho_\lambda) = 1 - \mathrm{i}\mathsf D(\lambda)$ has the rows $(\mathsf G w)_{x_j} = w_{x_j} - \mathrm{i}\lambda_j w_{y_j}$ and $(\mathsf G w)_{y_j} = w_{y_j} + \mathrm{i}\lambda_j w_{x_j}$ for every vector w.

Proof. $\rho_\lambda = \operatorname{diag}(p)$ with $p(s) = \prod_k p_k(s_k)$ and $p_j(s_j) = \frac{1}{2}(1 - \lambda_j(-1)^{s_j})$ (Definition 7.11). The factors $p_k(s_k)$, $k \neq j$, are common to $p(s)$ and $p(\mathrm{flip}_j s)$, so

$$p_j(s_j)\, p(\mathrm{flip}_j s) = p_j(1 - s_j)\, p(s) \qquad \text{for every } s, \tag{9.1}$$

that is, $(1 - \lambda_j Z_j)\operatorname{diag}(p \circ \mathrm{flip}_j) = \rho_\lambda(1 + \lambda_j Z_j)$, since $2p_j(s_j) = 1 - \lambda_j(-1)^{s_j}$ is the entry of $1 - \lambda_j Z_j$ and $2p_j(1 - s_j)$ that of $1 + \lambda_j Z_j$. As $Z_{<j}$ is diagonal and $X_j \operatorname{diag}(p) = \operatorname{diag}(p \circ \mathrm{flip}_j)X_j$ (Lemma 5.2),

$$(1 - \lambda_j Z_j)\, c_{x_j}\rho_\lambda = (1 - \lambda_j Z_j)Z_{<j} \operatorname{diag}(p \circ \mathrm{flip}_j)X_j = \rho_\lambda(1 + \lambda_j Z_j)Z_{<j}X_j = \rho_\lambda(1 + \lambda_j Z_j)\, c_{x_j}.$$

Since $Z_j c_{x_j} = -c_{x_j}Z_j = \mathrm{i}c_{y_j}$ (Lemma 5.2 and $c_{y_j} = \mathrm{i}Z_{<j}X_jZ_j$), $(1 \mp \lambda_j Z_j)c_{x_j} = c_{x_j} \mp \mathrm{i}\lambda_j c_{y_j}$, which is the first identity; the second is the same computation with c_{y_j} and $Z_j c_{y_j} = -c_{y_j}Z_j = -\mathrm{i}c_{x_j}$. The rows of G: $\Gamma(\rho_\lambda) = \mathsf{D}(\lambda)$ (Theorem 7.12) has the only nonzero entries $\pm\lambda_j$ at (x_j, y_j) and (y_j, x_j). □

Remark 9.9. For $|\lambda_j| < 1$, solving the relation for $c_a\rho_\lambda$ gives the KMS-type form $c_a\rho_\lambda = \rho_\lambda \sum_b \mathcal{K}_{ab}c_b$, where on (x_j, y_j)

$$\mathcal{K} = \begin{pmatrix} \cosh\varepsilon & -\mathrm{i}\sinh\varepsilon \\ \mathrm{i}\sinh\varepsilon & \cosh\varepsilon \end{pmatrix},$$

$\varepsilon = -2\operatorname{artanh}\lambda_j$, the block of $e^{-\mathrm{i}h}$ for $h = \mathsf{D}(\varepsilon)$; for $\lambda_j = -1$ it reads $2a_j\rho_\lambda = 2\rho_\lambda a_j^*$, both sides zero because the mode is empty, and for $\lambda_j = 1$ the same with a_j^*. The single identity covers all three cases; Theorem 9.10 needs no case distinction.

Theorem 9.10 (The product mixtures are Gaussian)**.** Lean ✓ *For every $\lambda \in [-1, 1]^n$ the product mixture ρ_λ is Gaussian:* $\operatorname{Tr}(\rho_\lambda \prod_i (v_i \cdot c)) = \mathsf{W}_{1-\mathrm{i}\mathsf{D}(\lambda)}(v_1, \ldots, v_m)$ *for all $v_1, \ldots, v_m$.*

Proof. We apply Lemma 9.6 with $\rho = \rho_\lambda$ and $\mathsf{G} = \mathsf{G}(\rho_\lambda) = 1 - \mathrm{i}\mathsf{D}(\lambda)$. Condition (i) is Lemma 9.7. For (ii) fix a mode j, a list $w_1, \ldots, w_k$ with k odd, $X = \prod_i (w_i \cdot c)$, and write $\langle Y \rangle = \operatorname{Tr}(\rho_\lambda Y)$, $A_a = \langle c_a X \rangle$, $B_a = \langle X c_a \rangle$ and $r_a = 2\sum_i (-1)^{i-1}(w_i)_a \langle X_{\hat{\imath}} \rangle$ for $a \in \{x_j, y_j\}$. Lemma 9.5 (second identity, $u = e_a$; k odd, so $(-1)^{k-i} = (-1)^{i-1}$) gives $B_a = r_a - A_a$. Multiplying the first identity of Lemma 9.8 by X on the right and taking the trace, with $\operatorname{Tr}(Y\rho_\lambda X) = \operatorname{Tr}(\rho_\lambda XY)$,

$$B_{x_j} - \mathrm{i}\lambda_j B_{y_j} = A_{x_j} + \mathrm{i}\lambda_j A_{y_j}.$$

Substituting $B_a = r_a - A_a$ for both labels, the terms $\pm\mathrm{i}\lambda_j A_{y_j}$ cancel and

$$2A_{x_j} = r_{x_j} - \mathrm{i}\lambda_j r_{y_j} = 2\sum_i (-1)^{i-1}\big((w_i)_{x_j} - \mathrm{i}\lambda_j (w_i)_{y_j}\big)\langle X_{\hat{\imath}} \rangle = 2\sum_i (-1)^{i-1}(e_{x_j}^{\mathsf{T}}\mathsf{G}w_i)\langle X_{\hat{\imath}} \rangle$$

by the row of G in Lemma 9.8: this is (ii) for $a = x_j$. The second identity of Lemma 9.8 gives in the same way $2A_{y_j} = r_{y_j} + \mathrm{i}\lambda_j r_{x_j}$, which is (ii) for $a = y_j$. □

Corollary 9.11 (The Gaussian families)**.** Lean ✓

(a) For $OO^{\mathsf{T}} = 1$, U implementing O and $\lambda \in [-1, 1]^n$, the state $U^*\rho_\lambda U$ (as in the proof of Theorem 7.19, which does not use $\det O = 1$) is Gaussian with $\mathsf{G} = O(1 - \mathrm{i}\mathsf{D}(\lambda))O^{\mathsf{T}}$.

(b) Every Gibbs state ρ_h, $h^{\mathsf{T}} = -h$, and every thermal state $\rho_{\tau h}$ is Gaussian.

(c) Every occupation state $|s\rangle\langle s|$ and every pure state $U^*|s\rangle\langle s|U$ with U implementing an orthogonal O is Gaussian.

Proof. (a) Theorems 9.10 and 9.4. (b) Theorem 8.3: $\rho_h = U^*\rho_\lambda U$ with $\lambda_j = -\tanh(\varepsilon_j/2)$ for a normal form $h = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$ and U implementing O; apply (a); $\rho_{\tau h}$ is the Gibbs state of τh, and $(\tau h)^{\mathsf{T}} = -\tau h$. (c) $|s\rangle\langle s| = \rho_\lambda$ with $\lambda_j = \pm 1$ (Theorem 7.12); apply (a). □

9.3 Moments determine the state

Definition 9.12 (Majorana monomials)**.** Order the labels $x_1 < \cdots < x_n < y_1 < \cdots < y_n$. For a set S of labels, c_S is the product of the c_a, $a \in S$, in increasing order; $c_\emptyset = 1$.

Lemma 9.13 (Traces of monomials)**.** Lean ✓

(a) $\operatorname{Tr}(c_S) = 0$ *for* $S \neq \emptyset$.

(b) $c_S^* c_S = 1$ *and, for* $S \neq T$, $c_S^* c_T = \pm c_{S \triangle T}$ *with* $S \triangle T \neq \emptyset$ *the symmetric difference. Hence* $\operatorname{Tr}(c_S^* c_T) = 2^n [S = T]$.

Proof. (a) If $|S|$ is odd, $\Pi_n c_S \Pi_n = -c_S$ by Lemma 9.7, so $\operatorname{Tr} c_S = \operatorname{Tr}(\Pi_n c_S \Pi_n) = -\operatorname{Tr} c_S$. If $|S| \geq 2$ is even, write $c_S = c_a c_{S'}$ with $a \in S$ and $|S'|$ odd; the labels of S' differ from a, so $c_{S'} c_a = (-1)^{|S'|} c_a c_{S'} = -c_a c_{S'}$ and $\operatorname{Tr}(c_S) = \operatorname{Tr}(c_{S'} c_a) = -\operatorname{Tr}(c_S)$. (b) Insertion rule: for a set T of labels and a label a, $c_a c_T = (-1)^{|\{t \in T : t < a\}|} c_{T \cup \{a\}}$ if $a \notin T$ and $c_a c_T = (-1)^{|\{t \in T : t < a\}|} c_{T \setminus \{a\}}$ if $a \in T$: move c_a to the right past the factors c_t with $t < a$ (one sign each, Theorem 5.4), where it either takes its place in the order or meets c_a and cancels by $c_a c_a = 1$. Hence, by induction on $|S|$, $c_S c_T = \pm c_{S \triangle T}$ for all S, T: for $S = \emptyset$ this is $c_T = c_T$; otherwise write $c_S = c_a c_{S'}$ with $a = \min S$, apply the hypothesis to $c_{S'} c_T = \pm c_{S' \triangle T}$ and then the insertion rule to $c_a c_{S' \triangle T}$, noting $S \triangle T = \{a\} \triangle (S' \triangle T)$. Since $c_S^* = \pm c_S$ (reversing $|S|$ distinct anticommuting factors, $c_a^* = c_a$), $c_S^* c_T = \pm c_{S \triangle T}$, with $c_S^* c_S = 1$ exactly: $c_S^* c_S = c_{a_m} \cdots c_{a_1} c_{a_1} \cdots c_{a_m}$ for $S = \{a_1 < \cdots < a_m\}$, and the innermost pair $c_{a_1} c_{a_1} = 1$ cancels, then the next, and so on. The trace then follows from (a), $S \triangle T \neq \emptyset$ for $S \neq T$, and $\operatorname{Tr} 1 = 2^n$. □

Theorem 9.14 (The monomial basis)**.** Lean ✓ *The* 4^n *monomials* c_S *are linearly independent, hence a basis of the* 4^n*-dimensional space of matrices on* $\mathcal{F}_n$*, and every matrix* A *satisfies*

$$A = 2^{-n} \sum_S \operatorname{Tr}(A c_S^*)\, c_S.$$

In particular, if $\operatorname{Tr}(A c_S) = \operatorname{Tr}(B c_S)$ *for all* S*, then* $A = B$.

Proof. If $\sum_S \alpha_S c_S = 0$, multiply by c_T^* on the left and take the trace: $2^n \alpha_T = 0$ by Lemma 9.13(b). There are $2^{2n} = (2^n)^2$ subsets and the matrices on the 2^n-dimensional space $\mathcal{F}_n$ form a $(2^n)^2$-dimensional space, so the independent family is a basis. Writing $A = \sum_S \alpha_S c_S$ and taking $\operatorname{Tr}(A c_T^*) = \sum_S \alpha_S \operatorname{Tr}(c_S c_T^*) = \sum_S \alpha_S \operatorname{Tr}(c_T^* c_S) = 2^n \alpha_T$ (cyclicity and Lemma 9.13(b)) gives the formula; the last statement follows since $c_S^* = \pm c_S$. □

Theorem 9.15 (A Gaussian state is determined by its covariance)**.** Lean ✓ *If* ρ *and* σ *are Gaussian and* $\Gamma(\rho) = \Gamma(\sigma)$*, then* $\rho = \sigma$.

Proof. For every $S = \{a_1 < \cdots < a_m\}$, $\operatorname{Tr}(\rho c_S) = \mathsf{W}_{\mathsf{G}(\rho)}(e_{a_1}, \ldots, e_{a_m})$ by Wick's rule, and $\mathsf{G}(\rho) = 1 - \mathrm{i}\Gamma(\rho) = \mathsf{G}(\sigma)$; so $\operatorname{Tr}(\rho c_S) = \operatorname{Tr}(\sigma c_S)$ for all S and Theorem 9.14 gives $\rho = \sigma$. □

For an admissible Γ write $\rho[\Gamma]$ for the Gaussian state with covariance Γ, unique by Theorem 9.15; it exists by Theorem 7.19 and Corollary 9.11(a).

Corollary 9.16 (Every Gaussian state is a rotated product mixture)**.** Lean ✓ *Let* ρ *be Gaussian with covariance* $\Gamma = O \mathsf{D}(\eta) O^{\mathsf{T}}$, $O O^{\mathsf{T}} = 1$, $\det O = 1$, $|\eta_j| \leq 1$ *(Theorem 7.13), and let* U *implement* O*. Then*

(a) $\rho = U^* \rho_\eta U$;

(b) if Γ *is pure,* $\rho = U^* |s\rangle\langle s| U$ *with* $s_j = [\eta_j = 1]$*, and* $\rho^2 = \rho$;

(c) if Γ *is strictly admissible,* $\rho = \rho_h$ *for some* $h^{\mathsf{T}} = -h$*; and for* $h^{\mathsf{T}} = -h$, $h'^{\mathsf{T}} = -h'$ *with* $\Gamma(\rho_h) = \Gamma(\rho_{h'})$, $\rho_h = \rho_{h'}$.

Proof. (a) $U^*\rho_\eta U$ is a density operator with covariance Γ (Theorem 7.19) and is Gaussian (Corollary 9.11(a)); apply Theorem 9.15. (b) For pure Γ, $|\eta_j| = 1$ for all j (Theorem 7.13) and $\rho_\eta = |s\rangle\langle s|$ (Theorem 7.12); $\rho^2 = U^*|s\rangle\langle s|UU^*|s\rangle\langle s|U = \rho$. (c) By Theorem 8.8 there is $h^{\mathsf{T}} = -h$ with $\Gamma(\rho_h) = \Gamma$; ρ_h is Gaussian (Corollary 9.11(b)); apply Theorem 9.15, and likewise for $\rho_h, \rho_{h'}$. □

For a pure covariance the Gaussian hypothesis of Theorem 9.15 can be dropped: two density operators with the same pure covariance are equal (Corollary 7.21).

10 The partial trace, the contraction of states, and the Gaussian fMPS

Section 2 defined the contraction $\Gamma\star\beta$ of an input covariance with a bond covariance and Section 3 built the covariance Γ_N of a periodic chain from it. This section performs the contraction on Fock space. Tracing the bond modes out of $\rho\,(1\otimes\sigma)$, for Gaussian ρ and σ with $1 - D\beta$ invertible, gives a Gaussian state whose covariance is $\Gamma(\rho)\star\Gamma(\sigma)$ (Theorem 10.10); the periodic chain is therefore a Gaussian state on Fock space with covariance Γ_N (Theorem 10.17). The tool is a single identity satisfied by every Gaussian state, $(\mathsf{G}^{\mathsf{T}}v\cdot c)\,\rho = \rho\,(\mathsf{G}v\cdot c)$, which characterizes the class among the density operators with vanishing odd moments.

The conventions of Section 9 are in force. Throughout, we split $n+m$ modes into the *retained* modes $1,\dots,n$ and the *eliminated* modes $n+1,\dots,n+m$; R and E denote their labels, and a matrix on the labels is written in blocks $\Gamma = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ with A on R and D on E. For a matrix β on E with $1 - D\beta$ invertible, $\Gamma\star\beta = A + B\beta(1-D\beta)^{-1}C$ (Definition 2.1); $1 - D\beta$ is invertible iff $1 - \beta D$ is, since $\det(1 - D\beta) = \det(1 - \beta D)$. The parity of $n+m$ modes is $\Pi_{n+m} = Z_1\cdots Z_{n+m}$ (Lemma 9.7). The blocks are those of Section 2 with the eliminated labels placed last; the orientation of the bond covariance β is the one fixed in Definition 2.7.

10.1 The partial trace

Definition 10.1 (Split, tensor product, partial traces)**.** Write a configuration $s \in \{0,1\}^{n+m}$ as (s_R, s_E) with s_R its first n and s_E its last m bits, so that a matrix M on $\mathcal{F}_{n+m}$ has entries $M_{(s_R,s_E),(t_R,t_E)}$. For X on $\mathcal{F}_n$ and Y on $\mathcal{F}_m$, $(X\otimes Y)_{(s_R,s_E),(t_R,t_E)} = X_{s_Rt_R}Y_{s_Et_E}$. The *partial trace over E* of M is the matrix on $\mathcal{F}_n$ with $(\mathrm{Tr}_E\, M)_{s_Rt_R} = \sum_u M_{(s_R,u),(t_R,u)}$, and the partial trace over R is the matrix on $\mathcal{F}_m$ with $(\mathrm{Tr}_R\, M)_{s_Et_E} = \sum_u M_{(u,s_E),(u,t_E)}$.

Lemma 10.2 (Majorana operators of a split)**.** Lean ✓ *Write $c_a^{(n)}$ and $c_e^{(m)}$ for the Majorana operators of $\mathcal{F}_n$ and $\mathcal{F}_m$. For a retained label a, $c_a = c_a^{(n)}\otimes 1$; for an eliminated label e of mode $n+j$, $c_e = \Pi_n\otimes c_{e'}^{(m)}$ with e' the label of mode j of $\mathcal{F}_m$. Consequently, for $v\in\mathbb{C}^{2n}$ extended by zero to $\bar v\in\mathbb{C}^{2(n+m)}$, $\bar v\cdot c = (v\cdot c^{(n)})\otimes 1$ and $\prod_i(\bar v_i\cdot c) = \big(\prod_i(v_i\cdot c^{(n)})\big)\otimes 1$; for $u\in\mathbb{C}^{2m}$ extended by zero to $\bar u$, $\bar u\cdot c = \Pi_n\otimes(u\cdot c^{(m)})$ and $\prod_{i=1}^k(\bar u_i\cdot c) = \Pi_n^k\otimes\prod_i(u_i\cdot c^{(m)})$; $\Pi_{n+m} = \Pi_n\otimes\Pi_m$; and $X\otimes 1$ commutes with $1\otimes Y$.*

Proof. X_j and Z_j act on bit j alone: for $j\le n$, $X_j = X_j^{(n)}\otimes 1$, $Z_j = Z_j^{(n)}\otimes 1$ and $Z_{<j} = Z_{<j}^{(n)}\otimes 1$, so $c_a = c_a^{(n)}\otimes 1$ by Definition 5.3. For $j > n$, $X_j = 1\otimes X_{j-n}^{(m)}$, $Z_j = 1\otimes Z_{j-n}^{(m)}$ and $Z_{<j} = Z_1\cdots Z_n\,Z_{n+1}\cdots Z_{j-1} = \Pi_n\otimes Z_{<j-n}^{(m)}$, so $c_{x_j} = Z_{<j}X_j = \Pi_n\otimes Z_{<j-n}^{(m)}X_{j-n}^{(m)} = \Pi_n\otimes c_{x_{j-n}}^{(m)}$, and likewise for c_{y_j}. The products follow from $(X\otimes Y)(X'\otimes Y') = XX'\otimes YY'$ and $\Pi_n^2 = 1$, the parity from $\Pi_{n+m} = Z_1\cdots Z_n\,Z_{n+1}\cdots Z_{n+m}$, and the commutation from $(X\otimes 1)(1\otimes Y) = X\otimes Y = (1\otimes Y)(X\otimes 1)$. □

Lemma 10.3 (Partial-trace calculus)**.** Lean ✓ *For matrices M, M' on $\mathcal{F}_{n+m}$, X on $\mathcal{F}_n$, and Y, V on $\mathcal{F}_m$:*

(a) Tr_E *is linear,* $\mathrm{Tr}(\mathrm{Tr}_E M) = \mathrm{Tr}\, M$, $\mathrm{Tr}_E(X \otimes Y) = (\mathrm{Tr}\, Y)X$, $\mathrm{Tr}_E\big((X \otimes 1)M\big) = X\ \mathrm{Tr}_E M$, $\mathrm{Tr}_E(M(X \otimes 1)) = (\mathrm{Tr}_E M)X$, $\mathrm{Tr}_E(M(1 \otimes V)) = \mathrm{Tr}_E\big((1 \otimes V)M\big)$, *and* $\mathrm{Tr}\big((\mathrm{Tr}_E M)X\big) = \mathrm{Tr}\big(M(X \otimes 1)\big)$.

(b) $(\mathrm{Tr}_E M)^* = \mathrm{Tr}_E(M^*)$*; if* $M \succeq 0$ *then* $\mathrm{Tr}_E M \succeq 0$*; if* M *is a density operator, so is* $\mathrm{Tr}_E M$.

(c) The same for Tr_R *with the roles of the two factors exchanged.*

Proof. (a) Entrywise from Definition 10.1:

$$\textstyle\sum_u (X \otimes Y)_{(s,u),(t,u)} = X_{st} \sum_u Y_{uu}, \qquad \big((X \otimes 1)M\big)_{(s,u),(t,u)} = \sum_{s'} X_{ss'} M_{(s',u),(t,u)},$$
$$\big(M(1 \otimes V)\big)_{(s,u),(t,u)} = \textstyle\sum_{u'} M_{(s,u),(t,u')} V_{u'u}, \qquad \big((1 \otimes V)M\big)_{(s,u),(t,u)} = \sum_{u'} V_{uu'} M_{(s,u'),(t,u)},$$

the second summed over u, and the last two summed over u both equal $\sum_{u,u'} M_{(s,u),(t,u')} V_{u'u}$; finally $\mathrm{Tr}((\mathrm{Tr}_E M)X) = \sum_{s,t,u} M_{(s,u),(t,u)} X_{ts} = \mathrm{Tr}(M(X \otimes 1))$. (b) $\overline{(\mathrm{Tr}_E M)_{ts}} = \sum_u \overline{M_{(t,u),(s,u)}} = \sum_u (M^*)_{(s,u),(t,u)}$. For $x \in \mathbb{C}^{2^n}$, $x^*(\mathrm{Tr}_E M)x = \sum_u x_u^* M x_u$ with $x_u = x \otimes e_u$, and each term is ≥ 0. The trace is preserved by (a). □

Theorem 10.4 (Marginals of a Gaussian state)**.** Lean ✓ *Let* ρ *be Gaussian on* $n+m$ *modes with covariance* $\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$*. Then* $\mathrm{Tr}_E \rho$ *is Gaussian on* n *modes with covariance* A*, and* $\mathrm{Tr}_R \rho$ *is Gaussian on* m *modes with covariance* D.

Proof. Both are density operators (Lemma 10.3(b)). For $v_1, \ldots, v_k \in \mathbb{C}^{2n}$ with extensions $\bar{v}_i$, Lemmas 10.3(a) and 10.2 give $\mathrm{Tr}\big(\mathrm{Tr}_E \rho \prod_i (v_i \cdot c^{(n)})\big) = \mathrm{Tr}\big(\rho \prod_i (\bar{v}_i \cdot c)\big) = \mathsf{W}_{\mathsf{G}(\rho)}(\bar{v}_1, \ldots, \bar{v}_k) = \mathsf{W}_{1-\mathrm{i}A}(v_1, \ldots, v_k)$, the last step because $\bar{v}_i^{\mathsf{T}} \mathsf{G}(\rho) \bar{v}_j = v_i^{\mathsf{T}}(1 - \mathrm{i}A)v_j$ and W depends on its matrix only through these pairings; and $1 - \mathrm{i}A = \mathsf{G}(\mathrm{Tr}_E \rho)$ by the two-point rule (Lemma 9.2(b)). For Tr_R and $u_1, \ldots, u_k \in \mathbb{C}^{2m}$ with extensions $\bar{u}_i$: if k is even, $1 \otimes \prod_i (u_i \cdot c^{(m)}) = \prod_i (\bar{u}_i \cdot c)$ (Lemma 10.2), so the moment is $\mathsf{W}_{\mathsf{G}(\rho)}(\bar{u}) = \mathsf{W}_{1-\mathrm{i}D}(u)$; if k is odd, $1 \otimes \prod_i (u_i \cdot c^{(m)}) = (\Pi_n \otimes 1) \prod_i (\bar{u}_i \cdot c)$, and $\Pi_n \otimes 1 = \prod_{j \leq n} (-\mathrm{i}\, c_{x_j} c_{y_j})$ is a product of $2n$ retained Majorana operators, so the moment is a moment of ρ of odd length $2n + k$, hence $0 = \mathsf{W}_{1-\mathrm{i}D}(u)$. □

10.2 The defining relation

Theorem 10.5 (The defining relation of a Gaussian state)**.** Lean ✓ *Let* ρ *be Gaussian with* $\mathsf{G} = \mathsf{G}(\rho)$*. Then* $(\mathsf{G}^{\mathsf{T}} v \cdot c)\, \rho = \rho\, (\mathsf{G} v \cdot c)$ *for every* $v \in \mathbb{C}^{2(n+m)}$*, and* $\Pi_{n+m} \rho \Pi_{n+m} = \rho$.

Proof. By Corollary 9.16(a), $\rho = U^* \rho_\lambda U$ with $\Gamma(\rho) = O\mathsf{D}(\lambda)O^{\mathsf{T}}$, $OO^{\mathsf{T}} = 1$ and U implementing O; write $\mathsf{G}_\lambda = 1 - \mathrm{i}\mathsf{D}(\lambda)$, so $\mathsf{G} = O\mathsf{G}_\lambda O^{\mathsf{T}}$. The two identities of Lemma 9.8 are $(\mathsf{G}_\lambda^{\mathsf{T}} e \cdot c)\rho_\lambda = \rho_\lambda(\mathsf{G}_\lambda e \cdot c)$ for $e = e_{x_j}, e_{y_j}$, since $\mathsf{D}(\lambda) e_{x_j} = -\lambda_j e_{y_j}$ and $\mathsf{D}(\lambda) e_{y_j} = \lambda_j e_{x_j}$. By linearity in e, the identity holds for every v in place of e. Conjugating, $U^*(w \cdot c)\rho_\lambda U = U^*(w \cdot c)U\, U^* \rho_\lambda U = (Ow \cdot c)\rho$ (Theorem 9.4, $U^*(w \cdot c)U = (Ow) \cdot c$), so $(O\mathsf{G}_\lambda^{\mathsf{T}} v \cdot c)\rho = \rho(O\mathsf{G}_\lambda v \cdot c)$; with $v = O^{\mathsf{T}} v'$ and $O^{\mathsf{T}} O = 1$, $O\mathsf{G}_\lambda^{\mathsf{T}} O^{\mathsf{T}} v' = \mathsf{G}^{\mathsf{T}} v'$ and $O\mathsf{G}_\lambda O^{\mathsf{T}} v' = \mathsf{G} v'$. Parity: $\Pi_{n+m} \rho \Pi_{n+m}$ is a density operator with $\mathrm{Tr}(\Pi_{n+m} \rho \Pi_{n+m} c_S) = \mathrm{Tr}(\rho\, \Pi_{n+m} c_S \Pi_{n+m}) = (-1)^{|S|} \mathrm{Tr}(\rho c_S) = \mathrm{Tr}(\rho c_S)$ for every monomial, the odd moments of ρ vanishing (Lemma 9.2(a)); so $\Pi_{n+m} \rho \Pi_{n+m} = \rho$ by Theorem 9.14. □

Remark 10.6. For $|\lambda_j| < 1$ the relation is the KMS-type relation of Remark 9.9 solved for $c_a \rho$. If $\mathsf{G} u = 0$ then $\mathsf{G}^{\mathsf{T}} u = (2 - \mathsf{G})u = 2u$ and the relation reads $(u \cdot c)\rho = 0$: $u \cdot c$ annihilates the state. The direction of the transposes matters: for $n = 1$ and $\rho_\lambda = \frac{1}{2}(1 - \lambda Z_1)$ with $\lambda \neq 0$, the reversed form $(\mathsf{G} v \cdot c)\rho = \rho(\mathsf{G}^{\mathsf{T}} v \cdot c)$ subtracted from the relation gives $c_{y_1}\rho_\lambda + \rho_\lambda c_{y_1} = 0$, whereas $c_{y_1} Z_1 = -Z_1 c_{y_1}$ gives $c_{y_1}\rho_\lambda + \rho_\lambda c_{y_1} = c_{y_1} \neq 0$. The relation is a condition on the state: for the non-Gaussian density operator $\rho = \frac{1}{4}(1 + c_{x_1} c_{y_1} c_{x_2} c_{y_2})$ of Remark 9.3, $\mathsf{G} = 1$, so the relation would say that ρ commutes with every c_a, but $c_{x_1}\rho - \rho c_{x_1} = \frac{1}{2} c_{y_1} c_{x_2} c_{y_2} \neq 0$.

Theorem 10.7 (The relation criterion)**.** Lean ✓ *Let ϱ be a matrix on $\mathcal{F}_{n+m}$ with $\operatorname{Tr}(\varrho X) = 0$ for every product X of an odd number of factors $w \cdot c$, and let G be a matrix on the labels with $\mathsf{G} + \mathsf{G}^{\mathsf{T}} = 2$ and $(\mathsf{G}^{\mathsf{T}} v \cdot c)\, \varrho = \varrho\, (\mathsf{G} v \cdot c)$ for every v. Then $\operatorname{Tr}\big(\varrho \prod_{i=1}^{k} (v_i \cdot c)\big) = (\operatorname{Tr} \varrho)\, \mathsf{W}_{\mathsf{G}}(v_1, \ldots, v_k)$ for every list. In particular, a density operator ρ with vanishing odd moments that satisfies the relation with $\mathsf{G} = \mathsf{G}(\rho)$ is Gaussian.*

Proof. Fix an odd list $w_1, \ldots, w_k$, $X = \prod_i (w_i \cdot c)$, and put $\alpha(v) = \operatorname{Tr}(\varrho (v \cdot c) X)$ and $p_i = \operatorname{Tr}(\varrho X_i)$; α is linear. By the relation and cyclicity, $\alpha(\mathsf{G} v) = \operatorname{Tr}\big((\mathsf{G}^{\mathsf{T}} v \cdot c) \varrho X\big) = \operatorname{Tr}\big(\varrho X (\mathsf{G}^{\mathsf{T}} v \cdot c)\big)$, and the second identity of Lemma 9.5 with $u = \mathsf{G}^{\mathsf{T}} v$ and k odd gives $\operatorname{Tr}(\varrho X (u \cdot c)) = \sum_i (-1)^{i-1} 2 (u^{\mathsf{T}} w_i) p_i - \alpha(u)$. Hence $\alpha(\mathsf{G} v) + \alpha(\mathsf{G}^{\mathsf{T}} v) = 2 \sum_i (-1)^{i-1} (v^{\mathsf{T}} \mathsf{G} w_i) p_i$, and the left side is $\alpha((\mathsf{G} + \mathsf{G}^{\mathsf{T}}) v) = 2\alpha(v)$. With $v = e_a$ this is the recursion (ii) of Lemma 9.6, $\operatorname{Tr}(\varrho c_a X) = \sum_i (-1)^{i-1} (e_a^{\mathsf{T}} \mathsf{G} w_i) \operatorname{Tr}(\varrho X_{\hat{\imath}})$. The induction in the proof of Lemma 9.6 uses only (i), (ii) and the linearity of the trace, so it applies to ϱ and G and gives $\operatorname{Tr}(\varrho \prod (v_i \cdot c)) = (\operatorname{Tr} \varrho) \mathsf{W}_{\mathsf{G}}(v)$, the factor $\operatorname{Tr} \varrho$ entering at the empty list. For a density operator, $\operatorname{Tr} \rho = 1$, and $\mathsf{G}(\rho) + \mathsf{G}(\rho)^{\mathsf{T}} = 2$ by antisymmetry of the covariance. □

10.3 The contraction of states

Definition 10.8 (Contraction of states)**.** For a matrix ρ on $\mathcal{F}_{n+m}$ and a matrix σ on $\mathcal{F}_m$,

$$\rho \star \sigma = \operatorname{Tr}_E\big(\rho\, (1 \otimes \sigma)\big), \qquad \mathcal{N}(\rho, \sigma) = \operatorname{Tr}(\rho \star \sigma) = \operatorname{Tr}\big(\rho\, (1 \otimes \sigma)\big),$$

a matrix on $\mathcal{F}_n$ and a number. When $\mathcal{N}(\rho, \sigma) \neq 0$, the *contracted state* is $\mathcal{N}(\rho, \sigma)^{-1} \rho \star \sigma$.

For a bond state σ on the eliminated modes, $\rho \star \sigma$ is the state of the retained modes after we trace the eliminated modes out against σ. For pure $\sigma = |\varphi\rangle\langle\varphi|$ the partial trace gives $\rho \star \sigma = (1 \otimes \langle\varphi|)\, \rho\, (1 \otimes |\varphi\rangle)$, the tensor-network contraction of ρ with φ, and the contracted state is this matrix divided by its trace $\mathcal{N}(\rho, \sigma)$. Theorem 10.10 gives the effect of the operation on the covariance.

Lemma 10.9 (Positivity)**.** Lean ✓ *If ρ and σ are density operators, then $\rho \star \sigma$ is Hermitian and positive semidefinite, $\mathcal{N}(\rho, \sigma) \geq 0$, and $\mathcal{N}(\rho, \sigma)^{-1} \rho \star \sigma$ is a density operator when $\mathcal{N}(\rho, \sigma) \neq 0$. If moreover $\Pi_{n+m} \rho \Pi_{n+m} = \rho$ and $\Pi_m \sigma \Pi_m = \sigma$, then $\operatorname{Tr}\big((\rho \star \sigma) X\big) = 0$ for every product X of an odd number of factors $w \cdot c^{(n)}$.*

Proof. Let $\sigma^{1/2} \succeq 0$ be the positive square root of σ. By Lemma 10.3(a), $\rho \star \sigma = \operatorname{Tr}_E\big((1 \otimes \sigma^{1/2}) \rho (1 \otimes \sigma^{1/2})\big)$, the partial trace of a positive semidefinite matrix, hence positive semidefinite (Lemma 10.3(b)); its trace is $\mathcal{N}(\rho, \sigma) \geq 0$. Parity: $\Pi_{n+m} = \Pi_n \otimes \Pi_m$ commutes with $1 \otimes \sigma$, and Π_n anticommutes with each $w \cdot c^{(n)}$, so with $\bar{X} = X \otimes 1$ and $\Pi = \Pi_{n+m}$, $\operatorname{Tr}(\rho (1 \otimes \sigma) \bar{X}) = \operatorname{Tr}\big(\Pi \rho \Pi\, \Pi (1 \otimes \sigma) \Pi\, \Pi \bar{X} \Pi\big) = -\operatorname{Tr}(\rho (1 \otimes \sigma) \bar{X})$, and by Lemma 10.3(a) the left side is $\operatorname{Tr}\big((\rho \star \sigma) X\big)$. □

Theorem 10.10 (Contraction of Gaussian states)**.** Lean ✓ *Let ρ be Gaussian on $n + m$ modes with covariance $\Gamma = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, let σ be Gaussian on m modes with covariance β, and let $1 - D\beta$ be invertible. Then, with $\varrho = \rho \star \sigma$ and $\widetilde{\mathsf{G}} = 1 - \mathrm{i}(\Gamma \star \beta)$,*

$$\operatorname{Tr}\Big(\varrho \prod_{i=1}^{k} (v_i \cdot c^{(n)})\Big) = \mathcal{N}(\rho, \sigma)\, \mathsf{W}_{\widetilde{\mathsf{G}}}(v_1, \ldots, v_k) \qquad \text{for all } v_1, \ldots, v_k \in \mathbb{C}^{2n}.$$

Proof. The defining relations of ρ and of σ (Theorem 10.5) each move a Majorana factor across $\tilde{\rho} = \rho (1 \otimes \sigma)$; eliminating the two functionals supported on the eliminated labels between them leaves the recursion of Lemma 9.6 on retained labels, with the Schur complement $\Gamma \star \beta$ as its pairing matrix. Steps (E1)–(E3) below record the three identities, and (I)–(III) solve them for the retained functional.

Write $\mathsf{G} = \mathsf{G}(\rho) = 1 - \mathrm{i}\Gamma$ with blocks $\mathsf{G}_{RR} = 1 - \mathrm{i}A$, $\mathsf{G}_{RE} = -\mathrm{i}B$, $\mathsf{G}_{ER} = -\mathrm{i}C$, $\mathsf{G}_{EE} = 1 - \mathrm{i}D$, and $\mathsf{G}_\sigma = 1 - \mathrm{i}\beta$; by antisymmetry $\mathsf{G}_{RE}^{\mathsf{T}} = \mathrm{i}C$, $\mathsf{G}_{ER}^{\mathsf{T}} = \mathrm{i}B$ and $\mathsf{G}_{RR} + \mathsf{G}_{RR}^{\mathsf{T}} = 2$. By Lemmas 10.3(a) and 10.2, $\operatorname{Tr}(\varrho \prod_i (v_i \cdot c^{(n)})) = \operatorname{Tr}(\tilde{\rho} \prod_i (\bar{v}_i \cdot c))$ with $\bar{v}_i$ the extensions by zero; so it suffices to establish

for $\tilde\rho$, on retained lists, the two hypotheses of the induction of Theorem 10.7 with the matrix $\widetilde{\mathsf{G}}$ (extended by zero to all labels; $\widetilde{\mathsf{G}}+\widetilde{\mathsf{G}}^{\mathsf{T}}=2$ on R by Theorem 2.3). Odd retained moments of $\tilde\rho$ vanish by Lemma 10.9, both states being even by Theorem 10.5. It remains to prove the recursion.

Fix a retained label a and an odd list $w_1,\dots,w_k$ of vectors supported on R, $X=\prod_i(w_i\cdot c)$, $p_i=\operatorname{Tr}(\tilde\rho X_{\hat\imath})$ and $\varkappa_i=(-1)^{i-1}p_i$. Define three linear functionals of $v\in\mathbb{C}^{2(n+m)}$,

$$\alpha(v)=\operatorname{Tr}\big(\tilde\rho\,(v\cdot c)X\big),\qquad b(v)=\operatorname{Tr}\big(\rho\,(v\cdot c)(1\otimes\sigma)X\big),\qquad r(v)=2\sum_i\varkappa_i\,(v^{\mathsf{T}}w_i),$$

and write α_R,α_E and b_R,b_E for their restrictions to vectors supported on R and on E. The relation of σ connects b and α on E, the relation of ρ connects them on all labels, and the moving lemma supplies r:

(E1) $b_R=\alpha_R$: a retained $v\cdot c$ commutes with $1\otimes\sigma$ (Lemma 10.2).

(E2) $b(\mathsf{G}_\sigma^{\mathsf{T}}u)=\alpha(\mathsf{G}_\sigma u)$ for u supported on E: Theorem 10.5 for σ gives $(\mathsf{G}_\sigma^{\mathsf{T}}u_E\cdot c^{(m)})\sigma=\sigma(\mathsf{G}_\sigma u_E\cdot c^{(m)})$; tensoring with Π_n on the left and using Lemma 10.2, $(\mathsf{G}_\sigma^{\mathsf{T}}u\cdot c)(1\otimes\sigma)=(1\otimes\sigma)(\mathsf{G}_\sigma u\cdot c)$.

(E3) $b(\mathsf{G}v)=r(\mathsf{G}^{\mathsf{T}}v)-\alpha(\mathsf{G}^{\mathsf{T}}v)$ for every v: Theorem 10.5 for ρ gives $b(\mathsf{G}v)=\operatorname{Tr}\big((\mathsf{G}^{\mathsf{T}}v\cdot c)\rho(1\otimes\sigma)X\big)=\operatorname{Tr}\big(\tilde\rho X(\mathsf{G}^{\mathsf{T}}v\cdot c)\big)$, and Lemma 9.5 with $u=\mathsf{G}^{\mathsf{T}}v$ and k odd gives $\operatorname{Tr}(\tilde\rho X(u\cdot c))=\sum_i(-1)^{i-1}2(u^{\mathsf{T}}w_i)p_i-\alpha(u)$, where $u^{\mathsf{T}}w_i=v^{\mathsf{T}}\mathsf{G}w_i$.

In (E3) split $v=v_R+v_E$ and use (E1); the cross terms $\alpha_R(\mathsf{G}_{RE}v_E)+\alpha_R(\mathsf{G}_{ER}^{\mathsf{T}}v_E)$ cancel since $\mathsf{G}_{RE}+\mathsf{G}_{ER}^{\mathsf{T}}=-\mathrm{i}B+\mathrm{i}B=0$. The v_R-part and the v_E-part read

$$\alpha_R(\mathsf{G}_{RR}v_R)+b_E(\mathsf{G}_{ER}v_R)+\alpha_R(\mathsf{G}_{RR}^{\mathsf{T}}v_R)+\alpha_E(\mathsf{G}_{RE}^{\mathsf{T}}v_R)=2\textstyle\sum_i\varkappa_i v_R^{\mathsf{T}}\mathsf{G}_{RR}w_i,$$
$$b_E(\mathsf{G}_{EE}v_E)+\alpha_E(\mathsf{G}_{EE}^{\mathsf{T}}v_E)=2\textstyle\sum_i\varkappa_i v_E^{\mathsf{T}}\mathsf{G}_{ER}w_i.$$

Writing a functional f on $\mathbb{C}^{2m}$ as the row vector with $f(v)=fv$, and putting $s=\alpha_E+b_E$, $d=\alpha_E-b_E$:

$$\text{(I)}\quad \alpha_E\mathsf{G}_{EE}^{\mathsf{T}}+b_E\mathsf{G}_{EE}=2\mathrm{i}\textstyle\sum_i\varkappa_i w_i^{\mathsf{T}}B,\qquad\text{i.e.}\qquad s+\mathrm{i}\,dD=2\mathrm{i}\textstyle\sum_i\varkappa_i w_i^{\mathsf{T}}B;$$
$$\text{(II)}\quad \alpha_E(1-\mathrm{i}\beta)=b_E(1+\mathrm{i}\beta),\qquad\text{i.e.}\qquad d=\mathrm{i}\,s\beta;$$
$$\text{(III)}\quad 2\alpha_R=2\textstyle\sum_i\varkappa_i w_i^{\mathsf{T}}\mathsf{G}_{RR}^{\mathsf{T}}-b_E\mathsf{G}_{ER}-\alpha_E\mathsf{G}_{RE}^{\mathsf{T}}=2\sum_i\varkappa_i w_i^{\mathsf{T}}\mathsf{G}_{RR}^{\mathsf{T}}-\mathrm{i}\,dC,$$

where (II) is (E2) in row form. Substituting (II) into (I), $s(1-\beta D)=2\mathrm{i}\sum_i\varkappa_i w_i^{\mathsf{T}}B$; since $1-\beta D$ is invertible, $s=2\mathrm{i}\sum_i\varkappa_i w_i^{\mathsf{T}}B(1-\beta D)^{-1}$ and $d=\mathrm{i}s\beta=-2\sum_i\varkappa_i w_i^{\mathsf{T}}B(1-\beta D)^{-1}\beta$. Then (III) gives

$$\alpha_R=\sum_i\varkappa_i w_i^{\mathsf{T}}\big(\mathsf{G}_{RR}^{\mathsf{T}}+\mathrm{i}B(1-\beta D)^{-1}\beta C\big)=\sum_i\varkappa_i w_i^{\mathsf{T}}\big(1+\mathrm{i}A+\mathrm{i}B\beta(1-D\beta)^{-1}C\big)=\sum_i\varkappa_i w_i^{\mathsf{T}}\widetilde{\mathsf{G}}^{\mathsf{T}},$$

using $B(1-\beta D)^{-1}\beta=B\beta(1-D\beta)^{-1}$ (Lemma 2.2) and $\widetilde{\mathsf{G}}^{\mathsf{T}}=1+\mathrm{i}(\Gamma\star\beta)$. Evaluating at $v=e_a$, $\operatorname{Tr}(\tilde\rho c_aX)=\sum_i(-1)^{i-1}(e_a^{\mathsf{T}}\widetilde{\mathsf{G}}w_i)\operatorname{Tr}(\tilde\rho X_{\hat\imath})$, the recursion on retained lists. The induction of Lemma 9.6 removes one factor at a time and never leaves the retained lists, so it applies to $\tilde\rho$ with the retained lists alone and gives $\operatorname{Tr}(\tilde\rho\prod_i(\bar v_i\cdot c))=(\operatorname{Tr}\tilde\rho)\mathsf{W}_{\widetilde{\mathsf{G}}}(v)$ with $\operatorname{Tr}\tilde\rho=\mathcal{N}(\rho,\sigma)$. □

The invertibility of $1-D\beta$ enters once, in solving (I)–(II) for s.

Corollary 10.11 (The contracted state)**.** Lean ✓ *Under the hypotheses of Theorem 10.10, if $\mathcal{N}(\rho,\sigma)\neq0$ then the contracted state $\mathcal{N}(\rho,\sigma)^{-1}\rho\star\sigma$ is Gaussian with covariance $\Gamma\star\beta$.*

Proof. It is a density operator (Lemma 10.9) with moments $\mathsf{W}_{\widetilde{\mathsf{G}}}(v)$ (Theorem 10.10). At $k=2$, $\operatorname{Tr}(\mathcal{N}(\rho,\sigma)^{-1}\varrho\,c_ac_b)=\widetilde{\mathsf{G}}_{ab}=\delta_{ab}-\mathrm{i}(\Gamma\star\beta)_{ab}$; as $\Gamma\star\beta$ is real antisymmetric (Theorem 2.3) and $\Gamma(\rho)_{ab}=\Re(\mathrm{i}\operatorname{Tr}(\rho c_ac_b))$ for $a\neq b$ (Theorem 5.6), the covariance is $\Gamma\star\beta$, and Wick's rule holds with respect to the state's own two-point matrix. □

10.4 Positivity of the normalization

Lemma 10.12 (Block elimination). Lean ✓ *Let D be a matrix on $I_1 \sqcup I_2$ with blocks D_{11}, D_{12}, D_{21}, D_{22}, and let $\beta = \beta_1 \oplus \beta_2$ be block diagonal with $1 - D_{22}\beta_2$ invertible. Then*

$$\det(1 - D\beta) = \det(1 - D_{22}\beta_2)\,\det\big(1 - (D \star \beta_2)\beta_1\big), \qquad D \star \beta_2 = D_{11} + D_{12}\beta_2(1 - D_{22}\beta_2)^{-1}D_{21},$$

the contraction of D with β_2 over I_2.

Proof. $1 - D\beta = \left(\begin{smallmatrix} 1-D_{11}\beta_1 & -D_{12}\beta_2 \\ -D_{21}\beta_1 & 1-D_{22}\beta_2 \end{smallmatrix}\right)$, and the block determinant formula around the invertible lower-right block gives $\det(1-D\beta) = \det(1-D_{22}\beta_2)\det\big(1-D_{11}\beta_1 - D_{12}\beta_2(1-D_{22}\beta_2)^{-1}D_{21}\beta_1\big)$. □

Lemma 10.13 (The kernel of the relations). Lean ✓ *Let F be a linear map on $\mathbb{C}^{2m}$ with $F\big((1 - \mathrm{i}D)v\big) = 0$ and $F\big((1 + \mathrm{i}\beta)u\big) = 0$ for all v, u, where $1 - \beta D$ is invertible. Then $F = 0$.*

Proof. $F(v) = F(\mathrm{i}Dv)$ and $F(u) = -F(\mathrm{i}\beta u)$ for all v, u; so $F(v) = F(\mathrm{i}Dv) = -F(\mathrm{i}\beta\,\mathrm{i}Dv) = F(\beta Dv)$, that is $F\big((1 - \beta D)v\big) = 0$ for all v, and $1 - \beta D$ is onto. □

Theorem 10.14 (Positivity of $\mathcal{N}$). Lean ✓ *Under the hypotheses of Theorem 10.10, $\mathcal{N}(\rho, \sigma) > 0$. The contracted state $\mathcal{N}(\rho,\sigma)^{-1}\rho \star \sigma$ therefore exists, and it is Gaussian with covariance $\Gamma \star \beta$.*

Proof. $\mathcal{N}(\rho,\sigma) \geq 0$ by Lemma 10.9; suppose $\mathcal{N}(\rho,\sigma) = 0$. Put $\rho_E = \mathrm{Tr}_R\,\rho$, Gaussian with covariance D (Theorem 10.4), so that $\mathcal{N}(\rho,\sigma) = \mathrm{Tr}(\rho_E\sigma)$ (Lemma 10.3(c)). Both matrices being positive semidefinite, $\mathrm{Tr}(\rho_E\sigma) = 0$ forces $\rho_E\sigma = 0$: with the positive square roots $\rho_E^{1/2}$ and $\sigma^{1/2}$, $\mathrm{Tr}(\rho_E\sigma) = \mathrm{Tr}\big((\rho_E^{1/2}\sigma^{1/2})^*(\rho_E^{1/2}\sigma^{1/2})\big)$, so $\rho_E^{1/2}\sigma^{1/2} = 0$ and $\rho_E\sigma = \rho_E^{1/2}(\rho_E^{1/2}\sigma^{1/2})\sigma^{1/2} = 0$.

Claim: $\rho_E\,X\,\sigma = 0$ for every product X of factors $w \cdot c$. We argue by induction on the number k of factors; $k = 0$ is $\rho_E\sigma = 0$. For $X = (w \cdot c)Y$ with Y a product of $k-1$ factors, $F(w) = \rho_E(w \cdot c)Y\sigma$ is linear in w and vanishes on the two ranges of Lemma 10.13, with $\mathsf{G}(\rho_E) = 1 - \mathrm{i}D$ and $\mathsf{G}(\sigma) = 1 - \mathrm{i}\beta$. For $w = \mathsf{G}(\rho_E)v$, Theorem 10.5 gives $\rho_E(\mathsf{G}(\rho_E)v \cdot c) = (\mathsf{G}(\rho_E)^{\mathsf{T}}v \cdot c)\rho_E$, so $F(w) = (\mathsf{G}(\rho_E)^{\mathsf{T}}v \cdot c)\,\rho_E Y\sigma = 0$ by the induction hypothesis. For $w = \mathsf{G}(\sigma)^{\mathsf{T}}u$, the first identity of Lemma 9.5 writes $(w \cdot c)Y = \sum_i \pm 2(w^{\mathsf{T}}y_i)Y_{\hat{\imath}} + (-1)^{k-1}Y(w \cdot c)$, where $\rho_E Y_{\hat{\imath}}\sigma = 0$ by the induction hypothesis and $\rho_E Y(w \cdot c)\sigma = \rho_E Y\sigma(\mathsf{G}(\sigma)u \cdot c) = 0$ by Theorem 10.5 for σ and the induction hypothesis. Lemma 10.13 applies since $\det(1 - \beta D) = \det(1 - D\beta) \neq 0$, so $F = 0$, which is the claim for k.

Hence $\rho_E c_S\sigma = 0$ for every monomial c_S, and since the monomials span the matrices (Theorem 9.14), $\rho_E M\sigma = 0$ for every M. With $M = |s\rangle\langle t|$, $(\rho_E)_{is}\,\sigma_{tj} = 0$ for all i, s, t, j; as $\rho_E \neq 0$ this gives $\sigma = 0$, contradicting $\mathrm{Tr}\,\sigma = 1$. The last sentence is Corollary 10.11. □

Remark 10.15. The closed form $\mathcal{N}(\rho,\sigma) = 2^{-m}\sqrt{\det(1 - D\beta)}$, with the positive square root, which is up to sign the Pfaffian of the antisymmetric matrix $\left(\begin{smallmatrix} -\beta & -1 \\ 1 & D \end{smallmatrix}\right)$ on $E \sqcup E$, whose determinant is $\det(1 - D\beta)$, is expected; for $n = 0$ it is the overlap $\mathrm{Tr}(\rho\sigma) = 2^{-m}\sqrt{\det(1 - \Gamma(\rho)\Gamma(\sigma))}$ of two Gaussian states [1]. We verify it numerically for $n + m \leq 4$ (Checks paragraph at the end of this section) and do not prove it here. One proof transports σ to a product mixture and eliminates one mode at a time, each mode contributing the factor $\frac{1}{2}(1 + \lambda_j g_j)$ with g_j the pivot of Lemma 10.12.

10.5 The Gaussian fermionic matrix product state

Definition 10.16 (Input state, bond state, Gaussian fMPS). In the setting of Definitions 3.1 and 3.2: N sites, P physical and $2M$ virtual modes per site, on-site covariances Γ_i, input matrix Γ_{in} on $R \sqcup E$ and bonds β_N on E. Identify R with the labels of NP modes and E with the labels of $2NM$ modes (the modes $Q = \mathbb{Z}/N \times \{1, \ldots, P\}$ and $L \sqcup L'$), and $R \sqcup E$ with the labels of $\mathcal{F}_{NP+2NM}$, retained modes first. When every Γ_i is admissible, the *input state* ρ_{in} is the Gaussian state on $\mathcal{F}_{NP+2NM}$ with covariance Γ_{in} and the *bond state* σ_N is the Gaussian state on $\mathcal{F}_{2NM}$

with covariance β_N; both exist and are unique (Theorem 7.19, Corollary 9.11(a), Theorem 9.15). When $1 - D\beta_N$ is invertible, with D the $E \times E$ block of Γ_{in}, the *Gaussian fermionic matrix product state* (Gaussian fMPS) [4] of the on-site states and bonds is $\rho_N = \mathcal{N}(\rho_{\mathrm{in}}, \sigma_N)^{-1} \rho_{\mathrm{in}} \star \sigma_N$.

The definition admits on-site covariances that are not pure; the Kitaev chain of Section 6.2 uses one.

Theorem 10.17 (The Gaussian fMPS is a Gaussian state)**.** Lean ✓ *Under the hypotheses of Definition 10.16, $\mathcal{N}(\rho_{\mathrm{in}}, \sigma_N) > 0$ and the Gaussian fMPS ρ_N is a Gaussian state on $\mathcal{F}_{NP}$ with covariance $\Gamma_N(\Gamma_\bullet) = \Gamma_{\mathrm{in}} \star \beta_N$.*

Proof. Γ_{in} is admissible (Lemma 3.3) and β_N is pure (Theorem 2.8), so both states exist. Theorem 10.14 gives $\mathcal{N}(\rho_{\mathrm{in}}, \sigma_N) > 0$ and Corollary 10.11 the rest, once we match the labels: a bijection of the labels preserving the split $R \sqcup E$ permutes the entries of A, B, C, D and of β in the same way, so $\Gamma \star \beta$ relabelled is the contraction of the relabelled matrices: a relabelling is conjugation by a permutation matrix, which commutes with products, sums and the inverse. □

Remark 10.18. For even on-site states, the tensor product with the modes reordered by a unitary implementing the label permutation of Definition 3.1 has the covariance Γ_{in}: this follows from Theorem 6.9 once one knows that a tensor product of even states has the direct-sum covariance, which these notes do not prove (permuting the tensor factors alone changes the Jordan–Wigner strings and does not suffice). If that state is Gaussian, it equals ρ_{in} by Theorem 9.15; these notes do not prove that a tensor product of Gaussian states is Gaussian. On one bond the bond state is $(|00\rangle + \mathrm{i}|11\rangle)/\sqrt{2}$ (Theorem 5.10).

Checks. The closed form of Remark 10.15 is compared with $\mathcal{N}(\rho, \sigma)$ by `checks/normalization_overlap.py` for every (n, m) with $n + m \le 4$ and $m \ge 1$ (for $m = 0$ both sides equal 1), on 138 pairs of thermal or ground states of random quadratic Hamiltonians and of bond states, in floating point with absolute tolerance 10^{-12}; the positive square root is the one compared, and pairs of pure states of opposite parity give 0 on both sides.

11 Purity of the Gaussian fMPS and the Kitaev ground state

Section 10 built the periodic chain as the state $\rho_N = \mathcal{N}(\rho_{\mathrm{in}}, \sigma_N)^{-1} \rho_{\mathrm{in}} \star \sigma_N$ with covariance Γ_N. When the on-site covariances are pure, ρ_N is a pure state and Γ_N is pure (Corollary 11.5): the contraction of two vectors is a rank-one computation, and a Gaussian state with $\rho^2 = \rho$ has a pure covariance (Theorem 11.2). For a pure covariance Γ the quadratic Hamiltonian $H_{-\Gamma}$ is the parent Hamiltonian of $\rho[\Gamma]$: its unique ground state is $\rho[\Gamma]$, separated from the rest of the spectrum by a gap of one unit (Theorem 11.8). Applied to K_N, the Gaussian fMPS of the Kitaev chain is the ground state of the Kitaev Hamiltonian H_{-K_N}.

The conventions of Sections 7 and 10 are in force; $\mathcal{E}_h(\rho) = \mathrm{Tr}(\rho H_h) = \frac{1}{4} \sum_{ab} h_{ab} \Gamma(\rho)_{ab}$ for $h^{\mathsf{T}} = -h$ (Lemma 7.3), and $\rho_\lambda = \mathrm{diag}(p)$ with $p(s) = \prod_j p_j(s_j)$, $p_j(s_j) = \frac{1}{2}(1 - \lambda_j(-1)^{s_j}) \in [0, 1]$ and $p_j(0) + p_j(1) = 1$ (Definition 7.11). For a vector $\psi \in \mathcal{F}_n$, $|\psi\rangle\langle\psi|$ is the matrix $\psi_s \overline{\psi_t}$.

11.1 Purity

Lemma 11.1 (Contraction of two vectors)**.** Lean ✓ *Let $\psi \in \mathcal{F}_{n+m}$ and $\varphi \in \mathcal{F}_m$, and let $\xi \in \mathcal{F}_n$ be the partial inner product $\xi_s = \sum_u \overline{\varphi_u}\, \psi_{(s,u)}$. Then $|\psi\rangle\langle\psi| \star |\varphi\rangle\langle\varphi| = |\xi\rangle\langle\xi|$ and $\mathcal{N}(|\psi\rangle\langle\psi|, |\varphi\rangle\langle\varphi|) = \|\xi\|^2$.*

Proof. With $M = |\psi\rangle\langle\psi|\,(1 \otimes |\varphi\rangle\langle\varphi|)$, $M_{(s,u),(t,u')} = \sum_{u''} \psi_{(s,u)} \overline{\psi_{(t,u'')}}\, \varphi_{u''} \overline{\varphi_{u'}} = \psi_{(s,u)}\, \overline{\xi_t}\, \overline{\varphi_{u'}}$, so $(\mathrm{Tr}_E\, M)_{st} = \sum_u \psi_{(s,u)} \overline{\varphi_u}\, \overline{\xi_t} = \xi_s \overline{\xi_t}$ (Definition 10.1); the trace is $\sum_s |\xi_s|^2$. □

Theorem 11.2 (An idempotent Gaussian state has a pure covariance)**.** Lean ✓ *A Gaussian state ρ satisfies $\rho^2 = \rho$ if and only if $\Gamma(\rho)$ is pure.*

Proof. If $\Gamma(\rho)$ is pure, $\rho = U^*|s\rangle\langle s|U$ (Corollary 9.16(b)) and $\rho^2 = \rho$. Conversely let $\rho^2 = \rho$ and write $\rho = U^*\rho_\lambda U$ with $\Gamma(\rho) = O\mathsf{D}(\lambda)O^{\mathsf{T}}$ (Corollary 9.16(a)). Then $\rho_\lambda^2 = U\rho^2U^* = U\rho U^* = \rho_\lambda$, so $p(s)^2 = p(s)$, that is $p(s) \in \{0,1\}$, for every configuration s. Suppose $|\lambda_j| < 1$ for some j. For $k \neq j$ choose s_k with $p_k(s_k) > 0$, possible since $p_k(0) + p_k(1) = 1$, and take any s_j; then $p_j(s_j) = \frac{1}{2}(1 \mp \lambda_j) \in (0,1)$ and $p(s) = p_j(s_j)\prod_{k\neq j} p_k(s_k)$ satisfies $0 < p(s) \leq p_j(s_j) < 1$, a contradiction. Hence $|\lambda_j| = 1$ for all j, $\mathsf{D}(\lambda)$ is pure, and so is $\Gamma(\rho) = O\mathsf{D}(\lambda)O^{\mathsf{T}}$ (Theorem 1.7). □

Outside the Gaussian class a projection need not have a pure covariance: for one mode, $\rho = \frac{1}{2}(1+c_{x_1})$ satisfies $\rho^2 = \frac{1}{4}(1+2c_{x_1}+c_{x_1}^2) = \rho$, while $\Gamma(\rho)_{x_1y_1} = \mathrm{i}\operatorname{Tr}(\rho c_{x_1}c_{y_1}) = \frac{\mathrm{i}}{2}\operatorname{Tr}(c_{x_1}c_{y_1}+c_{y_1}) = 0$ (Lemma 9.13), so $\Gamma(\rho) = 0$.

Theorem 11.3 (Contraction of pure Gaussian states)**.** Lean ✓ *Let ρ on $n+m$ modes and σ on m modes be Gaussian with pure covariances $\Gamma = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ and β, and let $1 - D\beta$ be invertible. Then $\rho = |\psi\rangle\langle\psi|$ and $\sigma = |\varphi\rangle\langle\varphi|$ for vectors ψ, φ; the contracted state is $\mathcal{N}(\rho,\sigma)^{-1}|\xi\rangle\langle\xi|$ with $\xi = (1 \otimes \langle\varphi|)\psi \neq 0$ and $\mathcal{N}(\rho,\sigma) = \|\xi\|^2$; it is a projection; and its covariance $\Gamma \star \beta$ is pure.*

Proof. $\rho = U^*|s\rangle\langle s|U = |\psi\rangle\langle\psi|$ with $\psi = U^*|s\rangle$ (Corollary 9.16(b)), likewise σ. Lemma 11.1 gives $\rho \star \sigma = |\xi\rangle\langle\xi|$ and $\mathcal{N}(\rho,\sigma) = \|\xi\|^2$; $\mathcal{N}(\rho,\sigma) > 0$ by Theorem 10.14, so $\xi \neq 0$ and $\mathcal{N}(\rho,\sigma)^{-1}|\xi\rangle\langle\xi|$ is a projection. It is Gaussian with covariance $\Gamma \star \beta$ (Corollary 10.11), so Theorem 11.2 gives purity. □

Corollary 11.4 (Purity of the contraction)**.** Lean ✓ *If Γ on $R \sqcup E$ and β on E are pure and $1 - D\beta$ is invertible, then $\Gamma \star \beta$ is pure.*

Proof. Take Gaussian states with covariances Γ and β (Theorem 7.19, Corollary 9.11(a)) and apply Theorem 11.3. □

Corollary 11.5 (The Gaussian fMPS of pure on-site covariances)**.** Lean ✓ *In the setting of Definition 10.16, let every on-site covariance Γ_i be pure and $1 - D\beta_N$ invertible. Then $\rho_N^2 = \rho_N$ and $\Gamma_N(\Gamma_\bullet)$ is pure.*

Proof. Γ_{in} is pure, being a relabelled direct sum of pure matrices (Lemma 3.3), and β_N is pure (Theorem 2.8); the input and bond states are Gaussian with these covariances. Theorem 11.3 and the relabelling of Theorem 10.17 complete the proof. □

11.2 Pure covariances are ground covariances

Theorem 11.6 (The ground state of $H_{-\Gamma}$ for pure Γ)**.** Lean ✓ *Let Γ be pure on n modes. Then the ground energy of $H_{-\Gamma}$ is $-n/2$; every density operator ρ with $\mathcal{E}_{-\Gamma}(\rho) = -n/2$ equals the Gaussian state $\rho[\Gamma]$ with covariance Γ; and every admissible Γ' satisfies $\frac{1}{4}\sum_{ab}(-\Gamma)_{ab}\Gamma'_{ab} \geq -n/2$, with equality iff $\Gamma' = \Gamma$.*

Proof. $-\Gamma$ is pure, so its normal form $-\Gamma = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$ ($OO^{\mathsf{T}} = 1$, $\det O = 1$) has $|\varepsilon_j| = 1$ for all j (Theorem 7.13), in particular $\varepsilon_j \neq 0$. We apply Theorem 7.20 with $h = -\Gamma$ and U implementing O: the ground energy is $-\frac{1}{2}\sum_j|\varepsilon_j| = -n/2$; every density operator of that energy is $\rho_{\mathrm{gs}} = U^*|s^\varepsilon\rangle\langle s^\varepsilon|U$; and an admissible Γ' has $\frac{1}{4}\sum_{ab}(-\Gamma)_{ab}\Gamma'_{ab} \geq -n/2$ with equality iff $\Gamma' = \Gamma(\rho_{\mathrm{gs}})$. Finally $\Gamma(\rho_{\mathrm{gs}}) = O\mathsf{D}(-\operatorname{sgn}\varepsilon)O^{\mathsf{T}} = -O\mathsf{D}(\varepsilon)O^{\mathsf{T}} = \Gamma$, since $\mathsf{D}(-\operatorname{sgn}\varepsilon) = -\mathsf{D}(\varepsilon)$ when $|\varepsilon_j| = 1$; and ρ_{gs} is Gaussian (Corollary 9.11(c)) with covariance Γ, hence $\rho_{\mathrm{gs}} = \rho[\Gamma]$ (Theorem 9.15). □

Remark 11.7. Since $\mathcal{E}_h(\rho) = -\frac{1}{4}\operatorname{Tr}(h\,\Gamma(\rho))$, the choice $h = -\Gamma$ gives $\mathcal{E}(\rho) = \frac{1}{4}\operatorname{Tr}(\Gamma\,\Gamma(\rho)) = -\frac{1}{4}\sum_a(\Gamma e_a)^{\mathsf{T}}(\Gamma(\rho)e_a)$. Each column of Γ is a unit vector and each column of $\Gamma(\rho)$ has norm at most 1 (Theorem 1.3), so by the Cauchy–Schwarz inequality $\mathcal{E}(\rho) \geq -\frac{1}{4}\cdot 2n = -n/2$, with

equality exactly when $\Gamma(\rho)e_a = \Gamma e_a$ for every a, that is at $\Gamma(\rho) = \Gamma$. For $H_{+\Gamma}$ the same bound reads $-\frac{1}{4}\operatorname{Tr}(\Gamma\,\Gamma(\rho)) \geq -n/2$ with equality at $\Gamma(\rho) = -\Gamma$, so the ground covariance of $H_{+\Gamma}$ is $-\Gamma$.

A gap that does not depend on Γ or on n separates the ground state from the rest of the spectrum. We call a density operator ρ *orthogonal* to the ground state when $\operatorname{Tr}(\rho\,\rho[\Gamma]) = 0$. Since $\rho \succeq 0$ and $\operatorname{Tr}(\rho\,\rho[\Gamma]) = \langle\psi_\Gamma|\rho|\psi_\Gamma\rangle$ for the ground vector ψ_Γ, this holds exactly when $\rho\psi_\Gamma = 0$, that is, when ρ is supported on the orthogonal complement of the ground vector; for a pure state $\rho = |\psi\rangle\langle\psi|$ it says that $\psi \perp \psi_\Gamma$.

Theorem 11.8 (The gap of $H_{-\Gamma}$)**.** Lean ✓ *Let Γ be pure on n modes and let ρ be a density operator with $\operatorname{Tr}(\rho\,\rho[\Gamma]) = 0$. Then $\mathcal{E}_{-\Gamma}(\rho) \geq -\frac{n}{2} + 1$. For $n \geq 1$ the bound is attained: some density operator ρ' has $\operatorname{Tr}(\rho'\rho[\Gamma]) = 0$ and $\mathcal{E}_{-\Gamma}(\rho') = -\frac{n}{2} + 1$.*

Proof. As in the proof of Theorem 11.6, write $-\Gamma = O\mathsf{D}(\varepsilon)O^{\mathsf{T}}$ with $|\varepsilon_j| = 1$, let U implement O, and recall that $\rho[\Gamma] = U^*|s^\varepsilon\rangle\langle s^\varepsilon|U$. Put $\rho'' = U\rho U^*$, a density operator with $\mathcal{E}_{-\Gamma}(\rho) = \mathcal{E}_{\mathsf{D}(\varepsilon)}(\rho'')$ (Lemma 7.3(c)). Since $H_{\mathsf{D}(\varepsilon)} = \operatorname{diag}(d_\varepsilon)$ (Lemma 7.4),

$$\mathcal{E}_{\mathsf{D}(\varepsilon)}(\rho'') = \sum_s \rho''_{ss}\, d_\varepsilon(s), \qquad \rho''_{ss} \geq 0, \qquad \sum_s \rho''_{ss} = 1,$$

and the weight of the ground configuration vanishes:

$$\rho''_{s^\varepsilon s^\varepsilon} = \operatorname{Tr}(\rho''\,|s^\varepsilon\rangle\langle s^\varepsilon|) = \operatorname{Tr}(\rho\,U^*|s^\varepsilon\rangle\langle s^\varepsilon|U) = \operatorname{Tr}(\rho\,\rho[\Gamma]) = 0.$$

Every other configuration differs from s^ε in at least one mode, and each such mode costs $|\varepsilon_j| = 1$ (Definition 7.6): $d_\varepsilon(s) \geq d_\varepsilon(s^\varepsilon) + 1 = -\frac{n}{2} + 1$. Hence $\mathcal{E}_{\mathsf{D}(\varepsilon)}(\rho'') \geq (-\frac{n}{2} + 1)\sum_{s \neq s^\varepsilon} \rho''_{ss} = -\frac{n}{2} + 1$. For the attainment let s' be s^ε with the occupation of mode 1 flipped and $\rho' = U^*|s'\rangle\langle s'|U$; then $\operatorname{Tr}(\rho'\rho[\Gamma]) = |\langle s'|s^\varepsilon\rangle|^2 = 0$ and $\mathcal{E}_{-\Gamma}(\rho') = d_\varepsilon(s') = -\frac{n}{2} + 1$. □

The same argument gives the gap $\min_j |\varepsilon_j|$ of a general quadratic Hamiltonian $H_{O\mathsf{D}(\varepsilon)O^{\mathsf{T}}}$.

Definition 11.9 (The Kitaev Hamiltonian)**.** For $N \geq 1$ the *Kitaev Hamiltonian* is H_{-K_N} on $\mathcal{F}_N$, the quadratic Hamiltonian of $h = -K_N$, with K_N the Kitaev chain covariance of Section 6.2.

Lemma 11.10 (H_{-K_N} as a sum over the pairs (y_i, x_{i+1}))**.** Lean ✓

$$H_{-K_N} = \tfrac{\mathrm{i}}{2} \sum_{i \in \mathbb{Z}/N} c_{x_{i+1}} c_{y_i} = -\tfrac{\mathrm{i}}{2} \sum_{i \in \mathbb{Z}/N} c_{y_i} c_{x_{i+1}}.$$

Proof. The nonzero entries of $-K_N$ are $(-K_N)_{y_i,x_{i+1}} = -1$ and $(-K_N)_{x_{i+1},y_i} = 1$, so $H_{-K_N} = \frac{\mathrm{i}}{4}\sum_i (c_{x_{i+1}}c_{y_i} - c_{y_i}c_{x_{i+1}}) = \frac{\mathrm{i}}{2}\sum_i c_{x_{i+1}}c_{y_i}$, since $x_{i+1} \neq y_i$ and distinct Majorana operators anticommute (Theorem 5.4); this holds also for $N = 1$, where $x_{i+1} = x_i$. □

Remark 11.11 (The sign of the hopping). Kitaev's Hamiltonian $H = \mathrm{i}t\sum_j c_{2j}c_{2j+1}$ reads, in the dictionary $c_{2j-1} = x_j$, $c_{2j} = y_j$, $\mathrm{i}t\sum_j c_{y_j}c_{x_{j+1}} = 2t\,H_{+K_N}$; so H_{-K_N} is Kitaev's Hamiltonian at $t = -\frac{1}{2}$, and for $t > 0$ the ground covariance is $-K_N$ (Remark 11.7). The sign of t is a choice of gauge, not of physics: the unitary $U = c_{x_1}c_{x_2}\cdots c_{x_N}$ satisfies $Uc_{x_j}U^* = (-1)^{N-1}c_{x_j}$ and $Uc_{y_j}U^* = (-1)^N c_{y_j}$ (each c_{x_j} anticommutes with the other $N-1$ factors, each c_{y_j} with all N), so U implements $O = \operatorname{diag}((-1)^{N-1}1_N, (-1)^N 1_N)$, which reverses the relative sign of the x and y labels and sends K_N to $OK_NO = -K_N$. Since O is diagonal with $O^2 = 1$, $\Gamma(U\rho U^*) = O\Gamma(\rho)O$ and $\mathcal{E}_{K_N}(U\rho U^*) = -\frac{1}{4}\operatorname{Tr}(K_N\,O\Gamma(\rho)O) = \mathcal{E}_{-K_N}(\rho)$. Hence ρ is the ground state of H_{-K_N} if and only if $U\rho U^*$ is the ground state of H_{+K_N}: the ground state of Kitaev's Hamiltonian for $t > 0$ is $U\rho_N U^* = \rho[-K_N]$, the Gaussian fMPS of Theorem 11.12 with the y Majoranas reflected. The pairing of y_i with x_{i+1} is the same for both signs of t.

Theorem 11.12 (The Gaussian fMPS of the Kitaev chain is the Kitaev ground state)**.** Lean ✓ *Let $N \geq 1$ and let ρ_N be the Gaussian fMPS of Definition 10.16 with the on-site covariance Γ_K of Definition 6.3 at every site. Then ρ_N is well defined, $\rho_N^2 = \rho_N$, $\Gamma(\rho_N) = K_N$, $\mathcal{E}_{-K_N}(\rho_N) = -N/2$ is the ground energy of H_{-K_N}, and every density operator ρ with $\mathcal{E}_{-K_N}(\rho) = -N/2$ equals ρ_N.*

Proof. $1 - D\beta_N$ is invertible and $\Gamma_N = K_N$ by Theorem 6.5, and K_N is pure (Theorem 6.5). By Theorem 10.17, ρ_N is Gaussian with covariance K_N, so $\rho_N = \rho[K_N]$ (Theorem 9.15) and $\rho_N^2 = \rho_N$ (Theorem 11.2). Theorem 11.6 with $\Gamma = K_N$ gives the rest. □

Corollary 11.13 (The gap of the Kitaev chain)**.** Lean ✓ *For $N \geq 1$, every density operator ρ on $\mathcal{F}_N$ with $\mathrm{Tr}(\rho\,\rho_N) = 0$ has $\mathcal{E}_{-K_N}(\rho) \geq -\frac{N}{2} + 1$, and the value is attained. The gap of H_{-K_N} is 1, independent of N.*

Proof. K_N is pure (Theorem 6.5) and $\rho_N = \rho[K_N]$ (Theorem 11.12); apply Theorem 11.8. □

In Kitaev's normalization $H = 2t\,H_{+K_N}$ the gap is $2|t|$, the energy of one broken pair.

Remark 11.14. The on-site covariance Γ_K is not pure (Lemma 6.4), so the input state is not pure, yet the Gaussian fMPS is a pure state, since $\Gamma_N = K_N$ is pure (Theorem 6.5) and Theorem 11.2 applies. The pure completion of Γ_K (Proposition 6.6) gives the same K_N for odd N and a singular $1 - D\beta_N$ for even N; the on-site covariance Γ_K, which is not pure, is the one valid for every N.

12 The entanglement rank of a Gaussian fMPS

A bond of M virtual modes cannot carry unlimited correlations across it. For a set S of sites cut off from the rest of the ring by $b(S)$ bonds, the block of Γ_N between the physical labels of S and of its complement has rank at most $2M\,b(S)$ (Theorem 12.4), whatever the on-site matrices are; and when Γ_N is pure, at most $M\,b(S)$ of the block's normal modes are entangled with the outside (Corollary 12.5). This is the fermionic form of the bound "Schmidt rank at most the bond dimension" for matrix product states. The proof runs through the contraction formula and a fact of linear algebra worth stating on its own: the off-diagonal block of an inverse has the rank of the off-diagonal block.

12.1 Two lemmas of linear algebra

Lemma 12.1 (Nullity lemma)**.** Lean ✓ *Let $W = \begin{pmatrix} W_{11} & W_{12} \\ W_{21} & W_{22} \end{pmatrix}$ be an invertible real matrix with square diagonal blocks of sizes p and q, and let $V = W^{-1} = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix}$ be partitioned the same way. Then* $\operatorname{rank} V_{12} = \operatorname{rank} W_{12}$.

Proof. Let $v \in \mathbb{R}^q$ with $W_{12}v = 0$. Then $W\binom{0}{v} = \binom{W_{12}v}{W_{22}v} = \binom{0}{W_{22}v}$, so $\binom{0}{v} = V\binom{0}{W_{22}v} = \binom{V_{12}W_{22}v}{V_{22}W_{22}v}$: $V_{12}(W_{22}v) = 0$ and $V_{22}(W_{22}v) = v$. The linear map $v \mapsto W_{22}v$ therefore sends $\ker W_{12}$ into $\ker V_{12}$, and it is injective there, since $W_{22}v = 0$ forces $v = V_{22}W_{22}v = 0$. Hence $\dim\ker W_{12} \leq \dim\ker V_{12}$. The same argument with W and V exchanged gives $\dim\ker V_{12} \leq \dim\ker W_{12}$. Both W_{12} and V_{12} are $p \times q$, so the ranks agree. □

The lemma holds when W_{11} and W_{22} are singular, which is the case for a chain whose on-site matrices have no virtual–virtual entries (the Kitaev chain, Section 6.2); this is why we can bound the cross block without a Schur complement of the on-site matrices.

Lemma 12.2 (Contraction against a pure pair covariance)**.** Lean ✓ *Let Γ_p be pure on E and D a matrix on E with $1 - D\Gamma_p$ invertible. Then $\Gamma_p + D$ is invertible and $\Gamma_p(1 - D\Gamma_p)^{-1} = -(\Gamma_p + D)^{-1}$.*

Proof. $(\Gamma_p + D)\Gamma_p = \Gamma_p^2 + D\Gamma_p = -(1 - D\Gamma_p)$, so $\Gamma_p + D = (1 - D\Gamma_p)\Gamma_p$ is a product of invertible matrices ($\Gamma_p^{-1} = -\Gamma_p$), and $(\Gamma_p + D)^{-1} = \Gamma_p^{-1}(1 - D\Gamma_p)^{-1} = -\Gamma_p(1 - D\Gamma_p)^{-1}$. □

12.2 The cross-block rank bound

We work in the setting of Definitions 3.1 and 3.2: N sites, P physical and $2M$ virtual modes per site, on-site matrices Γ_i, input matrix $\Gamma_{\text{in}} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ on $R \sqcup E$, bonds β_N on E, and $\Gamma_N = A + B\beta_N(1 - D\beta_N)^{-1}C$ when $1 - D\beta_N$ is invertible. Every label belongs to a site: a physical label (i, c) and a left virtual label $L(i, a)$ to site i, a right virtual label $L'(i, a)$ to site $i - 1$; the bond i joins $L(i, a)$ to $L'(i, a)$, that is, site i to site $i - 1$. Since Γ_{in} is a relabelling of $\operatorname{diag}(\Gamma_i)_i$, an entry of Γ_{in} whose two labels belong to different sites is zero.

Definition 12.3 (Site sets, blocks, cut bonds)**.** For $S \subseteq \mathbb{Z}/N$ let R_S be the physical labels of the sites in S, E_S the virtual labels belonging to sites in S, and $S^c = \mathbb{Z}/N \setminus S$. The *block covariance* of S is $\Gamma_S = (\Gamma_N)_{R_S R_S}$ and its *cross block* is $C_S = (\Gamma_N)_{R_S R_{S^c}}$. The bond i is *cut* by S when exactly one of $i-1, i$ lies in S; $b(S)$ is the number of cut bonds (the bonds crossing the boundary of S).

A set of k consecutive sites with $1 \le k \le N - 1$ has $b(S) = 2$; $b(\emptyset) = b(\mathbb{Z}/N) = 0$.

Theorem 12.4 (Cross-block rank)**.** Lean ✓ *Let the on-site matrices Γ_i be arbitrary real matrices with $1 - D\beta_N$ invertible. Then $\operatorname{rank} C_S \le 2M\, b(S)$ for every $S \subseteq \mathbb{Z}/N$.*

Proof. The correlations between S and S^c enter Γ_N only through the inverse $(\beta_N + D)^{-1}$, whose (E_S, E_{S^c}) block is hard to compute but easy to bound: by Lemma 12.1 its rank is that of the (E_S, E_{S^c}) block of $\beta_N + D$ itself, and there only the cut bonds contribute. Lemma 12.2 puts the contraction in the form that exposes this inverse.

By Lemma 12.2 (β_N is pure, Theorem 2.8), $\Gamma_N = A - B(\beta_N + D)^{-1}C$. Take the (R_S, R_{S^c}) block of each term. $A_{R_S R_{S^c}} = 0$, since a physical label of S and one of S^c lie at different sites. In the product $B(\beta_N + D)^{-1}C$, the row $a \in R_S$ of B is supported on the columns E_S and the column $a' \in R_{S^c}$ of C on the rows E_{S^c}, by the same-site support, so

$$C_S = -\,B_{R_S E_S}\left[(\beta_N + D)^{-1}\right]_{E_S E_{S^c}} C_{E_{S^c} R_{S^c}}, \qquad \operatorname{rank} C_S \le \operatorname{rank}\left[(\beta_N + D)^{-1}\right]_{E_S E_{S^c}}.$$

Partition $E = E_S \sqcup E_{S^c}$ and apply Lemma 12.1 to the invertible matrix $\beta_N + D$: the rank on the right equals $\operatorname{rank}(\beta_N + D)_{E_S E_{S^c}} = \operatorname{rank}(\beta_N)_{E_S E_{S^c}}$, because $D_{E_S E_{S^c}} = 0$. Finally, the nonzero entries of β_N are $\beta_{x_{L(i,a)} x_{L'(i,a)}} = 1$ and $\beta_{y_{L(i,a)} y_{L'(i,a)}} = -1$ with their transposes, one x-entry and one y-entry per bond i and $a \in \{1, \dots, M\}$, in distinct rows and columns. Such an entry lies in the (E_S, E_{S^c}) block only if its bond is cut, and a cut bond contributes at most $2M$ nonzero rows, so $\operatorname{rank}(\beta_N)_{E_S E_{S^c}} \le 2M\, b(S)$. □

Corollary 12.5 (Entanglement rank of a pure Gaussian fMPS)**.** Lean ✓ *If in addition Γ_N is pure (for instance all Γ_i pure, Corollary 11.5), then for every S: Γ_S is admissible, $1 + \Gamma_S^2 = C_S C_S^{\mathsf{T}}$, and with a normal form $\Gamma_S = O\mathsf{D}(\lambda)O^{\mathsf{T}}$ (Theorem 7.13; $|\lambda_j| \le 1$)*

$$\#\{j : |\lambda_j| < 1\} = \tfrac{1}{2}\operatorname{rank}(1 + \Gamma_S^2) \le M\, b(S).$$

Proof. Γ_S is a principal submatrix of the admissible Γ_N, hence admissible (Theorem 1.8). Order the physical labels as $R_S \sqcup R_{S^c}$; then $\Gamma_N = \begin{pmatrix} \Gamma_S & C_S \\ -C_S^{\mathsf{T}} & \Gamma_{S^c} \end{pmatrix}$ by antisymmetry, and the (R_S, R_S) block of $\Gamma_N^2 = -1$ reads $\Gamma_S^2 - C_S C_S^{\mathsf{T}} = -1$. So $1 + \Gamma_S^2 = C_S C_S^{\mathsf{T}}$, whose rank is $\operatorname{rank} C_S$ because $C_S C_S^{\mathsf{T}} v = 0$ forces $\|C_S^{\mathsf{T}} v\|^2 = v^{\mathsf{T}} C_S C_S^{\mathsf{T}} v = 0$, so $C_S C_S^{\mathsf{T}}$ and C_S^{T} have the same kernel. With $\Gamma_S = O\mathsf{D}(\lambda)O^{\mathsf{T}}$, $1 + \Gamma_S^2 = O\operatorname{diag}(1 - \lambda_j^2, 1 - \lambda_j^2)_j O^{\mathsf{T}}$ has rank $2\#\{j : \lambda_j^2 \ne 1\} = 2\#\{j : |\lambda_j| < 1\}$. Theorem 12.4 gives the bound. □

Remark 12.6. Corollary 12.5 counts the normal-form modes of Γ_S (the columns of O) with $|\lambda_j| < 1$, which we call the *entangled modes* of S: there are at most $M\, b(S)$ of them, so a bond of M virtual modes lets at most M modes be entangled across it.

The name comes from the following reading. If the reduced state of S is Gaussian (Theorem 10.4 proves this when S consists of the first or the last modes; for a general S it is not proved here), then in the normal-form modes of Γ_S it is a product over modes. A mode with $|\lambda_j| = 1$ is in a pure state; a pure factor of the marginal of the pure state with covariance Γ_N is a tensor factor of that state (a standard fact about pure states, not proved here), so such a mode is uncorrelated with S^c. The modes with $|\lambda_j| < 1$ are mixed, hence each is entangled with S^c.

Since each entangled mode can contribute a factor at most 2 to the Schmidt rank of the cut, one expects the Schmidt rank to be at most $2^{M\,b(S)}$, the counterpart of "Schmidt rank at most the bond dimension" for matrix product states with bond dimension 2^M; this is not proved here.

Remark 12.7. The following computation is not formalized. For the Kitaev chain ($M = P = 1$, $\Gamma_N = K_N$) a block S of k consecutive sites, $1 \le k \le N-1$, has exactly one entangled mode, with $\lambda = 0$: the $k-1$ pairs (y_i, x_{i+1}) inside S give modes with $\lambda = \pm 1$, while the boundary labels x_{first} and y_{last} are paired with labels outside S, so their rows and columns of Γ_S vanish and they form one mode with $\lambda = 0$. The Kitaev chain does not attain the bound $M\,b(S) = 2$.

For $M = P = 1$ the bound can be attained only if the block and its complement each contain at least two sites, that is $N \ge 4$ and $2 \le k \le N-2$: C_S has $2k$ rows and $2(N-k)$ columns, so by Corollary 12.5 at most $\min(k, N-k)$ modes are entangled. The theorem bounds a rank, not a norm: for a single site of a ring with $N \ge 2$, $b(S) = 2$ and the bound $M\,b(S) = 2M$ of Corollary 12.5 may exceed the site's own mode count P, in which case it says nothing.

That the condition above also suffices is observed numerically and not proved here: for $M = P = 1$, pure on-site covariances drawn at random attain the bound for every $N \ge 4$ and $2 \le k \le N-2$ in the computations named in the Checks paragraph below ($N \le 6$).

Checks. The statements of Remark 12.7 are computed in exact rational arithmetic by `checks/kitaev_block_modes.py`: for the Kitaev chain with $N = 2, \dots, 9$ and every block of consecutive sites, and for random pure on-site covariances with $(M, P) = (1, 1)$ and $N = 2, \dots, 6$.

Index of terms

Each term is defined where it first appears; the number is that of the definition, or of the statement, equation or section where the term is introduced. Terms marked * are introduced in these notes or used in a narrower sense than usual.

admissible* Def. 1.1
block covariance* Def. 12.3
bond covariance* Def. 2.7
bond state* Def. 10.16
bonds of a chain* Def. 3.2
canonical form $\mathsf{D}(\varepsilon)$* Def. 7.1
contracted state* Def. 10.8
contraction of covariances* Def. 2.1
contraction of states* Def. 10.8
covariance of the chain* Def. 3.2
cross block* Def. 12.3
cut bond* Def. 12.3
defining relation* Thm. 10.5
density operator Def. 5.5
direct sum of covariances §1.2
eliminated labels* §2
entangled modes* Rem. 12.6
Fock space §5
Fourier coefficients of the chain Def. 4.3
Gaussian fMPS Def. 10.16
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Gibbs state Def. 8.1
ground configuration* Def. 7.6
hopping gate Def. 6.7
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input state* Def. 10.16
Jordan–Wigner operators Def. 5.3
Kitaev chain covariance K_N^* §6.2
Kitaev Hamiltonian* Def. 11.9
Kitaev on-site covariance* Def. 6.3
Majorana covariance Def. 5.5
Majorana operators Def. 5.3
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normal form Thm. 7.13
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quadratic Hamiltonian Def. 7.2
relabelled matrix Γ^{e*} §3
retained labels* §2
rotation product* Def. 7.14
signed flip* Eq. (5.1)
site by site* §3
standard covariance Ω_n^* Eq. (1.3)
strictly admissible* Prop. 8.6
teleportation identity* Thm. 2.9
thermal state Def. 8.1
twisted on-site covariance* Def. 4.1
two-point matrix G* §9
Wick expansion* Def. 9.1
Wick's rule (Wick's theorem) Def. 9.1, §9

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Lean declarations

The linked Lean ✓ tag marks a theorem whose Lean proof has been checked and whose correspondence with the statement here has been reviewed. It refers to the stated result, not to every surrounding explanation. Definitions and equations are listed for navigation; only marked statements were reviewed. These entries identify the corresponding declarations in FermionicLean.

Lean names. For a reader of the library: the labels $x_1, \dots, x_n, y_1, \dots, y_n$ are `MajoranaIndex n = Fin n ⊕ Fin n` with `Sum.inl` the x block; the retained and eliminated labels R, E of a chain are `Retained N P` and `Eliminated N M`; the contraction $\star$ is `contract` and the hypothesis "$1 - D\Gamma_p$ invertible" is `ContractInvertible`; admissibility and purity are `IsAdmissible`, `IsPure` on `MajoranaIndex n` and `Admissible`, `Pure` on an arbitrary finite label type, which coincide on Majorana labels (`isAdmissible_iff_admissible`, `isPure_iff_pure`); the covariance of a state is `covariance`, Gaussian is `IsGaussian`, the energy is `energy`.

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2.8 Fermionic.pure_bond
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2.9 Fermionic.contract_teleportInput
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Fermionic.trace_ptraceR_mul
Fermionic.ptraceR_conjTranspose
Fermionic.ptraceR_posSemidef
Fermionic.isDensity_ptraceR
FermionicLean/Fock/PartialTrace.lean

10.4 Fermionic.wick_map_of_dotProduct
Fermionic.covariance_ptraceE
Fermionic.covariance_ptraceR
Fermionic.isGaussian_ptraceE
Fermionic.isGaussian_ptraceR
FermionicLean/Fock/PartialTrace.lean

10.5 Fermionic.relation_productMixture
Fermionic.conjTranspose_mul_lincomb_mul
Fermionic.IsGaussian.relation

Fermionic.IsGaussian.parity_conj
FermionicLean/Fock/Relation.lean

10.7 Fermionic.momentList_eq_trace_mul_wick_of_recursion
Fermionic.recursion_of_relation'
Fermionic.momentList_eq_trace_mul_wick_of_relation
Fermionic.isGaussian_of_relation
FermionicLean/Fock/Relation.lean

10.9 Fermionic.contractRaw_posSemidef
Fermionic.normalization_nonneg
Fermionic.isDensity_contractState
Fermionic.parity_mul_mul_tensor_one
Fermionic.momentList_contractRaw_eq_zero_of_odd
FermionicLean/Fock/GaussianMap.lean

10.10 Fermionic.row_system_solution
Fermionic.recursion_contractRaw
Fermionic.momentList_contractRaw_eq_wick
FermionicLean/Contraction/RowSolve.lean,FermionicLean/Fock/GaussianMap.lean

10.11 Fermionic.covariance_contractState
Fermionic.isGaussian_contractState
FermionicLean/Fock/GaussianMap.lean

10.12 Fermionic.det_one_sub_mul_fromBlocks
FermionicLean/Contraction/BlockDet.lean

10.13 Fermionic.linearMap_eq_zero_of_ranges
FermionicLean/Fock/GaussianMap.lean

10.14 Fermionic.mul_eq_zero_of_trace_mul_eq_zero
Fermionic.mul_prod_mul_eq_zero
Fermionic.normalization_ne_zero
Fermionic.normalization_pos
Fermionic.isGaussian_contractState'
FermionicLean/Fock/GaussianMap.lean

10.17 Fermionic.contractCov_inputCov_bondCov
Fermionic.normalization_chain_pos
Fermionic.isGaussian_chainState
Fermionic.covariance_chainState
FermionicLean/Chain/State.lean

11.1 Fermionic.partialInner
Fermionic.contractRaw_vecMulVec
Fermionic.normalization_vecMulVec
FermionicLean/Fock/Pure.lean

11.2 Fermionic.abs_eq_one_of_productMixture_mul_self
Fermionic.IsGaussian.isPure_covariance_of_mul_self
Fermionic.IsGaussian.mul_self_iff_isPure
FermionicLean/Fock/Pure.lean

11.3 Fermionic.conjTranspose_mul_occupation_mul_eq_vecMulVec
Fermionic.IsGaussian.exists_vecMulVec_eq_of_isPure
Fermionic.contractState_mul_self_of_isPure
FermionicLean/Fock/Pure.lean

11.4 `Fermionic.isPure_contractCov_of_isPure`
FermionicLean/Fock/Pure.lean

11.5 `Fermionic.pure_periodicInput_iff`
`Fermionic.isPure_inputCov`
`Fermionic.isPure_bondCov`
`Fermionic.chainState_mul_self_of_forall_pure`
`Fermionic.pure_periodicCovariance_of_forall_pure`
FermionicLean/Chain/Pure.lean

11.6 `Fermionic.groundEnergy_of_abs_eq_one`
`Fermionic.blockForm_neg_sgn_of_abs_eq_one`
`Fermionic.groundState_of_isPure`
`Fermionic.eq_of_energy_neg_eq_of_isPure`
`Fermionic.energy_neg_eq_of_isPure`
FermionicLean/Fock/PureGround.lean

11.8 `Fermionic.diag_conj_eq_trace_mul`
`Fermionic.groundEnergy_add_one_le_blockEnergy`
`Fermionic.blockEnergy_update_groundConfig`
`Fermionic.energy_neg_ge_of_trace_mul_eq_zero`
`Fermionic.exists_trace_mul_eq_zero_and_energy_neg_eq`
FermionicLean/Fock/Gap.lean

11.10 `Fermionic.kitaevCov`
`Fermionic.isPure_kitaevCov`
`Fermionic.kitaevHam`
`Fermionic.kitaevHam_eq`
FermionicLean/Chain/KitaevState.lean

11.12 `Fermionic.kitaevState`
`Fermionic.covariance_kitaevState`
`Fermionic.isGaussian_kitaevState`
`Fermionic.kitaevState_mul_self`
`Fermionic.energy_kitaevState`
`Fermionic.energy_kitaevState'`
`Fermionic.energy_kitaevHam_ge`
`Fermionic.eq_kitaevState_of_energy_eq`
FermionicLean/Chain/KitaevState.lean

11.13 `Fermionic.energy_kitaevHam_ge_of_trace_mul_eq_zero`
`Fermionic.exists_trace_mul_kitaevState_eq_zero_and_energy_eq`
FermionicLean/Chain/KitaevGap.lean

12.1 `Fermionic.finrank_ker_toBlocks12_le_inv`
`Fermionic.rank_toBlocks12_inv`
FermionicLean/Gaussian/Nullity.lean

12.2 `Fermionic.isUnit_det_add_of_pure`
`Fermionic.mul_inv_one_sub_mul_eq_neg_inv_add_of_pure`
FermionicLean/Contraction/PureInverse.lean

12.4 `Fermionic.siteOf`
`Fermionic.cutBonds`
`Fermionic.crossBlock`
`Fermionic.periodicInput_apply_of_siteOf_ne`
`Fermionic.crossBlock_mul_mul_eq`

Fermionic.rank_toBlocks12_periodicBond_submatrix_le
Fermionic.rank_crossBlock_periodicCovariance_le
FermionicLean/Chain/Cut.lean

12.5 Fermionic.one_add_blockCov_sq_eq
Fermionic.rank_one_add_blockCov_sq_le
Fermionic.rank_one_add_sq_normalForm
FermionicLean/Chain/Cut.lean